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Vision Impairment and Cognitive Decline Among Older Adults: A Systematic Review

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ABSTRACT

OBJECTIVES

There has been increasing epidemiological research examining the association between visual impairment (VI) and cognitive impairment and how poor vision may be a modifiable risk factor for cognitive decline. The objective of this systematic review is to synthesize the published literature on the association of VI with cognitive decline, cognitive impairment, or dementia, which may inform future research, develop interventions, and guide policies that address these important public health issues.

METHODS

A literature search was performed in Embase, Medline, and Cochrane library databases from inception to March 2020, and included abstracts and articles published in peer-reviewed journals in English. The review included publications that contained subjective and/or objective measures of vision and cognition, or a diagnosis of VI, cognitive impairment or dementia. Longitudinal or cross-sectional studies with ≥ 100 participants aged >50 years were included. The literature search identified 11,805 articles whose abstracts underwent independent screening by three teams of study authors. Data abstraction and quality assessment using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP) were performed by one author (NN). 10% of the articles underwent abstraction and appraisal by a second author (LA/VV), results were compared between both and were in agreement.

RESULTS

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110 full-text articles were selected for data extraction, of which 53 were cross-sectional, 43 longitudinal, and 14 were case-control studies. The mean age of participants was 73.0 years (range 50-93.1). Ninety-one (83%) of these studies reported that VI was associated with cognitive impairment.

CONCLUSION

Our systematic review indicates that a majority of studies examining the vision-cognition relationship report that VI is associated with cognitive decline, cognitive impairment, or dementia among older adults. This evidence synthesis supports the need for additional research to understand the mechanisms underlying the association between VI and cognitive impairment and to test interventions that mitigate the cognitive consequences of VI.

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Article Summary

Strengths and limitations of this study:

- There was heterogeneity in the measurement of cognitive and visual function among all included studies.
- The QA tool used for assessing quality of included studies penalized longitudinal studies that lost over 40% of participants due to dropouts/withdrawals, which is common in studies that span over many years.
- Majority of the included studies were cross-sectional, and these are prone to selection bias.

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Introduction

Dementia is among the most pressing public health challenges of the 21st century.[1] In 2015, 46.8 million people were living with dementia, and the number is expected to double every 20 years.[2] Vision impairment (VI), another major global health problem, affects at least 2.2 billion people worldwide,[3] most of whom are aged 50 years and older.[4] Both cognitive and VI are projected to affect an increasing number of people over time, primarily due to population ageing.[4,5]

Prior work has suggested that cognition and vision are associated,[6-8] and while there are shared risk factors, there is also longitudinal evidence that VI is associated with cognitive changes.[9, 123] The mechanisms underlying the vision-cognition relationship have not yet been fully characterized, but it is hypothesized that sensory loss, such as hearing and VI, may lead to increased cognitive load, structural and functional changes in the brain, and decreased emotional and social well-being, all of which could potentially increase the risk of cognitive impairment.[9,10] While the role of treating hearing loss in preventing cognitive impairment has been acknowledged, VI has not yet been recognized as a potentially modifiable risk factor for cognitive impairment.[1, 11, 127]

Since the majority of VI is due to correctable conditions, namely refractive error and cataract,[12] establishing the existence of an association between vision and cognitive impairment could present an additional opportunity to prevent cognitive impairment and dementia through interventions that optimize vision. In this systematic review of the literature, we examined the association between cognitive and vision impairment among older adults in existing observational studies. This qualitative review summarizes the existing research examining the

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vision-cognition relationship, providing insight on data gaps and areas for continued investigation, as well as highlighting differences in methodological approaches that may impact the interpretation of results across studies.

Methods/Literature Search

We followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (**e-supplement 1**).

Cross-sectional and longitudinal studies reporting a measure of association between visual function and cognitive impairment were included if they had ≥100 participants aged of ≥50 years (mean) at baseline.

An academic librarian searched: Ovid Medline, Embase, Cochrane, and PubMed from their inception through March 2020, and developed a search strategy that combined controlled vocabulary and keywords related to geriatrics, cognition, and vision (**e-supplement 2**-complete Ovid Medline search strategy). Searches were limited to human studies published in English. Conference and poster abstracts, and short oral presentations were also included.

Search results were exported to Covidence (Veritas Health Innovation, Melbourne, Australia). Three teams of two reviewers each worked independently and in duplicate to screen titles, abstracts, and full-text articles to determine inclusion (NN and MM; EC and YS; VV and LA). Disagreements were adjudicated by a member of the other study team.

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Data were extracted from the included publications by one author (NN), and another (LA) extracted data from a random sample of 10% to compare results. Data collected for each publication included: study design, participant characteristics, vision and cognition assessment methods, and the summary measure that described the vision-cognition association.

The methodological quality of included studies was assessed by one author (NN) using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP),[13] and the global quality ratings and findings were summarized qualitatively. A random 10% sample was reviewed by another author (VV) to ascertain consistency in quality assessment.

Results

Study Selection

e-supplement 3 is a PRISMA flowchart that describes the results of the search strategy of articles that examined the association between VI and cognitive impairment or decline. Of the 11,805 studies that were imported for screening, 110 articles were included in our final systematic review.[14-123] Reasons for exclusion of studies were: 1. Outcome measure was not vision or cognition, 2. Association between vision and cognition was not explored, 3. Sample size <100, 4. Publication not in English, 5. Mean age <50 years, 6. No cognitive measure, 7. No vision measure, 8. Outcome was not part of inclusion criteria.

Description of Included Studies

Table 1 describes the characteristics of the studies included. The total number of participants in this review was 9,799,329 (range: 112 to 7,210,535 per study), with a mean age of 73.0 years,

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(range: 50.0 to 93.1). Of the total 110 studies included, 53 were cross-sectional, 43 were longitudinal, and 14 had a case-control study design. The range of follow-up time for the longitudinal studies was 2 months to 10 years.

Of the 110 studies included, 51 reported findings from participants enrolled in population-based studies. There were five studies each from the following large population based longitudinal studies: English Longitudinal Study of Aging (ELSA) and The Three-City Study. Three studies each from Fujiwara-Kyo Study, Salisbury Eye Evaluation Study, Irish Longitudinal Study on Aging (ILSA), and Singapore Epidemiology of Eye Diseases (SEED). Two studies each came from The Newcastle 85+ study, Study of Osteoporotic Fractures (SOF), Blue Mountain Eye Study, Australian Longitudinal Study on Aging (ALSA), Health and Retirement Study (HRS), National Health and Aging Trends Study (NHATS), Leiden 85+ study, Health ABC study, and the Singapore Malay Eye Study (SiMES). Additionally, 10 studies used insurance claims data from different countries.

The studies in this review included participants from over 17 different countries (**Table 1**), 30 studies (27%) from the United States, followed by 25 studies from Europe (23%) including the United Kingdom, Germany, Ireland, Finland, Switzerland, France, and Netherlands. 90 of the studies were published between 2009 to 2020. All papers provided a description of sampling methods. 16 studies were included which were either conference abstracts or short oral presentations.[24, 32, 45, 49, 50, 51, 53, 56, 93, 94, 97, 113, 114, 115, 116, 117]

Assessment of Cognitive Function

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To assess cognitive function, 89 studies used objective assessments, 13 used other assessment methods such as self-report and diagnosis codes, and 7 studies used a combination of both **(Table 2);** one study did not provide information about cognitive function assessment.[93] Mini-Mental State Examination (MMSE) was the most commonly used objective method to assess cognitive function (42 studies). Other methods used to objectively measure cognition included: Montreal-Cognitive Assessment test (MoCA),[28, 32, 34, 53, 94] Addenbrooke's Cognitive Examination-Revised (ACE-R),[45, 61, 80] Cognitive Performance Scale (CPS),[42, 49, 66] Blessed-Orientation-Memory-Concentration test (BOMC),[25, 63] Abbreviated Mental Test (AMT),[73, 75, 90, 101, 111, 120] Blessed Dementia Scale (BDS), [87] Digit Symbol Substitution Test (DSST),[22, 123] and Cambridge Cognitive test (CAMCOG). [89] For the studies that used other assessment methods, 4 used self-reported cognitive measures, [18, 22, 39, 106] and 9 used diagnostic codes to define cognitive decline and/or dementia. [19, 20, 21, 51, 70, 100, 104, 116, 118] Among the studies that used a combination of objective and subjective methods, 4 used self-reported cognitive function along with an objective measure. [22, 30, 48, 55]

Assessment of Visual Function

In order to assess visual function, 66 studies used objective assessments, 34 used other assessment methods such as self-report and diagnosis codes, and 8 studies used a combination of both **(Table 2)**; no information was available from two studies.[24, 117] Visual acuity (VA) was the most commonly measured visual function (42 studies), of which the Snellen acuity chart was the most commonly used method (18 studies). VA was also measured in combination with other visual functions, including: visual fields (VF) (6 studies), [20, 51, 71, 91, 103, 119] contrast sensitivity (CS) (8 studies), [29, 52, 87, 94, 95, 103, 119, 123] macular pigment optical density (MPOD) (2 studies), [14,

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[34] and fundus photography (2 studies).[64, 119] Other methods used to objectively measure visual functions included: color vision,[17] VF only,[28] CS only,[35, 77] fundus photo with grading,[38, 101] and autorefraction.[75] Other assessment methods included: self-reported vision (24 studies) and diagnostic codes or patient records to define VI (10 studies). [19, 21, 46, 70, 84, 100, 105, 116, 118, 122] The studies that used a combination of methods, 8 studies used self-report along with an objective measure of visual function.[22, 40, 53, 59, 69, 72, 108, 114]

Quality of Studies

The methodological quality of included studies was assessed using the EPHPP.[13] The tool assessed each study on 5 domains: 1) Selection bias, 2) Study design, 3) Confounders, 4) Data collection methods, and 5) Analysis. For each included study the five relevant domains were ranked on a three-point Likert scale with three representing a low risk of bias ("strong"), two a possible risk of bias ("moderate"), and one a high risk of bias ("weak"). An overall rating was derived following the EPHPP methodology. A study consisting of at least one "weak" rating in a domain received an overall rating of "moderate," while those with two or more domains with "weak" ratings were automatically classified as "weak" overall. We present our studies in three different tables which is categorized based on the overall ratings, with "strong", "moderate", and "weak" studies in **Tables 3, 4, and 5,** respectively. In our sample, 17 studies received a rating of strong, 70 moderate, and 23 weak.

Study Findings

Of the 110 studies included (**Tables 3, 4 and 5**), 91 found a significant positive association between VI and cognitive decline, cognitive impairment or dementia, and 13 studies found no

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significant association. [26, 30, 43, 44, 60, 61, 69, 82, 83, 91, 95, 116, 118] There were 6 studies that were inconclusive. [23, 37, 68, 71, 80, 84] Of the 91 studies that found a significant association, 77 used objective methods to assess their vision or cognitive outcome. Of the 43 longitudinal studies, 38 found a significant association between VI and cognitive decline, cognitive impairment or dementia. The most commonly presented statistical measures were odds ratios (OR) and hazard ratios (HR). The random 10% of the study sample that was separately extracted by an independent author (LA) was found to be similar to elements from the primary extraction.

Discussion

In this systematic review, we evaluated the literature examining the association between VI and cognitive function among older adults and found strong agreement among published studies that VI is associated with cognitive impairment, cognitive decline, or dementia.

Ninety-one studies reported associations between decline in visual and cognitive functions. Garin et al [40], who received a "moderate" rating in the quality assessment (QA), performed a cross-sectional analysis in a representative sample of Spanish population and measured cognition objectively. They also measured distance and near vision and found that objective and subjectively measured poor distance and near VA were associated with worse cognitive functioning (Poor objective VA and Cognition (OR): Distance VA: 1.47, 95% CI 1.17-1.86, Near VA: 1.72, 95% CI 1.46-2.02 and Poor subjective VA and cognition (OR): Distance VA: 2.05, 95% CI 1.44-2.90, Near VA: 2.96, 95% CI 1.92-4.55). Lin et al [47], used data from a large longitudinal cohort study of older women and found that VI was associated with greater odds of

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cognitive and functional decline over 2 years (Cognitive decline: OR=1.78, 95% CI 1.21-2.61, Functional decline: OR=1.79, 95% CI 1.15-2.79). This study used objective measures of assessment for both vision and cognition and received a "strong" rating in the bias assessment. Luo et al [48], who received a "moderate" rating in QA, performed a cross-sectional analysis on a large population sample from China. They reported that those with VI and Dual Sensory Impairment (DSI) were more likely to have severe to extremely severe dementia compared to those without any sensory impairment (VI: OR=1.54, 95% CI 1.27-1.86, DSI: OR=1.63, 95% CI 1.25-2.11). Another longitudinal study that received a "moderate" rating in QA from Germany by Hajek et al [58] with a large sample size (n=2,394) showed that the onset of severe VI was associated with a decline in cognitive function scores (β= -0.38, p <0.001). Uhlmann et al [65] in their paired case-control study between VI and dementia patients concluded that VI was associated with both an increased risk and an increased clinical severity of AD (Near-VI: OR=2.5, 95% CI 1.1-10.5, Distance-VI: OR=1.9, 95% CI 0.8-4.6). Although Frost et al [38] found a strong association between early Age-related Macular Degeneration (AMD) and AD (OR=18.67, 95% CI 4.42-78.80), their study was cross-sectional, and the sample size was too low to derive an inference. Both these studies received "moderate" rating in QA. Davies-Kershaw et al [96] in their longitudinal analysis using the ELSA wave 2 and wave 7 data found that individuals in the younger group (50–69 years) and with moderate (HR=1.78, 95% CI=1.04–3.04) and poor (HR=3.60, 95%CI=1.10–11.78) self-rated vision were at greater risk of developing dementia than those with normal self-rated vision. Hamedani et al [100] used Medicare claims data from 2014 consisting of 472,871 participants and concluded that blindness/low vision was associated with a greater odd of Alzheimer's disease (AOR 1.44 95% CI: 1.415–1.464) and all-cause dementia (AOR 1.239, 95% CI: 1.223–1.254). Both these studies also

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received “moderate” rating in QA. Soto-Perez-De-Celis et al in their cross-sectional study case-control study found that DSI was significantly associated with possible CI (OR 3.2 95% CI: 1.3-8.1). However, the study received an overall “weak” rating in QA.

Of the 13 studies that found no association between VI and cognitive function, 6 were longitudinal, 5 were cross-sectional and 2 were case-control studies. Ihle et al [60] performed a cross-sectional analysis using a sample of 2,812 participants from Switzerland. They objectively measured cognition and vision and concluded that their data did not support an increased relation of cognitive and sensory abilities in old age. This study received a “weak” rating in QA. Hong T et al [83] used data from Blue Mountain Eye Study, a longitudinal study from Australia that studied associations between VI and a decline in MMSE scores over a duration of 10 years. The study concluded that VI was not associated with cognitive decline over 5 years (OR= 0.84, 95% CI 0.40–1.79) or 10 years (OR 1.09, 95% CI 0.52–2.30). Although the study included a large number of participants overall (n=2,334), only 152 individuals with VI were included in this analysis, suggesting that there may have been survival bias. Brenowitz et al [95] in their longitudinal study using the Health ABC data concluded that VA and CS independently were not significantly associated with incident dementia (VA: HR= 1.26, 95% CI 0.90-1.77, CS: HR=1.11 95% CI 0.88-1.38). However, Swenor et al [123] used data from the same study and found that impaired visual acuity, contrast sensitivity, and stereo acuity had a greater risk of incident cognitive impairment (VA impairment HR=1.55, 95% CI 1.12, 2.14 vs No VA impairment). This could be due to the different outcome measures assessed, i.e. dementia [95] vs cognitive impairment[123]. These three studies received a “moderate” rating in QA. Michalowsky et al [118], who received a “strong” rating in their case-control study concluded that VI was not significantly

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associated with dementia (OR = 0.97, 95% CI 0.97–1.02), a combination of both visual and hearing impairments was significantly associated with the risk of dementia (OR = 1.14, 95% CI 1.04–1.24).

There was considerable heterogeneity in the measurement and reporting of cognitive function. Studies measured cognitive function using a variety of instruments with the most common being MMSE. The MMSE is a paper-based test with a maximum score of 30, with lower scores indicating more severe cognitive impairment. A score of 24 is often used as a threshold to define "normal" cognitive function.[124] The MMSE has been found to be a valid and reliable tool as assesses by many studies.[124, 125] Several studies used self-report, diagnosis codes and data from existing records to define cognitive status. Similarly, visual function was also assessed by various methods including self-report. While VA was assessed most commonly, there was significant variation in the charts and tools used to assess it. The parameters used to define cognitive decline and VI may have impacted results across and within studies.

Our systematic review has found that there is a strong consensus in the literature that VI is associated with cognitive decline, cognitive impairment, or dementia. Two hypotheses may help explain this association. The first one is that a common pathological process (e.g. vascular disease) might be responsible for both the sensory and cognitive impairment in older adults. The second one is that by increasing cognitive load, sensory impairments such as VI might cause cognitive impairment. [126]

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Our review evaluated bias for all of the 110 included studies. The majority of studies included in our review were cross-sectional, and according to EPHPP guidelines, cross-sectional studies can only receive a low or moderate rating in the bias assessment. Cross-sectional studies are also prone to selection bias, thus yielding estimates that may not reflect true associations in the target population. Studies receiving a strong rating were all longitudinal. However, the tool penalizes longitudinal studies that lose >40% of participants due to dropouts/withdrawals. This may, perhaps unfairly, affect longer longitudinal studies to a greater extent since they collect data over many years and can have more dropouts due to deaths since they are conducted among older adults.

This review has several important implications. First, it highlights the need for standardized methods to assess and define both visual and cognitive function that will aid future research on these emerging public health issues. Second, it brings into focus the consistent association of VI with cognitive impairment in older adults and the need to better understand the mechanisms underlying this relationship. Third, as the longitudinal results support the sensory consequence theory, this points to a need for formulating preventive measures and vision rehabilitation models that could improve overall health and well-being of older adults.

Limitations

Given the large number of studies included in this review and the heterogeneity of measures used to assess the outcome, it was difficult to compare and meta-analyze results across studies. The studies included diverse populations, with different disease processes, and variation in definitions of both cognitive and visual impairment. However, despite the heterogeneity in

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studies, we synthesized the evidence qualitatively and by taking into account study quality assessed using a validated tool. While our search strategy was robust, it may have been limited by the exclusion of studies that were not published in English.

Conclusion

The number of older adults with VI and dementia is increasing globally, and therefore the elucidation of the relationship between vision and cognition is of particular public health importance. This systematic review found that the association of VI with cognitive decline, cognitive impairment, or dementia is largely consistent across studies using different measures of vision and cognition, as well as between countries and cohorts. This overall agreement in the literature suggests that poor visual and cognitive function are associated, and that additional research is now needed to move beyond documenting this relationship. The focus of this area of research should now turn to identifying the factors that mediate the vision-cognition relationship and identifying potential interventions that may mitigate the cognitive implications of VI.

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Table 1: Patient Demographics and Study Characteristics

Demographics and Study Characteristics	**No.**
Total Number of studies	110
Total number of participants	9,799,329
Mean age, y, IQR	73.0, (50-93.06)
Study design	
Cross-sectional	53 (48%)
Longitudinal	43 (39%)
Case-Control	14 (13%)
Country	
US	30 (27%)
Japan	6 (5%)
United Kingdom	6 (5%)
China	6 (5%)
Australia	5 (4.5%)
France	5 (4.5%)
Germany	5 (4.5%)
Singapore	5 (4.5%)
Ireland	3 (3%)
Canada	3 (3%)
Spain	2 (2%)
Taiwan	2 (2%)

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Sweden	1 (<1%)
Nordic Countries	1 (<1%)
New Zealand	1 (<1%)
Switzerland	1 (<1%)
Netherland	1 (<1%)
Others	27 (25%)

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Table 2: Measures of Vision and Cognition Assessed in Studies

Outcome Measures	Number of studies
Cognition (n, %)	
Objective assessment n=89 (81%)	
MMSE	42 (47%)
MoCA	5 (6%)
Global cognition scale	6 (7%)
Others	36 (40%)
Other assessment methods n=13 (12%)	
Self-report	4 (31%)
ICD diagnosis/from records	9 (69%)
Combination of Objective + Subjective (5%)	7
No info (<1%)	1
Vision (n, %)	
Objective n=66 (70%)	
Acuity	42 (65%)
Acuity + others	16 (23%)
Others	8 (12%)
Other assessment methods n=34 (22%)	
Self-report	24 (71%)

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ICD diagnosis/from records	10 (29%)
Combination of Objective + Subjective *(7.5%)*	 8
No info (<1%)	2

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Table 3: Studies with a "strong" rating

Table 4: Studies with a "moderate" rating

Table 5: Studies with a "weak" rating

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2 & 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	e-supplement 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency ($e.g., I^2$) for each meta-analysis.	NA

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7, 8 & 9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10, 11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11, 12, 13, 14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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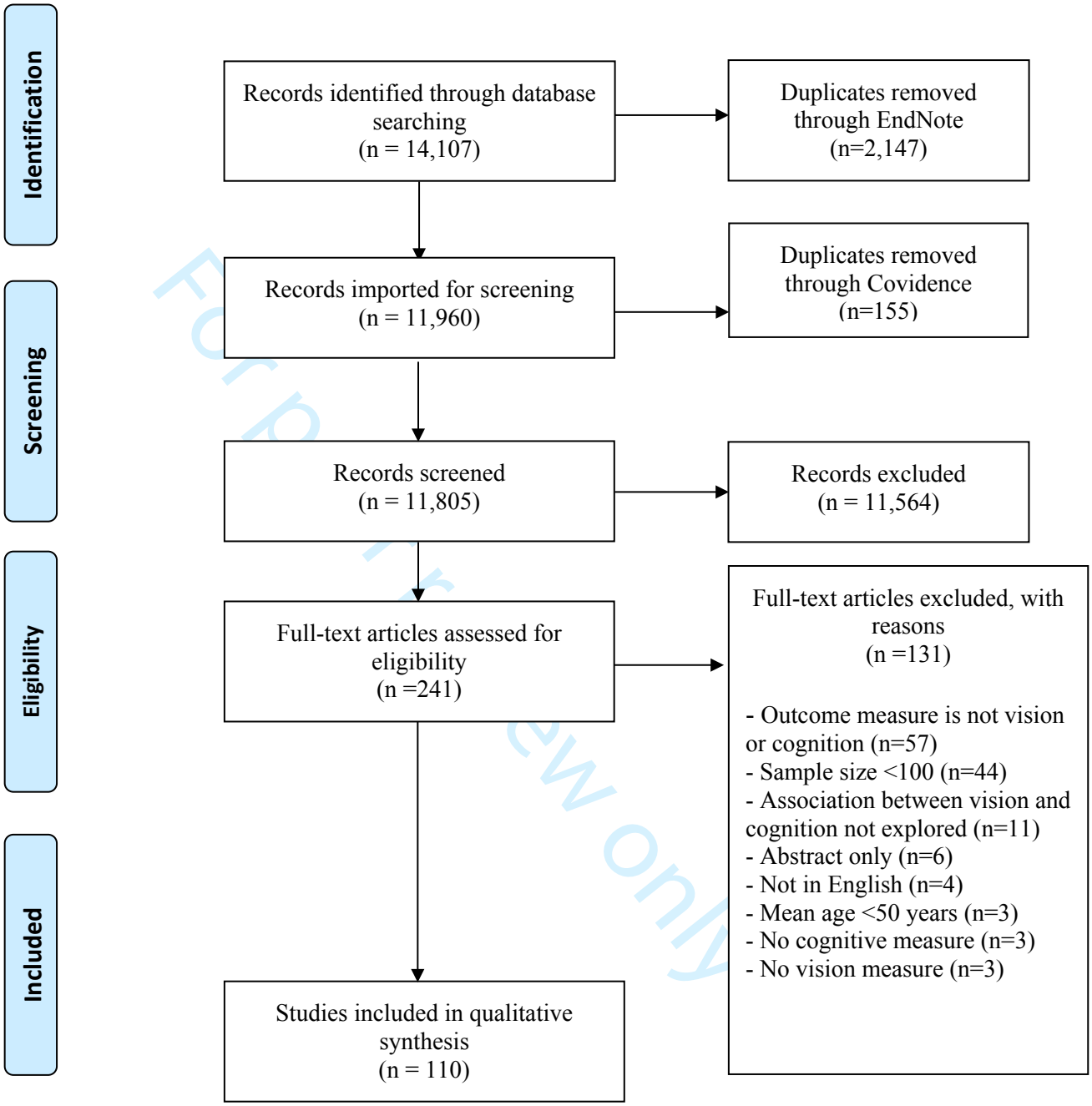
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e-Supplement 2 – Ovid Medline Search Strategy

1. exp Aged/ or exp Geriatrics/
2. (ageing or latelife or laterlife or "late life" or "later life" or elder* or Geron* or Geriatr* or senior*).tw.
3. (old* adj3 (age* or adult* or population* or communit* or people* or person*)).tw.
4. ((aging or aged) adj3 (adult* or population* or communit* or people* or person*)).tw.
5. 1 or 2 or 3 or 4
6. exp Cognitive Aging/ or Cognition/ or exp Cognition Disorders/ or exp Executive Function/ or Memory Disorders/ or exp Neuropsychological Tests/ or Dementia/ or exp Alzheimer Disease/
7. (cognitive or cognition or executive function* or executive control* or executive dysfunction* or executive impairment* or memory or dementia* or alzheimer*).tw.
8. 6 or 7
9. exp Vision, Ocular/
10. exp Vision Disorders/
11. exp Diagnostic Techniques, Ophthalmological/
12. exp Visual Perception/
13. exp Visually Impaired Persons/
14. (vision or visual or visually or occular or Ophthalmol*).tw.
15. 9 or 10 or 11 or 12 or 13 or 14
16. 5 and 8 and 15
17. Animals/ not humans/
18. 16 not 17
19. Epidemiologic Studies/ or exp Observational Study/ or exp Observational Studies as Topic/ or exp Cohort Studies/
20. ("epidemiologic study" or "epidemiologic studies" or "cohort study" or "cohort studies" or longitudinal or prospective* or "observational study" or "observational studies").tw.
21. 19 or 20
22. 18 and 21

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e-supplement 3: FIGURE DESCRIBING A PRISMA FLOWCHART



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BMJ Open

Vision Impairment and Cognitive Decline Among Older Adults: A Systematic Review

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ABSTRACT

OBJECTIVES

There has been increasing epidemiological research examining the association between visual impairment (VI) and cognitive impairment and how poor vision may be a modifiable risk factor for cognitive decline. The objective of this systematic review is to synthesize the published literature on the association of VI with cognitive decline, cognitive impairment, or dementia, to aid the development of interventions, and guide public policies pertaining to the relationship between vision and cognition.

METHODS

A literature search was performed with Embase, Medline, and Cochrane library databases from inception to March 2020, and included abstracts and articles published in peer-reviewed journals in English. Our inclusion criteria included publications that contained subjective/objective measures of vision and cognition, or a diagnosis of VI, cognitive impairment or dementia. Longitudinal or cross-sectional studies with ≥ 100 participants aged >50 years were included. The search identified 11,805 articles whose abstracts underwent screening by three teams of study authors. Data abstraction and quality assessment using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP) were performed by one author (NN). 10% of the articles underwent abstraction and appraisal by a second author (LA/VV), results were compared between both and were in agreement.

RESULTS

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110 full-text articles were selected for data extraction, of which 53 were cross-sectional, 43 longitudinal, and 14 were case-control studies. The mean age of participants was 73.0 years (range 50-93.1). Ninety-one (83%) of these studies reported that VI was associated with cognitive impairment.

CONCLUSION

Our systematic review indicates that a majority of studies examining the vision-cognition relationship report that VI is associated with more cognitive decline, cognitive impairment, or dementia among older adults. This synthesis supports the need for additional research to understand the mechanisms underlying the association between VI and cognitive impairment and to test interventions that mitigate the cognitive consequences of VI.

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Article Summary

Strengths and limitations of this study:

- There was heterogeneity in the measurement of cognitive and visual function among all included studies.
- The QA tool used for assessing quality of included studies penalized longitudinal studies that lost over 40% of participants due to dropouts/withdrawals, which is common in studies that span over many years.
- Majority of the included studies were cross-sectional, and these are prone to selection bias.

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Introduction

Dementia is among the most pressing public health challenges of the 21st century.[1] In 2015, 46.8 million people were living with dementia, and the number is expected to double every 20 years.[2] Vision impairment (VI), another major global health problem, affects at least 2.2 billion people worldwide,[3] most of whom are aged 50 years and older.[4] Both cognitive and VI are projected to affect an increasing number of people over time, primarily due to population ageing.[4,5]

Prior work has suggested that cognition and vision are associated,[6, 7] and while there are shared risk factors (neuropathological/vascular),[8] there is also longitudinal evidence that VI is associated with cognitive changes.[9] The mechanisms underlying the vision-cognition relationship have not yet been fully characterized, but it is hypothesized that sensory loss, such as hearing and VI, may lead to increased cognitive load, structural and functional changes in the brain, and decreased emotional and social well-being, all of which could potentially increase the risk of cognitive impairment.[9,10] While the role of treating hearing loss in preventing cognitive impairment has been acknowledged, VI has not yet been recognized as a potentially modifiable risk factor for cognitive impairment.[1, 11]

Since the majority of VI is due to correctable conditions, namely refractive error and cataract,[12] establishing the existence of an association between vision and cognitive impairment could present an additional opportunity to prevent cognitive impairment and dementia through interventions that optimize vision. In this systematic review of the literature, we examined the association between cognitive and vision impairment among older adults in existing observational studies. This qualitative review summarizes the existing research examining the

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vision-cognition relationship, providing insight on data gaps and areas for continued investigation, as well as highlighting differences in methodological approaches that may impact the interpretation of results across studies.

Methods/Literature Search

We followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (**e-supplement 1**).

Cross-sectional and longitudinal studies reporting a measure of association between visual function and cognitive impairment were included if they had ≥100 participants aged of ≥50 years (mean) at baseline. Reasons for exclusion of studies were: 1. Outcome measure was not vision or cognition, 2. Association between vision and cognition was not explored, 3. Sample size <100, 4. Publication not in English, 5. Mean age <50 years, 6. No cognitive measure, 7. No vision measure, 8. Outcome was not part of inclusion criteria.

An academic librarian searched: Ovid Medline, Embase, Cochrane, and PubMed from their inception through March 2020, and developed a search strategy that combined controlled vocabulary and keywords related to geriatrics, cognition, and vision (**e-supplement 2**-complete Ovid Medline search strategy). Searches were limited to human studies published in English. Conference and poster abstracts, and short oral presentations were also included.

Search results were exported to Covidence (Veritas Health Innovation, Melbourne, Australia). Three teams of two reviewers each worked independently and in duplicate to screen titles,

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abstracts, and full-text articles to determine inclusion (NN and MM; EC and YS; VV and LA). Disagreements were adjudicated by a member of the other study team.

Data were extracted from the included publications by one author (NN), and another (LA) extracted data from a random sample of 10% to compare results. Any discrepancies were adjudicated by a third author (VV). Data collected for each publication included: study design, participant characteristics, vision and cognition assessment methods, and the summary measure that described the vision-cognition association.

The methodological quality of included studies was assessed by one author (NN) using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP),[13] and the global quality ratings and findings were summarized qualitatively. A random 10% sample was reviewed by another author (VV) to ascertain consistency in quality assessment.

Results

Study Selection

e-supplement 3 is a PRISMA flowchart that describes the results of the search strategy of articles that examined the association between VI and cognitive impairment or decline. Of the 11,805 studies that were imported for screening, 110 articles were included in our final systematic review.[14-123]

Description of Included Studies

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Table 1 describes the characteristics of the studies included. The total number of participants in this review was 9,799,329 (range: 112 to 7,210,535 per study), with a mean age of 73.0 years, (range: 50.0 to 93.1). Of the total 110 studies included, 53 were cross-sectional, 43 were longitudinal, and 14 had a case-control study design. The range of follow-up time for the longitudinal studies was 2 months to 10 years.

Of the 110 studies included, 51 reported findings from participants enrolled in population-based studies. There were five studies each from the following large population based longitudinal studies: English Longitudinal Study of Aging (ELSA) and The Three-City Study. Three studies each from Fujiwara-Kyo Study, Salisbury Eye Evaluation Study, Irish Longitudinal Study on Aging (ILSA), and Singapore Epidemiology of Eye Diseases (SEED). Two studies each came from The Newcastle 85+ study, Study of Osteoporotic Fractures (SOF), Blue Mountain Eye Study, Australian Longitudinal Study on Aging (ALSA), Health and Retirement Study (HRS), National Health and Aging Trends Study (NHATS), Leiden 85+ study, Health ABC study, and the Singapore Malay Eye Study (SiMES). Additionally, 10 studies used insurance claims data from different countries.

The studies in this review included participants from over 17 different countries (**Table 1**), 30 studies (27%) from the United States, followed by 25 studies from Europe (23%) including the United Kingdom, Germany, Ireland, Finland, Switzerland, France, and Netherlands. 90 of the studies were published between 2009 to 2020. All papers provided a description of sampling methods. 16 studies were included which were either conference abstracts or short oral presentations.[24, 32, 45, 49, 50, 51, 53, 56, 93, 94, 97, 113, 114, 115, 116, 117]

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Assessment of Cognitive Function

To assess cognitive function, 89 studies used objective assessments, 13 used other assessment methods such as self-report and diagnosis codes, and 7 studies used a combination of both **(Table 2);** one study did not provide information about cognitive function assessment.[93] Mini-Mental State Examination (MMSE) was the most commonly used objective method to assess cognitive function (42 studies). Other methods used to objectively measure cognition included: Montreal-Cognitive Assessment test (MoCA),[28, 32, 34, 53, 94] Addenbrooke's Cognitive Examination-Revised (ACE-R),[45, 61, 80] Cognitive Performance Scale (CPS),[42, 49, 66] Blessed-Orientation-Memory-Concentration test (BOMC),[25, 63] Abbreviated Mental Test (AMT),[73, 75, 90, 101, 111, 120] Blessed Dementia Scale (BDS), [87] Digit Symbol Substitution Test (DSST),[22, 123] and Cambridge Cognitive test (CAMCOG). [89] For the studies that used other assessment methods, 4 used self-reported cognitive measures, [18, 22, 39, 106] and 9 used diagnostic codes to define cognitive decline and/or dementia. [19, 20, 21, 51, 70, 100, 104, 116, 118] Among the studies that used a combination of objective and subjective methods, 4 used self-reported cognitive function along with an objective measure. [22, 30, 48, 55]

Assessment of Visual Function

In order to assess visual function, 66 studies used objective assessments, 34 used other assessment methods such as self-report and diagnosis codes, and 8 studies used a combination of both **(Table 2)**; no information was available from two studies.[24, 117] Visual acuity (VA) was the most commonly measured visual function (42 studies), of which the Snellen acuity chart was the most commonly used method (18 studies). VA was also measured in combination with other

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visual functions, including: visual fields (VF) (6 studies), [20, 51, 71, 91, 103, 119] contrast sensitivity (CS) (8 studies), [29, 52, 87, 94, 95, 103, 119, 123] macular pigment optical density (MPOD) (2 studies), [14, 34] and fundus photography (2 studies).[64, 119] Other methods used to objectively measure visual functions included: color vision,[17] VF only,[28] CS only,[35, 77] fundus photo with grading,[38, 101] and autorefraction.[75] Other assessment methods included: self-reported vision (24 studies) and diagnostic codes or patient records to define VI (10 studies). [19, 21, 46, 70, 84, 100, 105, 116, 118, 122] The studies that used a combination of methods, 8 studies used self-report along with an objective measure of visual function.[22, 40, 53, 59, 69, 72, 108, 114]

Quality of Studies

The methodological quality of included studies was assessed using the EPHPP.[13] The tool assessed each study on 5 domains: 1) Selection bias, 2) Study design, 3) Confounders, 4) Data collection methods, and 5) Analysis. For each included study the five relevant domains were ranked on a three-point Likert scale with three representing a low risk of bias ("strong"), two a possible risk of bias ("moderate"), and one a high risk of bias ("weak"). An overall rating was derived following the EPHPP methodology. A study consisting of at least one "weak" rating in a domain received an overall rating of "moderate," while those with two or more domains with "weak" ratings were automatically classified as "weak" overall. We present our studies in three different tables which is categorized based on the overall ratings, with "strong", "moderate", and "weak" studies in **Tables 3, 4, and 5,** respectively. In our sample, 17 studies received a rating of strong, 70 moderate, and 23 weak.

Study Findings

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Of the 110 studies included (**Tables 3, 4 and 5**), 91 found a significant positive association between VI and cognitive decline, cognitive impairment or dementia, and 13 studies found no significant association. [26, 30, 43, 44, 60, 61, 69, 82, 83, 91, 95, 116, 118] There were 6 studies that were inconclusive. [23, 37, 68, 71, 80, 84] Of the 91 studies that found a significant association, 77 used objective methods to assess their vision or cognitive outcome. Of the 43 longitudinal studies, 35 found a significant association between VI and cognitive decline, cognitive impairment or dementia. The most commonly presented statistical measures were odds ratios (OR) and hazard ratios (HR). The random 10% of the study sample that was separately extracted by an independent author (LA) was found to be similar to elements from the primary extraction.

Discussion

In this systematic review, we evaluated the literature examining the association between VI and cognitive function among older adults and found strong agreement among published studies that VI is associated with cognitive impairment, cognitive decline, or dementia.

Ninety-one studies reported associations between decline in visual and cognitive functions. Garin et al [40], who received a "moderate" rating in the quality assessment (QA), performed a cross-sectional analysis in a representative sample of Spanish population and measured cognition objectively. They also measured distance and near vision and found that objective and subjectively measured poor distance and near VA were associated with worse cognitive functioning (Poor objective VA and Cognition (OR): Distance VA: 1.47, 95% CI 1.17-1.86, Near VA: 1.72, 95% CI 1.46-2.02 and Poor subjective VA and cognition (OR): Distance VA:

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2.05, 95% CI 1.44-2.90, Near VA: 2.96, 95% CI 1.92-4.55). Lin et al [47], used data from a large longitudinal cohort study of older women and found that VI was associated with greater odds of cognitive and functional decline over 2 years (Cognitive decline: OR=1.78, 95% CI 1.21-2.61, Functional decline: OR=1.79, 95% CI 1.15-2.79). This study used objective measures of assessment for both vision and cognition and received a "strong" rating in the bias assessment. Luo et al [48], who received a "moderate" rating in QA, performed a cross-sectional analysis on a large population sample from China. They reported that those with VI and Dual Sensory Impairment (DSI) were more likely to have severe to extremely severe dementia compared to those without any sensory impairment (VI: OR=1.54, 95% CI 1.27-1.86, DSI: OR=1.63, 95% CI 1.25-2.11). Another longitudinal study that received a "moderate" rating in QA from Germany by Hajek et al [58] with a large sample size (n=2,394) showed that the onset of severe VI was associated with a decline in cognitive function scores (β= -0.38, p <0.001). Uhlmann et al [65] in their paired case-control study between VI and dementia patients concluded that VI was associated with both an increased risk and an increased clinical severity of AD (Near-VI: OR=2.5, 95% CI 1.1-10.5, Distance-VI: OR=1.9, 95% CI 0.8-4.6). Although Frost et al [38] found a strong association between early Age-related Macular Degeneration (AMD) and AD (OR=18.67, 95% CI 4.42-78.80), their study was cross-sectional, and the sample size was too low to derive an inference. Both these studies received "moderate" rating in QA. Davies-Kershaw et al [96] in their longitudinal analysis using the ELSA wave 2 and wave 7 data found that individuals in the younger group (50–69 years) and with moderate (HR=1.78, 95% CI=1.04–3.04) and poor (HR=3.60, 95%CI=1.10–11.78) self-rated vision were at greater risk of developing dementia than those with normal self-rated vision. Hamedani et al [100] used Medicare claims data from 2014 consisting of 472,871 participants and concluded that blindness/low

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vision was associated with a greater odd of Alzheimer's disease (AOR 1.44 95% CI: 1.415–1.464) and all-cause dementia (AOR 1.239, 95% CI: 1.223–1.254). Both these studies also received "moderate" rating in QA. Soto-Perez-De-Celis et al in their cross-sectional study case-control study found that DSI was significantly associated with possible CI (OR 3.2 95% CI: 1.3-8.1). However, the study received an overall "weak" rating in QA.

Of the 91 studies that found an association between VI and cognitive function, 35 were longitudinal, 46 were cross-sectional and 10 were case-control studies. Of the 13 studies that found no association between VI and cognitive function, 6 were longitudinal, 5 were cross-sectional and 2 were case-control studies. Ihle et al [60] performed a cross-sectional analysis using a sample of 2,812 participants from Switzerland. They objectively measured cognition and vision and concluded that their data did not support an increased relation of cognitive and sensory abilities in old age. This study received a "weak" rating in QA. Hong T et al [83] used data from Blue Mountain Eye Study, a longitudinal study from Australia that studied associations between VI and a decline in MMSE scores over a duration of 10 years. The study concluded that VI was not associated with cognitive decline over 5 years (OR= 0.84, 95% CI 0.40–1.79) or 10 years (OR 1.09, 95% CI 0.52–2.30). Although the study included a large number of participants overall (n=2,334), only 152 individuals with VI were included in this analysis, suggesting that there may have been survival bias. Brenowitz et al [95] in their longitudinal study using the Health ABC data concluded that VA and CS independently were not significantly associated with incident dementia (VA: HR= 1.26, 95% CI 0.90-1.77, CS: HR=1.11 95% CI 0.88-1.38). However, Swenor et al [123] used data from the same study and found that impaired visual acuity, contrast sensitivity, and stereo acuity had a greater risk of incident cognitive impairment (VA

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impairment HR=1.55, 95% CI 1.12, 2.14 vs No VA impairment). This could be due to the different outcome measures assessed, i.e. dementia [95] vs cognitive impairment[123]. These three studies received a "moderate" rating in QA. Michalowsky et al [118], who received a "strong" rating in their case-control study concluded that VI was not significantly associated with dementia (OR = 0.97, 95% CI 0.97–1.02), a combination of both visual and hearing impairments was significantly associated with the risk of dementia (OR = 1.14, 95% CI 1.04–1.24).

There was considerable heterogeneity in the measurement and reporting of cognitive function. Studies measured cognitive function using a variety of instruments with the most common being MMSE. The MMSE is a paper-based test with a maximum score of 30, with lower scores indicating more severe cognitive impairment. A score of 24 is often used as a threshold to define "normal" cognitive function.[124] The MMSE has been found to be a valid and reliable tool as assesses by many studies.[124, 125] Several studies used self-report, diagnosis codes and data from existing records to define cognitive status. Similarly, visual function was also assessed by various methods including self-report. While VA was assessed most commonly, there was significant variation in the charts and tools used to assess it. The parameters used to define cognitive decline and VI may have impacted results across and within studies.

Our systematic review has found that there is a strong consensus in the literature that VI is associated with cognitive decline, cognitive impairment, or dementia. Two hypotheses may help explain this association. The first one is that a common pathological process (e.g. vascular disease) might be responsible for both the sensory and cognitive impairment in older adults. The second one is that by increasing cognitive load, sensory impairments such as VI might cause

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cognitive impairment. [126] Literature also suggests that vision rehabilitation in the form of cataract surgery slows the rate of cognitive decline, and therefore, early vision interventions could potentially reduce risk of dementia.[127]

Our review evaluated bias for all of the 110 included studies. The majority of studies included in our review were cross-sectional, and according to EPHPP guidelines, cross-sectional studies can only receive a low or moderate rating in the bias assessment. Cross-sectional studies are also prone to selection bias, thus yielding estimates that may not reflect true associations in the target population. Studies receiving a strong rating were all longitudinal. However, the tool penalizes longitudinal studies that lose >40% of participants due to dropouts/withdrawals. This may, perhaps unfairly, affect longer longitudinal studies to a greater extent since they collect data over many years and can have more dropouts due to deaths since they are conducted among older adults.

This review has several important implications. First, it highlights the need for standardized methods to assess and define both visual and cognitive function that will aid future research on these emerging public health issues. Second, it brings into focus the consistent association of VI with cognitive impairment in older adults and the need to better understand the mechanisms underlying this relationship. Third, as the longitudinal results support the sensory consequence theory, this points to a need for formulating preventive measures and vision rehabilitation models that could improve overall health and well-being of older adults.

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Limitations

Given the large number of studies included in this review and the heterogeneity of measures used to assess the outcome, it was not possible to compare and meta-analyze results across studies. The studies included diverse populations, with different disease processes, and variation in definitions of both cognitive and visual impairment. There is also potential bias associated with studies that used different protocols for cognitive and sensory measurements. The MMSE, which was the most commonly used assessment method for testing cognition is sensory dependent and therefore one can argue that the results may be confounded with vision impairment.[128] Further studies should examine the impact of using vision independent cognitive tests on the vision-cognition relationship. Our review examined all cause vision impairment and dementia, and further study is needed to examine the vision-cognition relationship by dementia subtype and by different vision pathology. However, despite the heterogeneity in studies and assessment methods, we synthesized the evidence qualitatively and by taking into account study quality assessed using a validated tool. While our search strategy was robust, it may have been limited by the exclusion of studies that were not published in English.

Conclusion

The number of older adults with VI and dementia is increasing globally, and therefore the elucidation of the relationship between vision and cognition is of particular public health importance. This systematic review found that the association of VI with cognitive decline,

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cognitive impairment, or dementia is largely consistent across studies using different measures of vision and cognition, as well as between countries and cohorts. This overall agreement in the literature suggests that poor visual and cognitive function are associated, and that additional research is now needed to move beyond documenting these associations. The focus of this area of research should now turn to identifying the factors that mediate the vision-cognition relationship and identifying potential interventions that may mitigate the cognitive implications of VI.

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Competing Interest:

a) Any associations with commercial entities that provided support for the work reported in the submitted manuscript (the time frame for disclosure in this section of the form is the lifespan of the work being reported): None
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1 **Table 1: Patient Demographics and Study Characteristics**

Demographics and Study Characteristics	**No.**
Total Number of studies	110
Total number of participants	9,799,329
Mean age, y, IQR	73.0, (50-93.06)
Study design	
Cross-sectional	53 (48%)
Longitudinal	43 (39%)
Case-Control	14 (13%)
Country	
US	30 (27%)
Japan	6 (5%)
United Kingdom	6 (5%)
China	6 (5%)
Australia	5 (4.5%)
France	5 (4.5%)
Germany	5 (4.5%)
Singapore	5 (4.5%)
Ireland	3 (3%)
Canada	3 (3%)
Spain	2 (2%)
Taiwan	2 (2%)

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Sweden	1 (<1%)
Nordic Countries	1 (<1%)
New Zealand	1 (<1%)
Switzerland	1 (<1%)
Netherland	1 (<1%)
Others	27 (25%)

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Table 2: Measures of Vision and Cognition Assessed in Studies

Outcome Measures	Number of studies
Cognition (n, %)	
Objective assessment n=89 (81%)	
MMSE	42 (47%)
MoCA	5 (6%)
Global cognition scale	6 (7%)
Others	36 (40%)
Other assessment methods n=13 (12%)	
Self-report	4 (31%)
ICD diagnosis/from records	9 (69%)
Combination of Objective + Subjective (5%)	7
No info (<1%)	1
Vision (n, %)	
Objective n=66 (70%)	
Acuity	42 (65%)
Acuity + others	16 (23%)
Others	8 (12%)
Other assessment methods n=34 (22%)	
Self-report	24 (71%)

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ICD diagnosis/from records	10 (29%)
Combination of Objective + Subjective *(7.5%)*	8
No info (<1%)	2

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Table 3: Studies with a "strong" rating

Author, Title & Year	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Dearborn, PJ	US (Maine-Syracuse Longitudinal Study (MSLS)	Longitudinal cohort, wave 6&7	655	Normal VA 60.34	40%/60%	Cognition	Visual-Spatial Organization and Memory, Scanning and Tracking, Verbal Episodic & Working memory	Acuity: Snellen eye test (log transformed)	Cross-sectional (CSA) and prospective analysis (PA) using multiple linear and multiple logistic regression	Poorer VA associated with lower cognitive function and 5-yr decline	Strong
2018	compared sixth [2001-2006] wave and			VI 70.6	40%/60%		Global composite score-loaded equally across domains, z-transformed		Global OR: CSA 1.512 (0.71-3.23), PA 1.539 (0.74-3.21), VSMO OR: CSA 1.28 (0.63-2.59), PA 2.26 (1.16-4.4)	in cognitive function over a range of domains incl the global measurements	
	seventh [2006-2010] wave)								Working OR: CSA 1.73 (0.85-3.51), PA 1.55 (0.78-3.08), VEM OR: CSA 0.55 (0.25-1.24), PA 1.43 (0.73-2.81)	Visual-spatial organization, verbal episodic memory	
Diniz-Filho, A.; Delano-Wood, L.; Daga, F. B.;	US	Longitudinal cohort	115	67.4		Cognitive Decline	Montreal Cognitive Assessment test (MoCA); assesses atten and concentration, executive functions	Standard Automated Perimetry (SAP) using 24-2 SITA	Univariate: 5-point decline in MoCA=0.18 dB increase of residuals of SAP MD, R2 = 4.3% (0.06-0.30)p0.003	Statistically significant association between change in	Strong
2017							memory, language, visuoconstructional skills, conceptual thinking, calculations and orientation	<33% fixation loss and <15% FP only included	Multivariate: 5-point decline in MoCA=0.23 dB inc of residuals of SAP MD, (0.11-0.35) p<0.001	MoCA scores and visual field variability over time	
Fischer, M. E	US (The Epidemiology of Hearing Loss Study)	Longitudinal cohort. Baseline info was from EHLS-2 (1998-2000), CI data from EHLS-3 (2003-05)and EHLS-4 (2009-10)	1884	66.7	40.9/49.1	Cognition	MMSE	Contrast sensitivity using Pelli-Robson letter charts. VI was defined as	Cox discrete time proportional hazard analyses performed to model relationship btwn CI and Sensory Imp.	Hearing, Visual and Olfactory impairment were independently	Strong
2016								score <1.55 log units in the better eye	Vision: HR = 2.05 (1.24, 3.38); Olfaction: HR = 3.92 (2.45, 6.26), Hearing: HR= 1.90 (1.11, 3.26)	associated with cognitive impairment risk	
Hall, TA	US (Impact of Cataract on Mobility (ICOM) study)	Longitudinal cohort study. 3 groups identified: No cataract, with cataract and undergoing surgery, with cataract and declined surgery	No Cat: 92	66.8	48.9/51.1	Cognition	Mattis Organic Mental Syndrome Screening Examination (MOMSSE)- evaluates 14 domains (abstraction,	Acuity: Dist VA ETDRS chart, Log Contrast senstivity: Pelli-Robson chart	Cat+no Sx & cat+ surgery grps signifcantly less CI (P <.001 and P =.009, respectively) than at baseline.	Cataract surgery does not affect cognitive function	Strong
2005			W Cat+Sx: 122	70.9	41.8/58.2		orientation,memory,speech,comprehension). Scored 0 to 28 (lower score higher functioning)	Ophthalmologist/optometrist graded cataract at clinic visit	No cataract group=no change in Cog status, association between change in VA in the better and worse eye and		
			W cat+ no SX: 87	71.1	59.8/40.2				change in MOMSSE observed (p=0.003 &p=0.03, respectively)		
Lin MY, Gutierrez PR	US (Study of Osteoporotic Fractures (SOF))	Longitudinal cohort	Vision testing sample:1668	75.9	0/100	Cognition	3MS-Modified version of MMSE	Acuity: Bailey Lovie Target	VI (worse than 20/40): Cog decline - 1.78 (1.21-2.61), Functional decline- 1.79 (1.15-2.79)	Vision impairment is associated with greater odds of cognitive and	Strong
2004			Overall 6112	76.1					DSI: Cog decline - 2.19 (1.26-3.81), Functional decline- 1.87 (1.01-3.47)	functional decline over time in older women	
Jefferis JM, Taylor JP	UK	Longitudinal cohort.	112	Normal cog: 80 (3.8)	20/80	Vision	Addenbrooke's Cognitive Examination (ACE-R; which includes the MMSE)	Acuity: LogMAR, Other: VFQ-25	Normal cog group: Mean at baseline 0.13 (0.09), 1-year FU 0 (0.09), 95% CI for difference: 0.09 to 0.16 <0.001	Patients with impaired cognition benefit from cataract surgery, but not to the same extent as patients with normal cognition. Cognitive impairment may, however, limit visual improvements	Strong
2015				Impaired cog: 81.2 (3.9)	30/70				Impaired cog group: Mean at baseline 0.18 (0.14), 1-year FU 0.06 (0.11), 95% CI for difference: 0.08 to 0.16	following cataract surgery	
Reyes-Ortiz C, Kuo YF	US (Hispanic Established Populations for Epidemiologic Studies of the Elderly)	Longitudinal cohort.	2140	Near VI: 73.5	39.3/60.7	Cognition	MMSE-blind	Acuity: subjects hold cards at least 7 inches from their eyes and asking them to read the numbers	Near vision impairment, -0.62 +/- 0.29, .03, Interaction btwn near VI & time= -0.13 +/- 0.07, .045	Results showed that near vision impairment was predictive of	Strong
2005				Distance VI: 75.3	35/65			Each card had 7-digit "telephone numbers" of three different type sizes: 7, 10, and 23 points	Distance vision impairment = -0.06 +/- 0.36, .87, Interaction btwn distance VI & time, -0.12 +/- 0.08, .14	cog decline in older Mexican Americans independent of other health factors	

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Haan M, Espeland MA	US (Women's Health Initiative MS-MRI)	Longitudinal cohort	511	69	0/100	Cognition	Modified Mini Mental State (3MS)	Acuity: Snellen with pinhole	Acuity slope in model of 3MSE score (β slope = 0.33, p = 0.06). Covariate-adjusted mean 3MSE scores for women	Retinopathy as a marker of small vessel disease is a risk factor for cerebrovascular disease that may influence cognitive performance and related brain changes	Strong
2012									with retinopathy compared to others throughout follow-up: mean (SE) difference 1.01 (0.43), p = 0.019	Very mild association between acuity and 3MSE scores	
Trick GL, Trick LR	US	Case-control. Visually and cognitively normal controls (n=61), and senile dementia of Alzheimer's type (SDAT) n=61	122	SDAT: 73.4	NA	Cognition	each patient was assigned a clinical dementia rating (CDR) based upon a scale developed and tested at Washington Uni	Acuity: ETDRS. Visual field analysis using 30-2 HFA	Patient groups, Foveal threshold, Mean deviation, Pattern standard deviation, Corrected pattern standard deviation.	We found that the frequency of potentially unreliable visual fields was significantly higher (chisquare,	Strong
1995				Control: 72.8			i.e CDR scale ranges from 0.0 (individual without dementia) to 3.0 (severely demented).		Control (n=44): 33.2+/-3.05, -2.81+/-3.45, 3.48+/-2.05, 2.58+/-2.30	p = 0.0232) in SDAT patients (44.3%) than in controls (27.9%)	
									SDAT (n=34): 30.79+4.00 (p=0.0027), -5.11+/-3.17(p=0.0057), 4.461+/-2.49(NS), 3.91+/-2.73(p=0.0408)		
									CDR=0.5(n=25): 31.12+/-3.54, -4.25+/-2.23, 4.47+/-2.60, 3.83+/-2.89		
Helmer C, Malet F	France (The Three-City-Alienor Cohort)	Longitudinal, cohort	812	79.7	35/65	Cognition	MMSE; Free and Cued Selective Reminding Test Isaacs Set Test the Trail Making Test parts A and B Then suspected cases were assessed by a neurologist	Acuity: No chart name provided, IOP: pneumotonometer	Association of OAG and incident dementia: Model 1: OR 4.3 95%CI 1.7-10.8 p 0.0020, Model 2: OR 4.2 95%CI 1.6-10.9 p 0.0030	Yes. Results show that OAG participants were at an increased risk for developing dementia during the 3-year follow-up time period.	Strong
2013									Model 3: OR 3.9 95%CI 1.5-10.4 p 0.0054, Association of Vertical cup:disk ratio and incident dementia: ≥0.65: OR 3.7 95%CI 1.4-9.7 p 0.0083 ≥0.70: OR4.4 95%CI 1.5-12.5 p 0.0055		
									Association of Minimal rim:disk ratio =< 0.1 and incident dementia: OR 2.7 95%CI 1.005-7.1 p 0.0489		
Javier de la Fuente	UK (English Longitudinal Study of Ageing)	Longitudinal, cohort	3508	69	43.2/56.8	Cognition	Four measured tests of verbal fluency, processing speed,	Three self-reported items covering eyesight in far, near, and general vision	Visual (β = 0.140, p < .001) difficulties predicted cognitive difficulties 8 years later	Yes. Visual difficulties were identified as predictors of	Strong
2018							and short-term and long-term memory	Dichotomized, collapsing "Excellent," "Very good," and "Good" as "Absence of difficulties".	The latent increase in cognitive difficulties was steeper in people with visual impairment (*d* = 0.52, p < .001)	subsequent cognitive decline in the old age	
Anni Hamalainen	Canada (Canadian Longitudinal Study on Aging)	Cross-sectional cohort	30029	No info	No info	Cognition	5 cognitive tests: Mental Alternation Test, Animal Fluency test, Controlled Oral Word Association Test, Stroop test	Vision was measured as the better-seeing eye pinhole-corrected visual acuity (VA; reported in logMAR)	VA was a predictor of only executive function (b=-0.785, p<0.001) but not memory	Yes, greater executive function scores were associated with better vision	Strong
2019							and Rey Auditory Verbal Learning Test with immediate and 5-minute recall				
Chuanying Huang	UK (English Longitudinal Study of Ageing)	Cross-sectional cohort	4197	Optimal:70.21	58.6/41.4	Cognition	Memory (immediate and delayed), verbal fluency, and time orientation	Self-report: 'Is your eyesight (using glasses or corrective lens as usual) excellent, very good, good, fair, poor, or registered blind?'	poor vision (= –1.309, p = 0.004), and poor dual sensory function (= –2.442, p = 0.013) was associated with worse cognition	Yes.	Strong
2019				Good: 71.84	51.2/48.8		Global cognition was the sum of scores for the 3 domains, and higher score indicates better cognitive performance	Answer of 'excellent' or 'very good'was defined as optimal vision. Answer of 'fair' or 'poor' was			
				Poor: 74.75	47.2/52.8			defined as poor vision. Thus vision was classified as optimal, good, and poor vision			
Virginie Naël	France (The Three-City-Alienor Cohort)	Longitudinal, cohort	7736	Incident dem:76.9	34.8/65.2	Cognition	3-step procedure: 1. MMSE, the Isaacs set test and the Benton Visual Retention Test	Near VA assessed using the Parinaud scale. Mild near VI was classifed by Parinaud 3 or 4 (Snellen equivalent 20/30–20/60)	Moderate to severe near VI was associated with an increased risk of dementia in the frst 2 years (HR2.0, 95% CI 1.2–3.3)	Yes. Near VI may represent an indicator of dementia risk at	Strong

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2019				No inci dem: 73.6	39.2/60.8		2.senior neurologist to establish a clinical diagnosis, 3.an independent committee of	and moderate to severe near VI by Parinaud>4 (Snellen equivalent<20/60)	and from 2 to 4 years (HR 1.8, 95% CI 1.1–3.1). distance VF loss was associated with an increased risk beyond 4 years (HR 1.5, 95% CI 1.1–2.0)	short and middle-term, mostly in depressedelderly people	
							neurologists and geriatricians reviewed all potential cases of dementia obtain a consensus on the diagnosis and etiology, according to the DSM-IV and the NINCDS-ADRDA criteria	Dist VA was selfreported, defued as an inability or diffculty in recognizing a familiar face at 4 m	but the association was no longer signifcant after taking into account baseline cognitive performances		
Zhi Wei Lim	Singapore (Singapore Epidemiology of Eye Diseases (SEED))	Longitudinal cohort	2478	67.6	50.7/49.3	Cognition	A locally validated Abbreviated Mental Test (AMT)	VA was measured at 4 m using the logMAR number chart	Baseline VI was associated with a decrease in AMT score over 6 years (β = −0.27; 95% CI, −0.37 to −0.17; P < .001)	Yes, , poor vision was independently associated with a decline in cognitive function	Strong
2020								(Lighthouse International).VI was defined as presenting VA worse than 20/40.	When change in vision over 6 years was evaluated, unchanged or deteriorated VI was associated		
									with a decrease in AMT score over 6 years (β = −0.29; 95% CI, −0.40 to −0.18; P < .001)		
Bernard Michalowsky	Germany (NA)	Case-control study	122708	81	39/61	Cognition	ICD-10 diagnosis code for dementia	ICD-10 codes for visual impairment	Visual impairment was not significantly associated with dementia (OR = 0.97, CI = 95% 0.97–1.02, p = 0.219)	No, but DSI was significantly associated with risk of dementia	Strong
2019									Combination of both visual and hearing impairments and the risk of dementia (OR = 1.14, CI = 95% 1.04–1.24, p = 0.005)		
Jung-Yu Liao	Taiwan (National Health Insurance Research database)	Case-control study	9200	NA	LOAD 36.7/63.3	Cognition	LOAD cases diagnosed as dementia and prescribed any acetylcholinesterase inhibtiors (AChEIs)	ICD-9 codes	Disorders of refraction and accomodation with LOAD OR for Year 1,2,3,4: 1.2, Year 6:1.3	Yes, in the path analysis model, disorders of ref and acc	Strong
2020	1997-2013				Without LOAD 36.7/63.3				Total effect reflects an association between prior diseases and LOAD incidence via all paths in the model: 0.042	had a significant positive effect on LOAD incidence	

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Table 4: Studies with a "moderate" rating

Author&Title	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Ajana,S	France (ALIENOR study, 3 City-Bordeaux cohort) France	Cross-sectional study	184	82.3	31.5%/68.5%	Cognition	MMSE, Isaacs Set test (IST15), Benton Visual Retention test, Free and Cues Selective Reminding Test	Acuity: Measured but no mention of method	MPOD 0.5 degree, Z score (b)=0.12 (0.01, 0.23). MPOD 1 degree, Z score (b)= 0.11 (0.01, 0.22)	Yes- Between MPOD and Global cognitive z score	Moderate
2018							Summarized by a composite global cognitive Z-score	MPOD: Measured using confocal scanning laser Ophthalmoscope	Regression models		
Anstey, K. J	Australia (Australian Longitudinal Study of Aging)	Longitudinal	1823	77.77	51.2%/49.8%	Cognition and vision	MMSE, Verbal ability, Processing speed and Memory	Acuity: Distance using Snellen chart at 3m, Near at 20cm	N/A	Yes- significant moderate-sized association between rates of	Moderate
2003						Analysis of change using growth curves		using chart containg short passages in font size 5 to 18. Left and Right eye were tested separately. Score was the smallest font size	Latent growth curve analysis	change in Memory and Vision	
Anstey, K. J.; Luszcz, M. A.	Australia (Australian Longitudinal Study of Aging)	Longitudinal	3766	76.34	Wave 1: 56%/44%	Cognition	Verbal ability, Processing speed and Memory	Acuity: Distance using Snellen chart at 3m	Sensory status x group interaction for memory comparing participants who declined on Vision vs not declined	Yes- Decline in visual performance had a significant	Moderate
2001					Wave 3: 33%/67%				Wilks Λ=0.98, F=2.94, p<0.05, η2=0.054	effect on decline in memory performance	
									Two-factor repeated measures MANCOVA (sensory status*wave)		
Arnaoutoglou, N. A	Greece	Longitudinal, Case-control	Total 103 AD group n=32	AD 73.3	38%/62%	Vision	MMSE, Dementia blood screening, Reisberg AD scale, MRI and NINDS-AIREN to diagnose VaD	Colour vision using Ischihara color deficiency test	Difference in ischihara scores between two groups F(1,161)=9.558, p=0.003. AUC 0.819 (0.70, 0.93)	Yes- Ischihara a promising test to differentiate AD and	Moderate
2017			VaD group n=36	VaD 75.6				Consists of 38 pseudoisochromaric plates	ANOVA to estimate any sig variations between three groups within their age, MMSE, Resiberg and	VaD	
			Healthy controls n=35	Controls 74.2				6 errors suggests definite color blindness	total Ischihara score		
Arrighi, McLaughlin	US (National Health interview survey (NHIS)2001-05)	Cross-sectional study using nationally available survey data	23474	With dementia 80.72	With dementia 36%/64%	Vision	Self reported surveys. Was categorized as With dementia if patient reported :senile"	Self reported vision problems	Prevalence odds ratio 1.46 vision problems with funtional limitations among americans aged >60	Yes- Higher odds of vision problems being prevalent	Moderate
2010			Person with dementia 443	Without dementia 72.3	Without 39%/61%				with and without limitations related to dementia	among people with funtional limitation	
			Without dementia 23031								
Bauer K, Schwarzkopf L	German statutory health insurance fund data (2006)	Cross-sectional study using claims data from insurance company	37753	80.1	28.4%/71.6%	Comorbidity complex	ICD codes with diagnoses of dementia	ICD codes with diagnoses of severe vision reduction	OR 0.59 persons with severe vision problems were less likely to have dementia	Yes- significantly lesser odds of vison problems among	Moderate
2014			Dementia group 9139	81.6	25.4%/74.6%	Patient having a diagnosis of vision prob				cases with dementia	

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			Without dementia 28614	79.6	29.4%/70.6%	Using ICD codes					
Chen, S-C; Chang, Y-P	Taiwan National Health Insurance	Case-control. Randomized sample data of one million patients who made claims from insurance	Eye disease 4097	50 y	AMD 60.9%/39.1%	Alzheimer's disease	ICD codes with diagnoses of Alzheimer's Disease	ICD codes with diagnoses of AMD, Diabetic retinopathy and Glaucoma	Cumulative IR of 1.22% ADE among eye disease group vs 0.04% among controls	AMD, DR and Glaucoma were asso with an increased	Moderate
2016	Research database (HIRD)		Controls 20745		DR 52.9%/47.1%				HR for DR 39.31 (4.79, 332.67), HR for AMD 36.94 (4.62, 295.46), HR for Glau 34.08 (13.37, 86.84)	risk of Alzheimer's disease	
					Glaucoma 45.9%/54.1%						
Chen, S. P.; Bhattacharya, J	US (NHANES and NHATS)	Cross-sectional analysis of two datasets: 1999-2002 NHANES data and 2011-2015 NHATS data	NHANES 2975	72	48%/52%	Cognition	NHANES: Cog testing done using Digit symbol substitution test (DSST). </= 28 indicated poor cognition	NHANES: Dist-Snellen VA chart, Near- 5-line near vision card	NHANES: Dist VI b=-5.1 (-8.6, -1.6), OR 2.8 (1.1, 6.7), Near VI b= -3.8*, OR 1.7*, Subj VI b=-5.3*, OR 2.7*	Yes, VI significantly associated with worse cognitive fn	Moderate
2017			NHATS 30202	NA	42%/58%		NHATS: Probable/possible dementia vs No dementia based on NHATS classification scheme (self-report)	NHATS: Vision tested via questionnaires	NHATS: Subj Near VI: OR 2.6 (2.2-3.1), Subj Dist VI: OR 3.9 (3.4-4.4)		
Chriqui, E	Canada	Longitudinal cohort	150	86.7	27%/73%	Vision	MMSE	Dist vision- Snellen chart, 6m. VI was Vn <6/12	Presenting VI: 37.3 (29.1-46.1)	37.3% of dementia participants had VI.	Moderate
2017								Near vision- Topcon paragraph/lighthouse	VI after refraction: 20.1 (13.7-27.9)	Just prevalence study, no correlation tested	
De Celis, E. S. P	US (Hurria et al. JCO 2011 & 2016)	Cross-sectional analysis of 2 prospective studies	750	Median age 72 (65-94)	No info	Cognition	Blessed OMC test (>11 on this test)	self-reported visual impairment (fair, poor or blind)	OR for association between VI and Cognition deficit.	Older pts with cancer and hearing/visual impairment	Moderate
2017									Univariate model: 2.0 (0.8-4.7), Multivariate model: 1.9 (0.75-5)	at higher risk of funtional and cognitive deficits	
de Kok, D. S	New Zealand (Life and Living in Advanced Age: A	Cross-sectional baseline data	661	Maori: 82.3	40%/60%	Cognitive decline	MMSE-blind version, cognitive test for the visually impaired	Acuity: ETDRS	Generalized linear models and structural equation modelling	No direct association found in this population	Moderate
2017	Cohort Study in New Zealand (LiLACS NZ)			Non-Maori: 84.6	47%/53%				Maori: Dis VA (logmar) b=0.103(-0.21,0.42), Self reported VI b= 0.110 (-0.030, 0.251)		
									Non-Maori: Dis VA (logmar) b=0.197 (-0.12, 0.51), Self-report VI b= 0.065 (-0.056, 0.187)		
Elliott, A. F	US	Cross-sectional cohort	238	No info	26%/74%	Vision	MMSE	Dist VA: ETDRS, Near VA: Lighthouse near VA card. VI < 20/40 one eye	β MMSE in Multiple linear Regression for Predicting Vn; Dist VA: -0.015, Near VA: -0.013 both(p 0.00)	Cognitive status contributed to the prediction of VI	Moderate
2015								Mars contrast sensitivity to assess CS. Score worse than 1.50 in one eye	Contrast sen: 0.17 (p0.00)		
Elliott, A. F.; McGwin, G., Jr	US	Longitudinal case-control	78	REC: 79.2	23.1%/76.9%	Cognition	Self-report and MMSE	Distance: ETDRS	Intervention group: MMSE baseline 20.2 (4.4), follow-up 19.4 (4.8), p=0.05	Vision-enhancing interventions did not lead to short	Moderate
2009			64	Delayed corrc: 78	25%/75%			Near: Lighthouse Near Visual Acuity Test	Control group: MMSE baseline 21.7 (4.5), follow-up 20.5 (5.5), p=0.015	term improvements in physical function/cognitive status in sample.	

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			30	Cataract Sx: 81	26.7%/73.3%				Pre&post-intervention measures of Phy fn &cognitive status compared within treatmnt groups paired t-tests		
			15	No cat sx: 87	13.3%/86.7%						
Elyashiv, S. M	US (The Religious Orders Study and The Memory snd Aging Project)	Longitudinal cohort study of aging and AD	2716	78.13	28.6%/71.4%	Cognition	Global cognitive scale (GCS), summary of raw scored of 19 tests incl episodic, semantic and working mem	Tested with both eyes open with a card held at 14 inches and measured	Mixed model for assoc with global cog score: VA b=-0.03312 (p<0.0001)	Significant positive correlation between GCS and VA	Moderate
2014							Scores converted, avergd and normalised-categorised into no,mild,mild CI+other diag, AD and AD+diag	at 7 increments from 20/40 or better to 20/400 or worse	Pearson and spearmann correlation coeff to assess relationship btwn GCS and VA		
Elliott, A. F.; McGwin, G., Jr	US	Cross-sectional	382	Range: 60 to <=100	No Impairment 26.8/73.2	Vision	MMSE	Acuity- Lighthouse near visual acuity test	Multiple linear regression used to estimate assoc btwn vision and cognition	VI affected HRQoL both with and without CI	Moderate
2009					VI only 5.9%/94.1%				NHVQoL, mean: No VI+CI=72.8, VI only=75.2, CI only=76.5, VI+CI=65.2. p=0.02		
					CI only 16.7%/83.3%						
					VI+CI: 20.4%/79.6%						
Feeney, J.; Finucane, C	Ireland (The Irish Longitudinal Study on Aging)	Cross-sectional cohort	4453	4th quartile 62.8	42.5/57.5	Cognition	MMSE, MoCA, Neuropsyh battery, CAPI for word-recall, CAMDEX-R for picture memory	Acuity:ETDRS, MPOD measured using Macular metrics densitometer	Multivariate regression to analyse assoc btwn Cog test performance and MPOD(continuous variable)	Lower MPOD associated with poorer performance in MMSE & MoCA	Moderate
2013				3rd quartile 62.4	46.3/53.7			Pt was asked if diagnosed with ARMD and Diabetes	MMSE: b= 0.48 (0.06, 0.90) p=0.026, MoCA: b=0.83 (0.15, 1.50) p=0.016, Memory: OR=2.24 (1.2, 4.15)p0.01		
				2nd quartile 62.2	53.9/46.1						
				1st quartile 62.3	53.8/46.2						
Formiga, F	Spain (NonaSantfeliu population-based study)	Cross-sectional design	186	93.06	23.5/76.5	Cognition	MEC (Spanish version of the MMSE), Functional status: Lawton-Brody Index (LI) and Barthel Index (BI)	Acuity: Snellen Eye test	T-test and chi-square test. MEC (± SD): group with visual deficit:16.5 ± 11, w/o visual deficit:23.6 ± 11, p<0.001	Functional status and cognition significantly asso with sensory	Moderate
2006									Lower LI was asso with: VI OR= 1.86 (1.44, 2.39) and combined impairment OR=1.995 (1.32, 3.01)	Impairment	
Friedman, D	US (The Salisbury Eye Evaluation in Nursing Home	Cross-sectional	656	83.4	25/75	Vision	MMSE	ETDRS charts or Lea symbols and grating acuity tests using Teller cards	Bland-Altman plot to measure how strongly grating acuity and recognition acuity relate to one another.	Grating acuity testing appears to be useful in	Moderate
2002	Groups (SEEING))								MMSE (5-point decline)with Teller acuity cards: OR= 2.67 (2.14, 3.31). Teller and recognition ICC=0.79	cognitively impaired individuals	
Frost S, Guymer R	Australia (Australian Imaging Biomarkers and Lifestyle	Cross-sectional case control	123	AD 70.2	59.1/40.9	Vision (ARMD)	MMSE and PET scan	Digital, non-stereoscopic retinal colour photographs, both eyes with a	Logistic regression: Medium Drusen: [N (%)]: CN: 13 (13), AD: 9 (41), OR: 4.69 (1.67-13.13)	A logistic model for early AMD found a	Moderate
2016	study of aging (AIBL))			CN 71.3	39.6/60.4			Canon CR-1 Non-Mydriatic retinal camera with digital Canon camera at the back	Early AMD: [N (%)]: CN: 3 (3.0), AD: 8 (36), OR: 18.67 (4.42-78.80)	significant association with AD diagnosis (p < 0.0001)	
								Grader (masked) evaluated images for signs of ARMD			

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Fuller, S. D	US (The American Community Survey (ACS))	Cross-sectional survey data	721053 5	61.31	47.1/ 52.9	Cognitio n	Self-report: Because of a physical, mental or emotional condition, does this person have difficulty	Self report: serious visual impairment- is this person blind or does	Cog Imp: NSI reference group, SHI: 45-64y: 3.55-3.67, 65-79y: 2.86-2.95, 80y+: 1.81-1.86, SVI alone: 45-64y:	Cognitive difficulty greatly associated with sensory impairment when	M od er ate
2018							concentrating, remembering or making decisions? Yes/No	he/she have serious difficulty seeing even when wearing glasses?yes/no	4.42-4.57, 65-79y: 4.71-4.91, 80y+: 2.33-2.44, DSI: 45-64y: 12.03-13.77, 65-79y: 11.19-12.84, 80y+:4.69-5.35	compared to NSI, esp greatest with SVI alone.	
Garin, N	Spain (COURAGE)	Cross-sectional cohort	50-64: 1760	57	47.7/ 52.3	Vision	Global standardised score from 5 performance tests: learning and short-term mem, attention & working	Obj: Snellen, Subjective: two questions to the participant, 1. how much	Asso btwn poor VA and Cognition (OR): Dist VA: 1.47 (1.17-1.86), Near VA: 1.72 (1.46-2.02)	Objective and subjective poor distance and near visual acuity	M od er ate
2014			>65: 1865	74.9	44.9 5/55. 05		mem and language. Lower score indicates worse cognitive functioning	diffculty seeing object or recognising at 20 m 2. at arm's length	Subjective distance VA: 2.05 (1.44-2.90), Subjective near VA: 2.96 (1.92-4.55)	were associated with worse cognitive functioning.	
Gaynes, B. I	US (Rush memory and aging project (1997-2007))	Cross-sectional cohort	714	VI presen t: 82	26.1/ 73.9	Cognitio n	Global cog fn summary contructed from 19 neuropsychological tests that assessed episodic, semantic,	Near VA: Rosenbaum Pocket Vision Screener positioned 14 inches from	X2 test, pooled t-test for equal variance. Divided analytic cohort into VI present and VI not present. Linear reg	Each unit higher in neuroticism level worsened association	M od er ate
2014				VI not presen t: 78.8	23.6/ 76.4		working memory, perceptual speed and visuospatial ability. Raw scores converted to Z-scores & averag	the eye. VI was defined as BCVA < 20/40	Global cog fn: VI b=-0.147, SE=0.086, p=0.09, Neuroticism b=-0.001, p=0.04, VIxNeurot b=-0.017, SE=0.005, p0.001	between vision impairment and lower global cognitive func	
Guthrie DM, Davidson JGS	Canada (Resident Assessment Instrument for Home Care)	Cross-sectional study	Home care: 291,824	82.8	38.9/ 61.1	Physical , social and emotion al	Cognitive performance scale (CPS) which has 4 terms pertaining to short-term memory, independence	Data from RAI-HC, particpants interviewed on the use of glasses	HC: CI+VI highest rate of procedural mem problems (56.1%) and highest prevalence of depression (28.2%)	CI+DSI had highest rate of functional impairments, communication	M od er ate
2018	(RAI-HC)		Longter m care: 110,578	86.9	30.5/ 69.5	function ing	in eating, expressive communication and decision making. Scale of 0-6, intact to very severe impairmnt	0 adequate Vn and 4 severely impaired(can see only lights and colours)	LTC: All three impairments showed highest rates on indicators of delirium.	problems and difficulty understanding others	
Hofer, SM	Nordic Research on Aging Study (NORA)	Cross-sectional study	1041	75 y	42.7/ 57.3	Cognitio n, Vision	WAIS-R- Digit Symbol test, Raven's Progressive Matrices- test of inductive reasoning,	Acuity: Computerized refractometer (Topcon RMA2300).	Of the correlations with the set of cognitive variables, the highest correlations were with corrected vision	no consistent associations were found across sensory, balance,	M od er ate
2003							Word Fluency Task,The Digit Span Forward and Backward test, Reaction time to vis/audi stimuli	Composite scores derived using avg of right and left vision scores	between .20 and .40 (visual choice reaction time), with the highest asso on visuomotor performance tests.	strength, and cognitive domains.	
Jefferis, J. M; Taylor, JP	Uk (Newcastle 85+ Cohort study)	Cross-sectional study	839	Diag catara ct: 85.5	30/7 0	Cognitio n	Standardized MMSE (sMMSE), Mmblind (blind version)	Review of family practice records, any pre-existing diagnosis of glauco,	Diagnosed Glau: 1.76 (1.05, 2.95), Diagnosed Glau or drops: 1.73 (1.05, 2.85)	Association between recorded glaucoma diagnosis and cognitive	M od er ate
2013				Diag Glauc oma: 85.4	36/6 4			cataract, cataract surgery (uni/ or bilateral) sight impaired registration		scores and record of previous cataract surgery & better cognition	
Luo Y, He P	China (Second National Sample Survey on Disability)	Cross-sectional study	250752	Total: 72.9	47.5/ 52.5	Cognitio n	Dementia ascertained using combo of self-report/family member report and on-site psychiatrist eval	Acuity: Trained Ophthalmologist assessed VI according to WHO BCVA	Dementia: Only VI- 1.54 (1.27-1.86), p< 0.001, Only HI- 1.04 (0.88-1.22), DSI- 1.63 (1.25-2.11), p<0.001	VI & DSI were morc likely to have severe to extremely severe	M od er ate
2018				With demen tia: 77.2	38.6/ 61.4		Questionnaire tested: poor memory, diff concentrating, diff controlling emotions, strange language/beha	criteria (low vision: 0.05 ≤ BCVA ≤ 0.29; blindness: no light perception		dementia than those without sensory impairment.	
							Positive on any one- then tyested by psychiatrist. Disablity diagnosis based on WHO DASII (score >52)	≤ BCVA < 0.05, visual field less than 10 degrees; the betterseeing eye)			
Rovner B, Casten R	US (AMD Trial)	Cross-sectional study	241	82.8	36.5/ 63.5	Vision function	Animal Fluency Test-brief assessment of cognitive function relevant to the completion of daily activities	Acuity: BCVA using ETDRS chart, NEI-VFQ near vision subscale	Predictors of Visual function: VA b= -1.93(−2.59 to −1.27), <.001, Animal fluency b=0.054 (0.019 to 0.090), .003	Visual ability was highly correlated with visual acuity. Multivariate	M od er ate
2011				Range 65 and older			Name as many animals as possible in 60secs. Requires semantic knowledge, speed mental processing	Contrast sensitivity using Pelli-Robson chart	CS b= 0.20 (−0.210 to 0.610), .33, PHQ-9 severity: −0.032 (−0.100 to 0.035), .35, Age b= −0.008 (−0.033 to 0.016)	model revealed coping startegies and cog fn contributed to Near Vn	

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Spier O, Fischer N	US	Cross-sectional study	190	81.6	30.5/69.5	Cognition	MMSE for the visually impaired (removed 8 items from the original MMSE) max score of 22	Corrected near VA was tested using the Jaeger chart. Refraction using	Asso VA & High MMSE blind score: Bad near VA (>J3): 1.0, good near VA (<J3); OR=3.18 (1.57-6.43) p0.001	Good VA and Wearing reading glasses were significantly associated	Moderate
2016				Range 75-101				Autorefractometer		with high MMSE-blind score	
Varin M, Kergoat MJ	Canada	Cross-sectional case-cohort	303	AMD 83.6	30.2/69.8	Total cognitive activities	Victoria Activity Questionnaire, which is a 70-item survey assessing older adults' participation in various	Acuity: ETDRS chart, Visual field: Humphrey frequency doubling	Total cognitive activities: Normal vision b=0 (ref), AMD: −4.19 (−5.96, −2.43) *P<0.05.	Patients with AMD and glaucoma participated in fewer cognitive	Moderate
2017				Glaucoma 77.5	40.9/59.1		cognitive activities over the past two years. Ex. Gardening, exercise,prep meals, doing housework,etc	technology full-threshold N-30 testing each eye. Medical chart reviewed	Glaucoma: −1.80 (−3.34, −0.26) *P<0.05.	activities compared to older adults with normal vision potentially	
				Normal 72.8	48.3/51.7		Cognition was tested with the MMSE-blind version	for diagnosis of AMD and any co-existing eye disease		putting them at risk for cognitive impairment	
Gussekloo J, de Craen AJ	Netherland (Leiden 85+ study)	Cross-sectional cohort	459	85	34/66	Cognition	MMSE, The 12-Word Learning Test was used as a visual test of long-term memory. Cognitive speed	Acuity: ETDRS chart	MMSE and VI: beta = -1.2 (-1.6 to -0.84), Immediate word-learning test beta= -1.3 (-1.8 to -0.69)	VI was associated with lower MMSE scores. Increasing VI was asso with	Moderate
2005							was measured with the Letter Digit Coding Test (processing speed)		Delayed word-learning test beta= -0.62 (-0.9 to -0.35), Letter digit coding test beta = -2.1 (-2.8 to -1.4)	woth poorer scored on memory and cog speed.	
									Stroop test beta = 4.7 (1.1 to 8.4)		
Hajek A, Brettschneider C	Germany (German Study on Ageing, Cognition and Dementia in Primary Care Patients)	Longitudinal cohort	2,394	82.5	34.2/65.8	Cognition	Cognitive activities included reading, writing, solving crossword puzzles, memory training, games	Patient graded their vision on three level Likert scale	Onset of mild visual impairment: beta (SE)= -0.065 (0.026)	Linear fixed-effects regression showed that the onset of severe	Moderate
2016							(e.g. card games, board games), playing music, and social engagement (e.g., in church, volunteering, in a club)	(with optical aid if necessary) as no impairment, mild, severe or profound VI	Onset of severe or profound visual impairment: beta (SE)= -0.376 (0.062)	VI was associated with a decline in physical & cognitive function score in the total sample	
Hidalgo J, Martinez I	Spain	Cross-sectional cohort	1160	73.3	44.1/55.9	Vision	Short portable Mental Status questionnaire	Acuity: Essilor visiotest instrument with snellen optotypes	Visual acuity < 6/18 in the better eye −0.344 (−0.392 to −0.295), Self-reported vision: average/poor/very poor	Using multiple regression analysis, the variables associated with VF status	Moderate
2009								Self reported vision as very good, good, average, poor and very poor.	−0.077 (−0.101 to −0.054), Dependence in activities of daily living: −0.121 (−0.160 to −0.081), Cognitive impairment −0.073 (−0.117 to −0.029)	were: visual impairment, self-reported poor vision, dependence in daily activities, cognitive	
										impairment, depressive symptoms, female gender and older age	
Thuan QP, Kiffley A	Australia (The Blue Mountains Eye Study)	Cross-sectional case control	3509	Late AMD: 79	34/66	Cognition	Modified MMSE: omitted vision-dependent items, with a max score of 22.	Acuity: LogMAR chart, Macula photo using Zeiss FF3 fundus camera	OR for MMSE 24-27 and: NO AMD 1, Late AMD 2.3 (1.1-5.0), Early AMD 1.0 (0.7-1.4)	Significant association found between Late AMD and cognitive impairment	Moderate
2006				Early AMD: 75.1	38.1/61.9			Amd lesions were assesed from the photos based on Wisconsin AMD grading system	For MMSE 0-23 and: NO AMD 1, Late AMD3.7 (1.3-10.6), Early AMD 1.4 (0.8-2.6)		
				No AMD: 65.7	43.7/56.3						
Uhlmann RF, Larson EB	US	Cross-sectional case control	174	77	42/58	Cognition	MMSE	Acuity: was measured with the Snellen and Rosenbaum methods for far and near vision.	Relative Odds for Dementia of Far-vision Impairment: 1.9 (95% CI = 0.8-4.6), RR of dementia and near Vn:	Visual impairment is associated with both an increased	Moderate
1991									20/100-20/200 1.5 (0.5-4.7), >=20/200 1.8 (0.5-6.1), RR of dementia and far Vn: 20/70-20/100 1.5 (0.3-6.1),	risk and an increased clinical severity of Alzheimer's disease	
									>=20/200 1.5 (0.3-6.0)	Consistent with the hypotheses that VI exacerbates Cog dysfucntion in dementia	

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Guthrie DM, Declercq A	4 countries	Cross-sectional	Canada: Comparison: 138 650, DSI 48 210	NA	F 67.2, 67.9	Cognition	Cognitive performance scale (CPS), captures issues with memory, independence in eating and decision-making	inter RAI assessment inleudes items measuring the presence of impairments	individuals with DSI were more likely than all others to have moderate to severe cognitive impairment (between 3.8% and 9.0% higher rate),	Yes. DSI significantly associated with CI	Moderate
2016			US: comparison 60 971, DSI 6577		F 66.8, 68.6		(Established validity against MMSE)	in vision. The vision item captures the person's ability to see close objects in adequate light while using their typical assistive device. Scored on a five-point scale where 0 = adequate, 1 = impaired, 2 = moderately impaired, 3 = highly impaired and 4 = severely impaired	highly compromised independence with ADLs (5.2% to 11.4% higher), instrumental ADLs (8.8% to 21.4% higher). Those with DSI were not more likely to experience social isolation		
			Belgium: comparison 500, DSI 256		F 74.9, 78.9						
			Finland: comparison 4765 DSI 1358		F 70.9, 76.2						
Harrabi H, Kergoat MJ	Canada	Cross-sectional cohort	420	AMD 82.5	25/75	Cognition	MMSE	Acuity: ETDRS chart with illuminated light box at 2m, at 1m if patient could	Model 1*: Binocular Logmar acuity, per 0.1 unit, -0.11 (-0.18, -0.04),	People with AMD, Fuch's corneal dystrophy, and glaucoma had lower cognitive	Moderate
2015				Fuch's 78.5	17/83			not see at 2m. Letter by letter scoring done, converted to logMAR	Model 2*: Log contrast sensitivity, per 0.1 unit, 1.40, (0.72, 2.13)	scores, on average, than controls (P < 0.05).	
				Glaucoma 76.6	42/58			CS measured using Pelli-Robinson chart 1m	Model 3*: Visual field in better eye, per 1 dB, 0.04, (-0.003, 0.08)	Scores were between 0.7 and 0.8 units lower than the control group	
				Control 74.0	41/59			Visual field: Humphrey frequency doubling technology (FDT)			
Holmen K, Anderssonn L.	Sweden (Kungsholmen project)	Case-control	Cog imapired 55	82		Loneliness, but measured interaction between	MMSE	Acuity: Snellen, self-report questions: Do you have any problems with your sight? Yes or No" was used to assess the problems	There were no significant differences in the cognitive groups between subjects with slight to moderate	NO. In a multiple regression analysis, higher MMSE score and visual improvement were significantly related to lower levels of self-reported loneliness among the elderly with their cognition intact, but not	Moderate
1994			Cog intact 92	83		Cognition & vision			habitual visual acuity (i.e. those who could read newspapers with glasses and visually orient themselves	among the subjects with impaired cognition.	
									without difficulty), and subjects with pronounced to severe visual impairment with residual vision (x2 = 2.28, p = 0.13). The same differences were found		
									when corrected visual acuity was compared analytically between the cognitive groups (x2 = 1.15, p=0.28)		
Su C, Lin C	Taiwan (Taiwan National Health Insurance program)	Cross-sectional cohort	Glaucoma 6509	59.4	45.5/56.5	Cognition	Dementia patients enrolled using ICD codes	Glaucoma cohort formed enrolling patients who were newly diagnosed with it	HR=1.13 (1.01-1.27)	The patients with glaucoma exhibited a significantly higher risk of	Moderate
2016			Control 26036	59.2	42.4/54.6					dementia than the individuals without glaucoma did (hazard ratio	
										[HR]¼1.13, 95% confidence interval [CI]¼1.01-1.27)	
Miyata K, Yoshikawa T	Japan (Fujiwara-kyo Study)	Cross-sectional cohort	2764	76.3	52.6/47.4	Cognition	MMSE	Acuity, Landolt ring chart at 5m, converted to logMAR. refractive errors were measured by an autoref/keratometer	OR for MCI: no cataract surgery 1.00 (ref) / cataract surgery 0.79 (0.64-0.97) p 0.025	The subjects who had prior cataract surgery had significantly lower odds ratio (OR) of having MCI	Moderate
2018								A prior cataract surgery was determined by the selfadministered questionnaire	OR for dementia: no cataract surgery 1.00 (ref) / cataract surgery 1.20 (0.64-0.97) p 0.36	Cataract surgery may be important in reducing the risk of developing MCI but not for dementia independently	
Ong SY, Ikram MK	Singapore (Singapore Malay Eye Study (SiMES))	Cross-sectional cohort	1032	Cognitive dysfunction 70.5	24/76	Cognition	Abbreviated Mental Test (AMT)	Acuity: number chart at a distance of 4m converted to logMAR. VI logMAR > 0.3 in the better eye	Cog Dysfunction OR: Model 2 :Myopia 1.78 (1.02-3.10) Hyperopia 1.11 (0.68-1.80) Emmetropia 1.0	Yes. Compared with individuals with emmetropia, persons with myopia were	Moderate

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2013				No Cog dysfunction 67.7	66.5/33.5		[cognitive dysfunction was defined as a score less than or equal to 6 of 10 for those with 0 to 6 years of formal education		Model 3 : Myopia 1.86 (1.01-3.42) Hyperopia 0.92 (0.54-1.55) Emmetropia 1.0	almost twice as likely to have cognitive dysfunction	
							and less than or equal to 8 of 10 for those with more than 6 years of formal education.]				
Soler V, Sourdet S	France	Cross-sectional cohort	1648	82.6	35.6/64.4	Vision	MMSE	Acuity. Snellen decimal chart for distant vision, Parinaud chart for near vision	model 1 for distant vision: MMSE OR (CI) 0.94 (0.92-0.97) p<0.001	Yes. Results show that visual impairment is independently associated	Moderate
2016							ADL, IADL, Fried score, SPPB class (Short Physical Performance Battery),	Amsler grid testing	model 2 for near vision: MMSE OR (CI) 0.92 (0.89-0.94) p<0.001	with lower educational level, cognitive impairment, and lower ADL-assessed autonomy	
Sun HP, Liu H	China	Cross-sectional cohort	4123	67.5	48.8/51.2	Cognition	Abbreviated Mental Test (AMT)	Objective refraction was measured using an autorefractor (Canon RK-5)	OR of cog dysfunction for myopia: Before propensity score matching, OR 1.98 (1.61- 2.44)	Results of this PSM analysis support a small, but positive association of	Moderate
2016								Spherical equivalent (SE) [Myopia: a SE of less than -0.50 dioptre (D) in the right eye. High myopia: a SE of less than -6.00 D.]	After matching: OR 1.31 (1.00- 1.71). Multivariate logistic regression	myopia with cognitive dysfunction among elderly Chinese in China	
									model on the association between myopia and cognitive dysfunction, OR (95%CI):		
									Myopia (Yes vs No) 1.52 (1.23-2.06), Age-related cataract (Yes vs No) 2.06 (1.57-2.98)		
Zheng DD, Swenor BK	US (The Salisbury Eye Evaluation Study)	Longitudinal cohort	2520	73.5	42/58	Cognition	MMSE	Acuity. ETDRS chart refraction was performed on participantswith	Worse baseline VA was associated with worse baseline MMSE score	Worse baseline VA was associated with worse baseline MMSE score	Moderate
2018								20/32 visual acuity orworse using a forced-choice procedure.	(r = −0.226; 95%CI, −0.291 to −0.16; P < .001). The rate of worsening VA was associated with	The rate of worsening VA was associated with a significant	
								Presenting binocular distanceVAwas converted to logMAR with a higher score indicating worse VA.	the rate of declining MMSE score (r = −0.139; 95%CI, −0.261 to −0.017; P = .03).	the rate of declining MMSE score	
Ward ME, Gelfand JM	US (Study of Osteoporotic Fractures (SOF))	Longitudinal cohort	1352	77.7	0/100	Vision	Global cognition (3MS), verbal memory (California Verbal Learning Test II, short form [CVLT]),	Contrast sensitivity, using a VISTECH (Hartford, CT) VCTS 6500 wall chart and light mete	OR of MCI/dementia for lowest quartile of baseline contrast sensitivity	These data support an association between impaired contrast sensitivity and future MCI/ dementia.	Moderate
2018							executive function (Trails B, Digit Span Forward, Digit Span Backward), and semantic memory (category and verbal fluency)	Self-reported age-related macular degeneration (AMD), glaucoma, and cataract	compared with women in the highest quartile: OR 2.16 (95% CI =1.58-2.96)		
							Clinical adjudication if person was positive in the above tests.		OR of MCI/dementia for every 10 point decrease in contrast sensitivity:		
									OR=1.21, 95% CI=1.13-1.31		
Mitoku K, Masaki N	Japan (Gujo City Long-Term Care Insurance database)	Cross-sectional cohort	1754	80.89/82.42	34.5/65.5	Cognition	The assessment measures include communication, short-term memory,	Acuity: No chart name provided	Logistic regression: OR of CI was 1.389 (95% CI 1.04-1.85) in those with VI	Yes. Supports the relationship between DSI and CI, increased risk of mortality in thise with DSI & CI.	Moderate
2016							location awareness and understanding of daily tasks, remembering own name and date of birth, and recognizing the	Other: Vision at baseline 5 levels, "normal sight", "able to see chart at 1m"	1.581 (95 % CI 1.24-2.01) in those with HI, and 2.349 (95 % CI 1.79-3.09) in those with DSI		
							season of year in addition to 18 problem behaviors	able to see chart in front, "very little sight", "indeterminable"			
							CI classified as none, mild, moderate, severe				
Jefferis JM, Clarke MP	UK	Longitudinal, cohort	112	80.7	45/55	Cognition	The ACE-R (which includes the MMSE), is scored from 0 (worst cognition) to 100 (best cognition);	logMAR visual acuity in the better eye corrected with up-to-date refraction and/or pinhole	Group, Mean+/-SD:Baseline, Postop, 1 Year. P Value*: 3-Way†, Baseline Vs Postop, Baseline Vs 1 Year, Postop Vs 1 Year	Significant improvements in ACE-R scores were seen between baseline and 1 year	Moderate
2015							38 of these points pertain to vision-dependent items and 62 to vision-independent items		Normal (n =46): 92.5+/-2.8, 92.6+/-3.8, 93.7+/-3.4, 0.018, 0.9, 0.009, 0.004::: Mildly impaired cognition (n =22): 84.5+/-1.4, 84.8+/-4.3, 88.0+/-3.5, <0.001, 0.75, <0.001 0.002	postoperatively (95% confidence interval for improvement, 0.5-2.8; P Z .005). Improved	

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									Moderately impaired cognition (n =23): 72.6+/-8.7, 73.6+/-10.4, 73.3+/-14.4, 0.75, --, --,--	cognition did not correlate with improved visual acuity (r Z 0.13, P Z .22)	
									All participants (N=91): 85.6+/-9.5, 85.9+/-10.0, 87.2+/-11.4, 0.004, 0.45, 0.005, 0.006, *P < .05 statistically significant		
Lindenberger U	Germany (BASE (Berlin Aging Study))	Longitudinal, cohort	516	85	NA	Cognition	MMSE	Acuity: Snellen decimal units. Close visual acuity (close vision) was measured separately for the left and the right eye with a standard reading table presented at reading distance. Distance visual acuity (distant vision) was assessed binocularly with a reading table presented to the participants at a minimum distance of 2.5 meters	Multivariate Multilevel Modeling Results With Control for Age, Time to Death, and Risk of Dementia Construct, Mean (Level, Linear change, Quadratic change), Variance (Level, Linear change), Age, Dementia status, Age X Linear Change, Residual variance.	Contrary to expectations, the correlations between cognitive and sensory declines were only moderate in size, underscoring the need to delineate both domain-general and function-specific mechanisms ofbehavioral when controlling for age at first measurement, distance to death, and risk of dementia senescence	Moderate
2009									Close vision: 47.33 (0.30), -1.37 (0.11), ---, 34.17 (2.95), 0.83 (0.16), -0.44 (0.04), ---, -0.03 (0.01), 25.87 (1.42)		
									Distant vision: 48.47 (0.32), -0.56 (0.11), ---, 28.14 (3.17), 1.20 (0.25), -0.59 (0.04), ---, ---, 44.79 (2.45)		
Hong T, Mitchell P	Australia (Blue Mountain Eye Study)	Longitudinal, cohort	2334	No Sensory impairment 66.9 (7.4),	No SI 58.4/ 41.6	Cognition	MMSE	Acuity: Distance VA at 244 cm followed by pinhole acuity (VA was recorded as the number of letters read correctly) If no letters could be read count fingers, hand movements, light perception, or no light perception	association between Decline >=3 MMSE blind score and visual impairment (adjusted for age and sex). OR after 5 years: Visual impairment 0.84 95%CI 0.40-1.79 Dual sensory impairment 1.41 95%CI 0.54-3.72 after 10 years: Visual impairment 1.09 95%CI 0.52-2.30 Dual sensory impairment 1.15 95%CI 0.28-4.73	No.The presence of VI, HL or DSI was not associated with possible cognitive decline over 5 years or 10years.	Moderate
2016				Visual impairment 74.3 (8.4),	VI 65.8/ 34.2					There were no changes to these findings after adjustment for other potential confounder	
				Dual impairment 80.4 (7.0)	Dual impairment 57/43						
Mandas A, Mereu RM	Italy	Case-Control	1168	78.4	29/71	Cognition	MMSE	Acuity: Snellen Chart	Total(n =1168), Control1 (n =436), All types of dementia (n =732), MCI (n =181), AD (n =230), MD (n =126), VD (n =195)		Moderate
2014								IOP, fundus examination under slit-lamp and fundu photograph	All vision disorders: No, n=297, n=433, n=117, n=135, n=68, n=113 yes, n=139, n=299, n=64, n=95, n=58, n=82 χ2, 9.37, 0.70, 5.87, 8.60,,6.12 P-value, 0.002, 0.40, 0.015,,0.003, 0.013 OD (95%CI), 1.5(1.1-1.9), 1.2(0.8-1.7), 1.5(1.1-2.1), 1.8(1.2-2.7), 1.5(1.1-2.2)	Subjects with any type of age-related vision disorders(n D438), cataract,	
										macular degeneration,glaucoma,and diabetic retinopathy,were more likely be depressed	
										and had significantly lower mini mental state examination scores than Control 2	
Mendola JD, Cronin-Golomb A	US	Case-Control	188	NA	NA	Vision	Blessed Dementia Scale (BDS)	Acuity: Snellen, Contrast sensitivity: two-alternative forced-choice format,	Test, Prevalence in AD, Correlation with BDS, % variance from BDS.	The results indicate that visual dysfunction, especially on Backward Masking, is a common sign of AD.	Moderate
1995								Stereopsis: binocular disparity stereopsis, as a clue to depth perception(Stereo Optical)	Backward Pattern Masking: 58% r=0.22; p=0.106; df=55, 13%, Gollin Incomplete Pictures: 30%, r= 0.33; p= 0.03; df= 17, 6%		
								funduscopy, evaluation of fixation, pupillary function, extraocular movement,	Luria Mental Rotation: 39%, rb= 0.74; p= 0.004; df= 36, 13%		
								Backward Masking, Gollin Incomplete Pictures. City University Colour Vision.	Log Contrast Sensitivity at 1 cpd: 39%, rb= 0.69; p= 0.002; df= 17, 52%, Log Contrast Sensitivity at 2cpd: 6%, rb= 0.55; p= 0.022; df= 17, 7%		
								Luria Mental Rotation. Moeny Road Map. Local speed discrimination and Global motion detection. Critical Flicker fusion.	Money Road Map: 29%, rb= 0.37; p=0.028 ; df= 35, 17%		
Magalhães, M. O. D. C	Brazil	Cross-sectional cohort	466	71.4	44.4/ 55.6	Cognition	CAMCOG (Cambridge Cognitive Examination)	Self-report	logistic regression model for dementia:	Yes. Study showed strong association between dementia and sight-impairment	Moderate

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2008							Dementia was diagnosed in individuals who fulfilled CAMDEX criteria for definitive, probable or possible dementia through a combination of clinical criteria, organicity indices ≥5 and CAMCogG<80.		sight impairment OR (95%CI): 1.83 (1.15-2.91)		
									Pvalue 0.011		
Ong SY, Cheung CY	Singapore (Singapore Malay Eye Study (SiMES))	Cross-sectional cohort	1179	with cognitive dysfunction : 71.2	21/79	Cognition	The Abbreviated Mental Test (AMT) is a 10-question test of general cognitive function	Acuity: LogMAR number chart (lighthouse international)	Age-related eye disease, model 1, model 2, model 3.		Moderate
2012				Without cognitive dysfunction : 68.0	67.7/33.3		Items assess orientation (3 points), semantic knowledge (1point), episodic memory (3 points), delayed recall	Cataracts were assessed from lens photographs, Glaucoma diagnosed and classified using International Society Geographical and Epidemiological Ophthalmology scheme based on gonioscopy	Cataract: 1.52 (1.02-2.27), 1.42 (0.93-2.16), 1.44 (0.94-2.22), AMD: 1.71 (0.89-3.29), 1.44 (0.71-2.91), 1.36 (0.34-5.45)	Yes. VI due to cataract (OR=2.75; 95% CI, 1.35-5.63)	
							(1 point), picture naming (1 point), and attention (1 point)		Moderate or severe DR: 2.43 (1.13-5.20), 2.26 (1.02-5.00), 5.57 (1.56-19.91)	were more likely to have cognitive dysfunction. Only moderate to severe diabetic retinopathy was independently associated with cognitive dysfunction	
									Glaucoma: 1.68 (0.81-3.46), 1.61 (0.75-3.45), 2.09 (0.52-8.31)		
Mangione CM	US	Cross-sectional cohort	472	73	40/60	Cognition	TICS questionnaire	Acuity: Snellen (results were transformed into percentages of functional central	multivariate linear regression , percent increase in ODDS of abnormal TICS score (95% CI):		Moderate
1993								vision loss following the formula outlined in the Physician's Desk Reference for Ophthalmology)	binocular vision loss(per 10% decline in vision) -9 (-23,9), percent field loss (per 1 category decline) 13 (-13,48) cataract (per eye) 18 (-12,58) ocular miotics 37 (-17,126)	No. An association between low TICS score and visual disorders were not seen	
								Visual field: Humphrey field analyser			
Walter Wittich	Canada (COMPASS-ND study)	Longitudinal, cohort	109	72.94	No info	Cognition	Montreal Cognitive Assessment (MoCA)	Reading acuity (MNRead) and Contrast Sensitivity (Mars Test)	Individuals with AD had significantly lower contrast sensitivity than those with MCI and SCI (p = .04, $\omega 2$ = .04;), after adjusting for age, sex and education. No differences in visual acuity were observed p = .46, $\omega 2$ = .004.	Yes. Declines seen in contrast sensitivity in individuals with AD relative to those with MCI and SCI.	Moderate
2019										Study also observed higher rates of reduced reading acuity in the AD and MCI groups relative to those reported the general population.	
Brenowitz	US (Health ABC Study)	Longitudinal, cohort	1810	76.7	39.4/60/6	Cognition	Dementia was defined as meeting one or more of the following criteria through study	Bailey-Lovie distance VA test, Pelli-Robson contrast sensitivity	Multivariable model for incident dementia with VA: IRR= 1.26 (0.90-1.77), CS: HR=1.11 (0.88-1.38)	No. Vision impairment and contrast sensitivity independently was not significantly associated with incident dementia	Moderate
2019							Year 15 (2011–2012):(a) hospitalization with dementia as a primary or secondary diagnosis,				
							(b) documented use of dementia medication, or (c) clinically meaningful Modified Mini-Mental State Examination (3MS)				
							decline from Health ABC baseline (1.5 SD, race-stratified)				
Davies-Kershaw	UK (English Longitudinal Study of Ageing)	Longitudinal cohort	7685	NA	44/56	Cognition	3 methods to identify individuals with dementia: a physician diagnosis of dementia	Rate their eyesight (using glasses or corrective lenses as usual)	Cross-sectional analysis using Wave 7 measures adjusted for covariates: compared to normal VA, moderate OR = 2.04 (1.36–3.07)	Yes. Older adults with vision impairment have higher rates of dementia crosssectionally (all ages) and are at greater risk of incident dementia longitudinally (<70 only)	Moderate
2918	Wave 7						that the participant or a caregiver reported between Wave 3 (2006–7) and 7 (2014– 15):	as excellent, very good, good, fair, poor, or registered blind	poor OR = 4.02 (2.64–6.13) p<0.001		
							a score less than 3.5 on the adaptive Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE)	Combined groups into 3 categories for analysis (excellent or very good=normal,	Longitudinal analysis using Wave 2 measures individuals in the younger group (50–69) and with moderate (HR=1.78, 95% CI=1.04–3.04) and poor (HR=3.60, 95%CI=1.10–11.78) selfrated vision were at greater risk of developing dementia than those with normal self-rated vision		

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							or prescriptions for anticholinesterase inhibitors, N-methyl-D-aspartic acid receptor antagonists,	good or fair=moderate, poor or registered or legally blind=poor or blind)			
							and other relevant medication (galantamine, rivastigmine, memantine, donepezil, tacrine) to indicate dementia	Self-rated vision measure from Wave 7 (2014–15) for cross-sectional analysis and from Wave 2 (2004–05) for the longitudinal analysis			
Gui-Ying Cao	China (China Health and Retirement Longitudinal Study)	Longitudinal, cohort	7269	60.2	56.2/43.8	Cognition	Episodic memory (measured as the average of immediate and delayed recall scores of ten Chinese nouns),	VI was assessed by two self-reported questions, about distance visual acuity for seeing faces on the other side of the street	Compared with older people with no VI, (1) those with DVI only were associated with poor episodic memory (β=–0·076, p<0·0001)	Yes. VI is associated with an increased risk of poor cognitive function	Moderate
2018							mental intactness (measured using some components of the Telephone Interview of Cognitive Status (TICS) battery),	and about near visual acuity for reading newspapers. Individuals were assigned to one of four categories: no VI,	mental intactness (β=–0·074, p<0·0001) and global cognition ((β=–0·089, p<0·0001)		
							and global cognition (the sum of episodic memory and mental intactness scores)—was evaluated and followed up every 2 years	distance vision impairment (DVI), near vision impairment (NVI), or both distance and near vision impairment (DNVI)	Those with NVI only were associated with poor mental intactness (β=–0·031, p=0·0001) and global cognition (β=–0·032, p=0·0224)		
									Those with DNVI were associated with poor episodic memory (β=–0·106, p<0·0001), mental intactness (β=–0·107, p<0·0001), global cognition (β=–0·105, p<0·0001)		
Ali. G. Hamedani	US (Medicare Claims data from 2014)	Cross-sectional cohort	472,871	>65	34/66	Cognition	ICD-9 diagnosis codes for dementia or cognitive impairment	ICD-9 diagnosis codes for blindness/ low vision	Among the low vision/blindness group, the prevalence of dementia or CI was 50%.	Yes. There was a significant association between low vision/blindness and dementia or AD	Moderate
2019									Blindness/low vision was also associated with a greater odds of Alzheimer's disease (AOR 1.44 95% CI: 1.415–1.464)		
									and all-cause dementia (AOR 1.239, 95% CI: 1.223–1.254)		
Preeti Gupta	Singapore (Singapore Epidemiology of Eye Disease Study)	Longitudinal, cohort	682	67.3	55.6/44.4	Cognition	CI was assessed using the validated Abbreviated Mental Test (AMT), defined as scores of	Fundus images using Canon DGI camera, graded by trained graders at University of Sydney	Those with any DR had higher odds of incident CI (OR=2.32, 95% CI 1.07 to 5.03), and those with moderate or worse DR also had a higher likelihood of	Yes, DR, particularly at the more severe stages, is associated with	Moderate
2019							≤6 and ≤8 for those with 0–6 and >6 years of formal education, respectively	using the modified Airlie House classification system and categorised as none (Early Treatment of Diabetic Retinopathy Study (ETDRS) level 10),	developing CI (OR=3.41, 95% confidence interval 1.06 to 11.00, p-trend=0.021), compared with those without DR	increased risk of developing CI, independent of vision and other risk factors	
								minimal/mild (level 20–35) and moderate or worse DR, (level 43–90) using data from the better eye			
Jost. B. Jonas	China (The Beijing Eye Study)	Cross-sectional cohort	3127	64.2	43.4/56.6	Cognition	MMSE, assessed as cognitive function score (cfs). Categorised as mild (cfs 23-19), moderate (cfs 18-10)	Autorefractometry, Presenting, uncorrected and best corrected VA,	Better cognition (i.e., higher CFS) was significantly associated with better best corrected visual acuity (r2=0.38),	Yes	Moderate
2018							severe (cfs <9)	fundus photo, SD-OCT	smaller amount of undercorrected visual acuity, lower prevalence of primary angle-closure glaucoma, and		
									thicker subfoveal choroidal thickness		
Allen T.C. Lee	China	Longitudinal, cohort	15576	74.5	36.2/63.8	Cognition	ICD-10 codes or Clinical dementia rating 1 to 3	Acuity: Snellen E chart, converted to LogMAR	Participants with incident dementia had poorer visual acuity at baseline than those without (adjusted HR= 5.88, 95% CI=4.04–8.57).	Yes, Moderate-to-severe visual impairment could be a	Moderate
2020									Incident Dementia at years 4- 6 of Follow-up in Participants With Mild (HR=1.19, 95% CI 0.86–1.65, p= 0.31)	potential predictor and possibly a risk factor for dementia	
									Moderate (HR= 2.09, 95% CI=1.47–2.97) or Severe VI (HR= 8.66, 95% CI=4.60–16.30) at baseline		
Cecilia. S. Lee	US (Adult Changes in Thought (ACT))	Longitudinal, cohort	3877	NA	42/58	Cognition	Cognitive Abilities Screening Instrument scores 85 underwent a standardized diagnostic evaluation, including physical and neurologic examinations and a neuropsychological test battery	ICD-9 codes for diagnosis of glaucoma, AMD, DR and cataract	The recent and established hazard ratio were 1.46 (P 5.01) and 0.87 (P 5.19) for glaucoma,	Yes, Increased AD risk was found for recent glaucoma diagnoses,	Moderate

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2019							Dementia diagnoses using NINCDS-ARDRA criteria, Our primary outcome was either probable or possible late-onset clinical AD		1.20 (P 5.12) and 1.50 (P .001) for AMD, and 1.50 (P 5 .045) and 1.50 (P 5 .03) for DR	established AMD diagnoses, and both recent and established DR	
Moon Jeong Lee	US (National Health and Nutrition Examination Survey (1999-2006 cycles))	Cross-sectional cohort	5795	Non-VI 70.3	45/55	Cognition	Memory or confusion complaints were defined as present if participants responded "yes" to the question—	VA was assessed using the built-in chart in an auto-refractor (NIDEK ARK-760; Nidek Co Ltd, Tokyo, Japan)	Individuals with VI were more likely (OR = 1.3, p = .049) to report cognitive complaints as compared with those without VI	Yes	Moderate
2019				VI 77.8	39/61		"(Are you/is survey participant) limited in any way because of difficulty	VI was defined as autorefractor corrected VA worse than 20/40 in the better-seeing eye			
							remembering or because (you/s/he) experience(s) periods of confusion?"				
Ruby Yu	China	Longitudinal, cohort	1949	76.1	23.1/76.9	Cognition	5-item Abbreviated Memory Inventory for Chinese (AMIC). Scored range from 0–5 (1 point for each item; 0—best to 5—worse). An AMIC score ≥ 3 is predictive of MCI	Questionnaire: "Do you have any difficulty seeing things?""very good" and "good" = "robust"; "fair", "not too well", "poor", and "very poor" = "poor"	Poor vision (OR 2.2 95% CI 1.8–2.7) at baseline was significantly associated with an increased risk of at least 3 SMCs at follow-up	Yes	Moderate
2019											
Michio Maruta	Japan	Longitudinal cohort	2190	78.9	20.6/79.4	Cognition	Dementia scale labelled Level 0 to Level IV and level M based on symptoms and the	VA assessed at 5 levels: normal vision (there is no hindrance in daily life),' 'able to see vision testing chart at a distance of about 1 m,'	DSI associated with a higher cumulative dementia incidence compared to no sensory impairment (log-rank χ2 = 39.92; P < 0.001)	Yes.	Moderate
2020							necessity for care. Level II and greater during the 8-year follow-up period was considered	'able to see vision testing chart at a distance of in front,' 'very poor eyesight,'	DSI is the greatest risk factor for developing dementia among sensory impairments (HR, 1.45; 95% CI, 1.22–1.71; P < 0.001)	Older adults with sensory impairments have a high incidence of dementia, with DSI presenting the greatest risk	
							"incident dementia"	and 'undecidable due to difficulty in communication.	Older adults with VI were found to be more likely to have day-night reversal symptoms when dementia occurs		
Ann E. M. Liljas	UK (English Longitudinal Study of Ageing)	Longitudinal cohort	4621	64.9	45/55	Cognition	Working memory, Executive function. Results from the three cognitive tests available were summed,	VA: Self-report asking participants whether their eyesight was excellent, very good, good, fair, or poor using glasses or corrective lens if they normally do so	Compared with people with good vision, poor vision was asso with worse cognitive function	Yes, aging adults with individual and combined impairments in hearing and vision had greater risks of worse cognitive performance at 6-year follow-up compared with those with good sensory function	Moderate
2020							providing a cognitive score (range 0-80), with a lower score indicating worse cognitive function		B = 1.61, 95% CI = [0.92, 2.29] adjusted for age, sex, baseline cognition		
									Compared with no sensory impairment, dual sensory impairment was associated with worse cognition (B = 2.30, 95% CI = [1.21, 3.39]		
Stephanie Chen	US (National Health and Aging Trends Study (NHATS))	Longitudinal cohort	7075	NA	40/60	Cognition	1. Doctor told the sample person that he/she had dementia or AD, 2. A score that indicates probable dementia on the AD8 Dementia Screening Interview	Self-reported distance and near VI	Participants with self-reported visual impairment were at significantly higher risk of developing	Yes, self-reported visual impairment in the US Medicare	Moderate
2019							3. Cognitive tests that evaluate the sample person's memory and orientation		probable or possible dementia over subsequent followup (HR=4.4, CI: 3.9-5.0, p<0.001), compared to those without visual impairment	population may be associated with greater risk of cognitive decline	
									This association persisted after full adjustments for covariates (HR=2.1, CI: 1.8-2.5, p<0.001)		
Ali G. Hamedani	US (Medicare beneficiaries)	Cross-sectional cohort	773905	NA	NA	Cognition	ICD-9 codes and procedure code-based algorithms	ICD-9 codes and procedure code-based algorithms	Low vision was associated with greater odds of incident hip fracture (AOR 1.13, 95% CI: 1.04–1.22	No.	Moderate
2019									and incident anxiety (AOR 1.11, 95% CI: 1.05–1.18 but not incident depression or dementia.		

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TIEN Y. WONG	Singapore (SEED)	Cross-sectional cohort	10020	58.9	49.3/ 50.7	Vision	Abbreviated Mental Test (AMT), which consists of 10 questions of general cognitive function	VA ETDRS chart at 4m converted to LogMAR	For best-corrected bilateral visual loss, cognitive impairment (OR=2.07; 95% CI=1.60–2.68)	Yes	Moderate
2019									was a significantly associated risk factor. CI was also significantly associated with higher risk of presenting bilateral VI or blindness (OR=2.15; 95% CI= 1.75-2.63)		
Bonnielin Swenor	US (Health ABC study)	Longitudinal cohort	2444	74	47.8/ 52.2	Cognition	Digit symbol substitution test and 3MS. Incident cognitive impairment was defined as	VA: Bailey-love chart, CS: Pelli-Robson chart	VA impairment HR=1.55, 95% CI 1.12, 2.14 vs No VA impairment, CS impairment HR=1.33	Yes, visual acuity, contrast sensitivity, and stereo acuity are	Moderate
2019							a 3MS score <80 or a decline in 3MS >5 points following Year 3		95% CI 1.13, 1.55 vs No CS impairment	risk factors for cognitive decline	

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Table 5: Studies with a "weak" rating

Author&Title	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Bayer, A. U.; Ferrari, F	Germany	Case control	228	With AD+ Glau 72.9	10%/ 19%	Glaucoma	Diagnosis of Alzheimer's Disease according to the NINADS-ADRDA classification	Acuity - Snellen	25.9% occurrence rate of probable glaucoma among AD	Yes- Significantly higher occurrence rate of glaucoma	Weak
2002			AD 112	With AD+ no Glau 71.4	28%/ 55%			Visual fields - Humphrey Field Analyser	5.2% occurrence rate of probable glaucoma among healthy controls	among AD patients	
			Control 116	W/o AD + No glau 68.1	38%/ 72%			Optic disc evaluation			
				W/o AD + Glau 70.1	2%/4 %						
Cigolle, C	US (1995-2010 of Health and Retirement Study)	Longitudinal cohort, Adults >65y	8847	No info	No info	Cognition	Performance based measure (Telephone interview) to determine cognitive function (0-27)	No info	Vison and hearing predicted cognitive decline (p<0.001)	Cognitive function declines in an accelerating fashion, with older age	Weak
2013	Abstract Only									and visual impairment predicting decline over time	
Feeney, J.; Finucane, C	Ireland (The Irish Longitudinal study on Ageing)	Cross-sectional study	4281	No info	No info	Cognition	MMSE, MoCA, CTT, CRT, SART, Prospective memory, picture memory, word recall and visual reasoning	heterochromatic flicker photometry (HFP) - a non-invasive method of	Linear, Negative binomial and logistic regression. MPOD men=0.20, SD=0.15.	MPOD is significantly associated with cognitive function	Weak
2013	Abstract only							gauging the density of Macular Pigment	One SD inc in MPOD assoc with few errors on MoCA: b=-0.03 p<0.01 & MMSE: b= -0.05 p=<0.05	among older adults.	
									Faster time to complete CTT2: b= -0.02 p<0.01, faster CRT: b= -0.06 p<0.01, better word recall:b=.07 p<0.05		
									Fewer SRT omission errors: b= -0.04 p<0.05, success on prospective memory task: b= 0.14 p<0.01		
Jefferis, J. M; Clarke MP	UK	Cross-sectional study	112	80.7	NA	Cognition	Revised Addenbrook's Cognitive examination (ACE-R)	Acuity: LogMar chart, Cataract density graded using LOC III system	ACE-R scores were split into those items requiring vision and those items not requiring vision, both subscores	Better general cognition (ACE-R total score) was associated with	Weak
2013	Abstract Only								were significantly and similarly associated with visual acuity (p=0.008 and p=0.001 respectively).	better vision (p=0.001)	
Maharani A, Pendleton N	Survey of Health, Ageing and Retirement in Europe 2002-2014	Cross-sectional study	24116	NA	NA	Cognition	Using a summary cognitive score from the questions on episodic memory, verbal fluency, and numeracy.	Self-reported hearing and visual quality. Sensory function recoded into 2:	Older adults with single SI: b=-0.064 (-0.072 to -0.056), with DSI: b=-0.229 (-0.247 to -0.211)	Older adults with single and dual SI showed significantly lower	Weak
2017	Conference abstract							Excellent,v good & good into Good sens fn and fair & poor into Poor sens		cognitive function compared to those without SI	
Miyata K, Mine M	Japan (Fujiwara-Kyo study)	Cross-sectional study	668	76.3	NA	Cognition	MMSE	Cataract surgery history- obtained using a self-reported questionnaire	CI for Cataract surgery group OR: 0.82 (0.68, 0.99) p=0.042 when compared to no cataract sx group	Cat Sx group significantly lower odds for CI than the no Cat Sx group	Weak
2016	Conference abstract										
Nael V, Delcourt C	France (Three-City Study)	Longitudinal cohort study	7722	Range: 65 & older	NA	Cognition	Incident dementia over the 12-year follow-up was actively screened for (diagnosis code)	Dist VF:loss self-reported, as inability/difficulty recognizing a face@ 4 m.	Cox regression models: Near VI: HR=1.30 (1.05, 1.61), Dist VF loss: HR= 1.47 (1.02, 2.11) compared to those	Both near VI and distant VF loss were associated with an increased	Weak
2017	Short oral presentation							Near VI:presenting binoc visual acuity <20/30 @ reading distance of 33cm.	without visual loss	risk of dementia	

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Setti A, Lough man J	Ireland (The Irish Longitudinal Study on Ageing (TILDA))	Cross-sectional study	5021	NA	NA	Cog nitio n	MMSE and MoCA	Vision measured subjectively and objectively (no other info)	Poor hearing and immediate recall of orally presented words: b = -0.46, poor hearing and delayed recall b= -0.60	Vision was a significant predictor of visually presented tasks	We ak
2013	Abstract only								Poor hearing and category fluency b =-1.96, poor hearing & memory (OR 0.23)& absent-mindedness (OR2.18).	but also of category fluency.	
Zheng DD, Christ SL	US (The Salisbury Eye Evaluation Study)	Longitu dinal cohort	2520	Range 65-84		Visi on	MMSE	Acuity: ETDRS chart	VA and MMSE score worsened over time (VA (Log MAR) intercept=.004, slope=.022 for VA; MMSE intercept=27.3, slope=-.59; all P<.001)	Both presenting VA and MMSE score worsened over time	We ak
2017	Conference abstract								The intercept of VA trajectory is statistically significantly associated with the intercept of MMSE trajectory (r=-.267, se=.029, p <.001)		
									suggesting that worse baseline VA is associated with worse baseline MMSE score.		
Ihle A, Oris M	Switzerland (LIVES study)	Cross-sectional cohort	2812	77.9	52.7/ 47.3	Cog nitio n	Processing speed- Was assessed by the Trail Making Test part A, Cognitive flexibility- Was assessed by the Trail Making Test part B	Self-report on a 3-point Likert-type scale whether their	Stability in (inter)relations of the other cognitive and the sensory abilities across the age tranches in old age were not moderated by general cognitive ability, educational level, nor general health status (all ps > .225)	No.Present data do not support the view of a generally increased relation of cognitive and sensory abilities in old age.	We ak
2015				range 65-101			Verbal abilities- Was assessed by administering the Mill Hill vocabulary scale	current eyesight allowed them to read a newspaper by choosing one of the following			
								0 D 'no'; 1 D 'yes, but with difficulties'; or 2 D 'yes, without difficulties.'			
Soto-Perez-de-Celis E	US	Cross-sectional case control	750	72	56/44	Cog nitio n	Score of 11 in the Blessed Orientation-Memory-Concentration test (BOMC).	Self-reported visual impairment based on their rating of their eyesight	OR for VI and: IADL Dependence: 1.9 (1.2-3.2) <.01, Poor Physical Function: 1.9 (1.1-3.3) .03 , Possible CI: 1.9	VI was associated with IADL dependence, poor physical function and	We ak
2018				Range 65-94				(with glasses) as "excellent," "good," "fair," "poor", "totally blind"	(0.7-4.8) 0.20, Depression: 2.5 (1.4-4.3) <.01, Anxiety: 1.4 (0.8-2.4) .19. OR for DSI and: IADL dep 2.8 (1.5-5.3) <0.01	depression. Dual SI was assicated with IADL dependence, anxiety, depression and CI	
									Poor Physical Function: 1.7 (0.9-3.4) .10, Possible CI: 3.2 (1.3-8.1) .01, Depression: 2.5 (1.3-4.8) <.01, Anxiety: 2.3 (1.2-4.2) .01		
Mahar ani A, Dawes P	HRS, ELSA, SHARE	Cross-sectional cohort	13,1 23 HRS	67.8/3 2.2	41.8/ 58.2	Cog nitio n	Episodic memory: In all surveys, the interviewer reads a list of 10 common nouns to the respondent	In HRS and ELSA, self-reported vision quality was collected in all seven waves using the question: 'Is your eyesight [using glasses or corrective lens as usual] excellent (1), very good (2), good (3), fair (4) or poor (5)?'.	HRS: Single impairment β −0.15 (0.02)*	Yes. Those with sensory impairment are at a greater risk of developing cognitive impairment	We ak
2018			11,4 17 ELS A	64.8/3 5.2	45.6/ 54.4		then asks the respondent to recall as many words as possible from the list in any order twice: immediately after the respondent heard the complete list (immediate recall) and at the end of the cognitive function module	In SHARE, we used the two self-reported measures of visual function	Dual impairment β −0.25 (0.04)*	and may show a faster trajectory of cognitive decline that those without sensory impairment	
			21,2 65 SHA RE	64.8/3 5.2	45.5/ 54.5		(delayed recall).	that are present in all waves: distance eyesight and reading eyesight.	ELSA: Single impairment β−0.14(0.02)*		
									Dual impairment β −0.35 (0.05)*		
									SHARE: Single impairment β −0.26 (0.01)*		
									Dual impairment β −0.68 (0.03)*		
Jefferi s JM, Coller ton J	UK (Newcastle 85+ study)	Cross-sectional cohort	839	no info	no info	Cog nitio n	sMMSE	Data collection of general practice records on whether the participant was	Median (inter-quartile range) sMMSE scores were 25 (22-29) for SI and 28 (25-29) for non-SI participants (P= 0.006).	It is important to consider the possibility of vision impairment in older	We ak
2012								registered as blind (severly sight impaired) or partially sighted (sight impaired), or neither (by a consultant ophthalmologist)	SI participants had lower subscale scores on tasks requiring vision (P< 0.001 for each) but also for some subscale	people when carrying out the MMSE and to consider using the MMblind.	
									scores not obviously requiring vision: orientation (P= 0.018) and repetition (P= 0.030).	Beyond this however, visual impairment may be a marker for cognitive impairment.	
MacD onald SWS	Canada	Longitu dinal, cohort	408	NIC 75.44	NIC 35.5/ 64.5	Cog nitio n	Episodic memory-The word recall task (Hultsch, Hertzog, & Dixon, 1990) was used to assess episodic memory	Acuity: Vision-Binocular-corrected distance visual acuity was measured at 3 meters using the Snellen chart.	Age-related change in sensory function.	Decline in visual acuity was also a notable predictor	We ak
2018				SA-MCI :76.71	SA-MCI : 40.5/ 59.5		Inductive reasoning-The Letter Series test (Thurstone, 1962) was used to assess inductive reasoning.	Snellen acuity fractions ranged from 0.20 to 1.	Variables, Intercept γ00, Slope γ10, SE Slope, p	of being classified as SA-MCI or MA-MCI	

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				MA-MCI: 75.68	MA-MCI : 48.2/ 51.8		Perceptual speed-The WAIS-R Digit Symbol Substitution task (Wechsler, 1958) was used to assess perceptual processing speed		Olfaction: 7.05, -0.043, 0.008, < .001		
							Verbal fluency-The Controlled Associations test from the ETS kit of factor-referenced cognitive tests (Ekstrom, French, Harman, & Dermen, 1976) was used to assess verbal fluency.		Distance Vision: 0.946, -0.004, 0.001, < .001		
							Vocabulary-A recognition vocabulary test, combining three 18-item multiple-choice tests from the ETS kit of factor-referenced cognitive tests (Ekstrom et al., 1976), was used to assess vocabulary		Audition: 32.84, 0.796, 0.059, < .001, γ00 = Average sensory function centered at the grand mean of age (74.17 years; SD=9.20); γ10 = slope reflecting the average rate of linear change per additional year of age		
Miyata K, Obayashi K	Japan (HEIJO-KYO Cohort)	Cross-sectional cohort	945	71.7	46.8/ 53.2	Cognition	MMSE	Acuity: using the Landolt ring chart. A better value of LogMAR was used for analysis.	Logistic Regression Analysis for the Association Between Ocular Status and Cognitive Impairment	Yes. ORs for cognitive impairment were significantly lower in the pseudophakic group than in the phakic group	Weak
2015								Questionnaire asking participants about pseudophakia/phakia	Age-adjusted, Multivariate OR for cognitive impairment: Model 1, Model 2, Model 3		
								Confirmed by ophthalmologist using slit-lamp	Phakia (no previous cataract surgery): 1.00 (ref), 1.00 (ref), 1.00 (ref), 1.00 (ref),		
									Pseudophakia (previous cataract surgery): 0.66 (0.45, 0.98), 0.66 (0.44, 0.98), 0.64 (0.43, 0.95), 0.64 (0.43, 0.96)		
									p: 0.038, 0.039, 0.026, 0.031		
Mine M, Miyata K	Japan (Fujiwara-kyo Study)	Cross-sectional cohort	2818	76.3	52.7/ 47.3	Cognition	MMSE; analyzed the MMSE excluding the following vision-related five items: "naming two objects," "following a 3-step command," "reading and following instruction," "writing a sentence," and "visual reconstruction" and the maximum score for this was 22 points.	Acuity: Landolt ring chart at 5m (converted to logMAR) [mild visual impairments (>0.2 logMAR)]	Associations Between Visual Acuity and Cognitive Impairment (multivariate regression model):	Yes. Subjects with mild visual impairments had 2.4 times higher odds of	Weak
2016									OR (CI) for different BCVA in the better eye (logMAR) groups:	having cognitive impairment than those without visual impairment	
									≤0: 1.0 (reference), 0-0.1: 1.8 (1.2-2.7) p 0.007 0.1-0.2: 1.7 (0.8-3.5) p 0.187 >0.2: 3.3 (2.1-5.4) p 0.005		
Steffi G. Riedel-Heller	Germany (LEILA 75+ and AgeCoDe)	Longitudinal, cohort	3,199	79.3	34.7/ 65.3	Cognition	No data	Self-report at baseline	Vision impairment (HR 1.19, 95%CI 1.01-1.42, p=0.043) and the combination of both (HR 1.47, 95%CI 1.18-1.83, p=0.001)	Yes. Vision impairment is independently associated with incident dementia. Looking at the combination we could show that individuals suffering from both are at highest risk for developing dementia	Weak
2019									on dementia incidence adjusted for baseline gender, age, education, marital status, depression, diabetes and cardiovascular risks (smoking, hypertension)		
Asri Maharani	US (Health and Retirement Study)	Longitudinal cohort	19618	57.8	44.6/ 55.4	Cognition	Telephone Interview for Cognitive Status (TICS): include episodic memory, serial of 7 subtraction, and counting backward tests	Self report: "Is your eyesight [using glasses or corrective lens as usual] excellent (1), very good (2), good (3), fair (4) or poor (5)?"	VI and risk of possible CIND HR 1.351 (1.267, 1.441) p< 0.001 when compared to no VI. VI and risk of probable	Yes. Self-assessed sensory impairment is independently associated with cognitive decline and incident possible CIND and probable dementia	Weak
2019	Waves 3 (1996) to 12 (2014)						categorized those scoring 0 to 6 points on the 27-point TICS scale as having probable dementia	Further categorized sensory impairment in the simultaneous model into no impairment,	dementia HR 1.255 (1.074, 1.466) p=0.004 when compared to no VI		

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							7 to 11 as having possible CIND, and 12 to 27 as having normal cognitive function	single (vision or hearing) sensory impairment, and dual sensory impairment (impairment in both senses).			
Phillip Hwang	US (GEM study)	Longitudinal cohort	2827	NA	NA	Cognition	Incident dementia over seven years of follow-up was based on a clinical diagnosis of dementia using DSM-IV criteria	Self-report	For all-cause dementia, the adjusted hazard ratio was 1.27 (95% CI: 1.02, 1.59) for single sensory visual or hearing impairment,	Yes, Dual visual and hearing impairment was strongly associated with increased risk of all-cause dementia and AD	Weak
2019							Alzheimer's disease was determined using NINCDS-ADRDA criteria		and 1.70 (95% CI: 1.18, 2.45) for dual visual and hearing impairment, compared to no sensory impairment		
Virginie Naël	France (The Three-City-Alienor Cohort)	Longitudinal cohort	7460	NA	NA	Cognition	4 cognitive domains (global cognition (MMSE and MMSE-blind), verbal fluency (IST),	At baseline, near VI was measured using the Parinaud scale	Participants with near VI and distance VF loss had lower baseline performances in verbal fluency,	Yes but only baseline. Participants with	Weak
2019							executive function (TMT) and visuospatial abilities (BVRT)) were assessed up to 6 times over 12 years of follow-up	at a standardized reading distance of 33 cm and distance Visual Function (VF) loss was	global cognition, executive function and visuospatial abilities. Regarding changes over time,	visual loss had lower baseline performances in several cognitive tests	
								self-reported, defined as inability or difficulty in recognizing a familiar face at 4m	MMSE where participants with mild near VI exhibited a faster cognitive decline (b= -0.02, p=0.04)		
M.Q.Li	China	Cross-sectional cohort	10116	NA	NA	Vision	AD8 questionnaire	No info	Risk factor for cognitive impairment: visual acuity loss 1.383 (1.188, 1.610)	Yes, The old people who suffered from a variety of chronic diseases including, poor eyesight were prone to cognitive impairment	Weak
2019											
Melanie Varin	US	Cross-sectional cohort	365	AMD: 83.1	32.4/67.6	Cognition	6 cognitive tests orally: the 1-minute verbal fluency test (letter and category versions)	Binocular visual acuity was measured using the ETDRS chart at 2 m	People with glaucoma showed lower scores on 3 cognitive tests than the group with normal vision:	Yes	Weak
2019				Glaucoma:78.1	43.7/56.3		the digit span test (forward and backward versions), and the logical memory test (immediate and 30-minute delayed recall)	VF measured using HFA, CS measured using the Pelli Robson chart at 1 m	the digit span forward and backward tests (b = -0.8, 95% CI -1.5, -0.2 and b= -0.7, 95% CI -1.3, -0.1, respectively)	People with glaucoma showed lower scores on cognitive tests	
Peiyuan Qiu	China (Chinese Longitudinal Healthy Longevity Survey)	Longitudinal cohort	3859	74.5	48.7/51.3	Cognition	Chinese Mini-Mental State Examination (C-MMSE)	Self-report	NA	Yes, visual impairment was a risk factor for cognitive decline	Weak
2019											

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2 & 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	e-supplement 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7, 8 & 9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10, 11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11, 12, 13, 14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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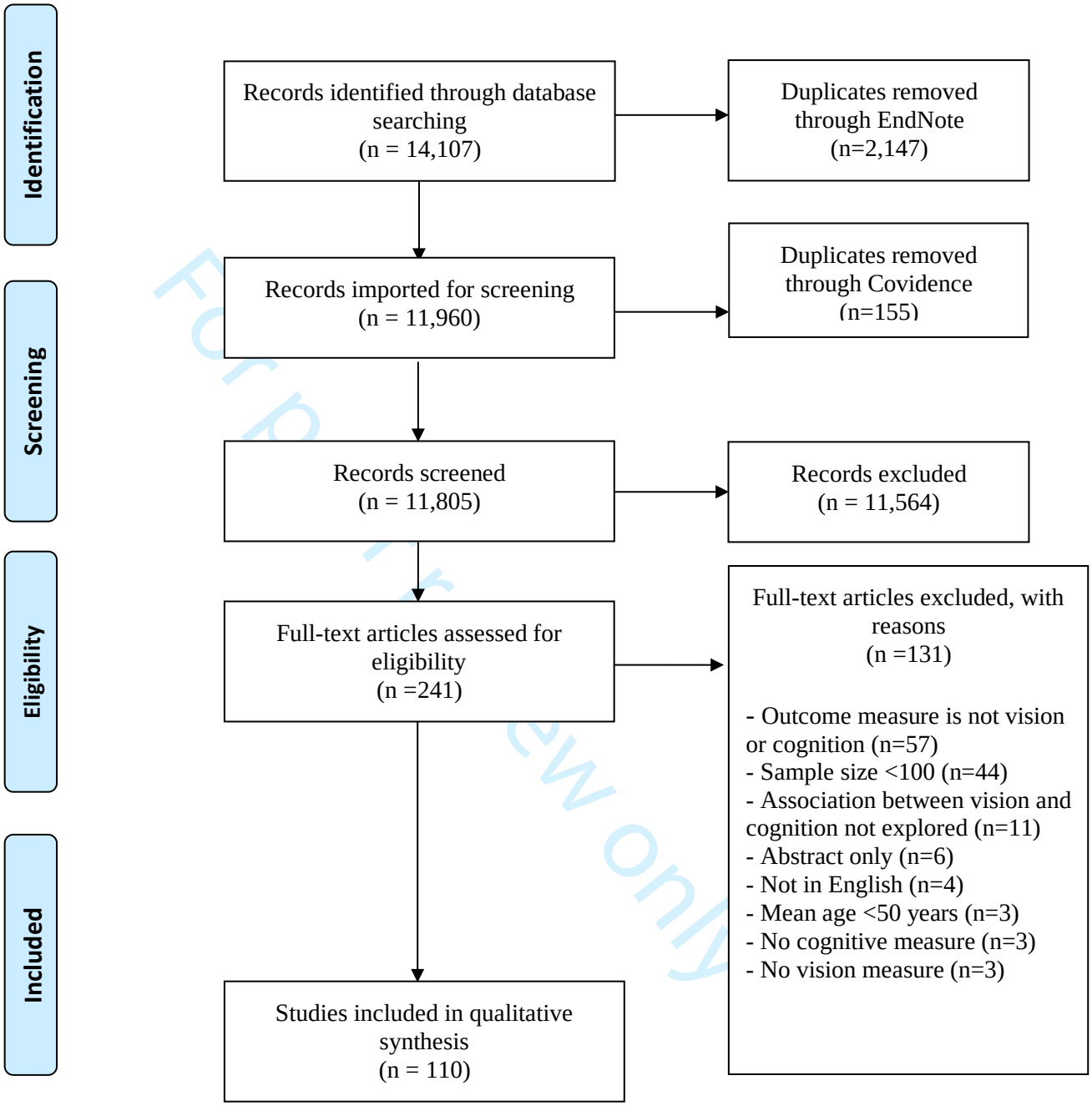
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e-Supplement 2 – Ovid Medline Search Strategy

1. exp Aged/ or exp Geriatrics/
2. (ageing or latelife or laterlife or "late life" or "later life" or elder* or Geron* or Geriatr* or senior*).tw.
3. (old* adj3 (age* or adult* or population* or communit* or people* or person*)).tw.
4. ((aging or aged) adj3 (adult* or population* or communit* or people* or person*)).tw.
5. 1 or 2 or 3 or 4
6. exp Cognitive Aging/ or Cognition/ or exp Cognition Disorders/ or exp Executive Function/ or Memory Disorders/ or exp Neuropsychological Tests/ or Dementia/ or exp Alzheimer Disease/
7. (cognitive or cognition or executive function* or executive control* or executive dysfunction* or executive impairment* or memory or dementia* or alzheimer*).tw.
8. 6 or 7
9. exp Vision, Ocular/
10. exp Vision Disorders/
11. exp Diagnostic Techniques, Ophthalmological/
12. exp Visual Perception/
13. exp Visually Impaired Persons/
14. (vision or visual or visually or occular or Ophthalmol*).tw.
15. 9 or 10 or 11 or 12 or 13 or 14
16. 5 and 8 and 15
17. Animals/ not humans/
18. 16 not 17
19. Epidemiologic Studies/ or exp Observational Study/ or exp Observational Studies as Topic/ or exp Cohort Studies/
20. ("epidemiologic study" or "epidemiologic studies" or "cohort study" or "cohort studies" or longitudinal or prospective* or "observational study" or "observational studies").tw.
21. 19 or 20
22. 18 and 21

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e-supplement 3: FIGURE DESCRIBING A PRISMA FLOWCHART



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The PRISMA for Abstracts Checklist

TITLE	CHECKLIST ITEM	REPORTED ON PAGE #
1. Title:	Identify the report as a systematic review, meta-analysis, or both.	1
BACKGROUND		
2. Objectives:	The research question including components such as participants, interventions, comparators, and outcomes.	2
METHODS		
3. Eligibility criteria:	Study and report characteristics used as criteria for inclusion.	2
4. Information sources:	Key databases searched and search dates.	
5. Risk of bias:	Methods of assessing risk of bias.	
RESULTS		
6. Included studies:	Number and type of included studies and participants and relevant characteristics of studies.	3
7. Synthesis of results:	Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals.	
8. Description of the effect:	Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients.	
DISCUSSION		
9. Strengths and Limitations of evidence:	Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence)	4
10. Interpretation:	General interpretation of the results and important implications	
OTHER		
11. Funding:	Primary source of funding for the review.	4
12. Registration:	Registration number and registry name.	

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Vision Impairment and Cognitive Decline Among Older Adults: A Systematic Review

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Title: Vision Impairment and Cognitive Decline Among Older Adults: A Systematic Review

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ABSTRACT

OBJECTIVES

There has been increasing epidemiological research examining the association between visual impairment (VI) and cognitive impairment and how poor vision may be a modifiable risk factor for cognitive decline. The objective of this systematic review is to synthesize the published literature on the association of VI with cognitive decline, cognitive impairment, or dementia, to aid the development of interventions, and guide public policies pertaining to the relationship between vision and cognition.

METHODS

A literature search was performed with Embase, Medline, and Cochrane library databases from inception to March 2020, and included abstracts and articles published in peer-reviewed journals in English. Our inclusion criteria included publications that contained subjective/objective measures of vision and cognition, or a diagnosis of VI, cognitive impairment or dementia. Longitudinal or cross-sectional studies with ≥ 100 participants aged >50 years were included. The search identified 11,805 articles whose abstracts underwent screening by three teams of study authors. Data abstraction and quality assessment using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP) were performed by one author (NN). 10% of the articles underwent abstraction and appraisal by a second author (LA/VV), results were compared between both and were in agreement.

RESULTS

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110 full-text articles were selected for data extraction, of which 53 were cross-sectional, 43 longitudinal, and 14 were case-control studies. The mean age of participants was 73.0 years (range 50-93.1). Ninety-one (83%) of these studies reported that VI was associated with cognitive impairment.

CONCLUSION

Our systematic review indicates that a majority of studies examining the vision-cognition relationship report that VI is associated with more cognitive decline, cognitive impairment, or dementia among older adults. This synthesis supports the need for additional research to understand the mechanisms underlying the association between VI and cognitive impairment and to test interventions that mitigate the cognitive consequences of VI.

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Article Summary

Strengths and limitations of this study:

- There was heterogeneity in the measurement of cognitive and visual function among all included studies.
- The QA tool used for assessing quality of included studies penalized longitudinal studies that lost over 40% of participants due to dropouts/withdrawals, which is common in studies that span over many years.
- Majority of the included studies were cross-sectional, and these are prone to selection bias.

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Introduction

Dementia is among the most pressing public health challenges of the 21st century.[1] In 2015, 46.8 million people were living with dementia, and the number is expected to double every 20 years.[2] Vision impairment (VI), another major global health problem, affects at least 2.2 billion people worldwide,[3] most of whom are aged 50 years and older.[4] Both cognitive and VI are projected to affect an increasing number of people over time, primarily due to population ageing.[4,5]

Prior work has suggested that cognition and vision are associated,[6, 7] and while there are shared risk factors (neuropathological/vascular),[8] there is also longitudinal evidence that VI is associated with cognitive changes.[9] The mechanisms underlying the vision-cognition relationship have not yet been fully characterized, but it is hypothesized that sensory loss, such as hearing and VI, may lead to increased cognitive load, structural and functional changes in the brain, and decreased emotional and social well-being, all of which could potentially increase the risk of cognitive impairment.[9,10] While the role of treating hearing loss in preventing cognitive impairment has been acknowledged, VI has not yet been recognized as a potentially modifiable risk factor for cognitive impairment.[1, 11]

Since the majority of VI is due to correctable conditions, namely refractive error and cataract,[12] establishing the existence of an association between vision and cognitive impairment could present an additional opportunity to prevent cognitive impairment and dementia through interventions that optimize vision. In this systematic review of the literature, we examined the association between cognitive and vision impairment among older adults in existing observational studies. This qualitative review summarizes the existing research examining the

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vision-cognition relationship, providing insight on data gaps and areas for continued investigation, as well as highlighting differences in methodological approaches that may impact the interpretation of results across studies.

Methods/Literature Search

We followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (**e-supplement 1**).

Cross-sectional and longitudinal studies reporting a measure of association between visual function and cognitive impairment were included if they had ≥100 participants aged of ≥50 years (mean) at baseline. Reasons for exclusion of studies were: 1. Outcome measure was not vision or cognition, 2. Association between vision and cognition was not explored, 3. Sample size <100, 4. Publication not in English, 5. Mean age <50 years, 6. No cognitive measure, 7. No vision measure, 8. Outcome was not part of inclusion criteria.

An academic librarian searched: Ovid Medline, Embase, Cochrane, and PubMed from their inception through March 2020, and developed a search strategy that combined controlled vocabulary and keywords related to geriatrics, cognition, and vision (**e-supplement 2**-complete Ovid Medline search strategy). Searches were limited to human studies published in English. Conference and poster abstracts, and short oral presentations were also included.

Search results were exported to Covidence (Veritas Health Innovation, Melbourne, Australia). Three teams of two reviewers each worked independently and in duplicate to screen titles,

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abstracts, and full-text articles to determine inclusion (NN and MM; EC and YS; VV and LA). Disagreements were adjudicated by a member of the other study team.

Data were extracted from the included publications by one author (NN), and another (LA) extracted data from a random sample of 10% to compare results. Any discrepancies were adjudicated by a third author (VV). Data collected for each publication included: study design, participant characteristics, vision and cognition assessment methods, and the summary measure that described the vision-cognition association.

The methodological quality of included studies was assessed by one author (NN) using the Effective Public Health Practice Project Quality Assessment Tool (EPHPP),[13] and the global quality ratings and findings were summarized qualitatively. A random 10% sample was reviewed by another author (VV) to ascertain consistency in quality assessment.

Results

Study Selection

e-supplement 3 is a PRISMA flowchart that describes the results of the search strategy of articles that examined the association between VI and cognitive impairment or decline. Of the 11,805 studies that were imported for screening, 110 articles were included in our final systematic review.[14-123]

Description of Included Studies

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Table 1 describes the characteristics of the studies included. The total number of participants in this review was 9,799,329 (range: 112 to 7,210,535 per study), with a mean age of 73.0 years, (range: 50.0 to 93.1). Of the total 110 studies included, 53 were cross-sectional, 43 were longitudinal, and 14 had a case-control study design. The range of follow-up time for the longitudinal studies was 2 months to 10 years.

Of the 110 studies included, 51 reported findings from participants enrolled in population-based studies. There were five studies each from the following large population based longitudinal studies: English Longitudinal Study of Aging (ELSA) and The Three-City Study. Three studies each from Fujiwara-Kyo Study, Salisbury Eye Evaluation Study, Irish Longitudinal Study on Aging (ILSA), and Singapore Epidemiology of Eye Diseases (SEED). Two studies each came from The Newcastle 85+ study, Study of Osteoporotic Fractures (SOF), Blue Mountain Eye Study, Australian Longitudinal Study on Aging (ALSA), Health and Retirement Study (HRS), National Health and Aging Trends Study (NHATS), Leiden 85+ study, Health ABC study, and the Singapore Malay Eye Study (SiMES). Additionally, 10 studies used insurance claims data from different countries.

The studies in this review included participants from over 17 different countries (**Table 1**), 30 studies (27%) from the United States, followed by 25 studies from Europe (23%) including the United Kingdom, Germany, Ireland, Finland, Switzerland, France, and Netherlands. 90 of the studies were published between 2009 to 2020. All papers provided a description of sampling methods. 16 studies were included which were either conference abstracts or short oral presentations.[24, 32, 45, 49, 50, 51, 53, 56, 93, 94, 97, 113, 114, 115, 116, 117]

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Assessment of Cognitive Function

To assess cognitive function, 89 studies used objective assessments, 13 used other assessment methods such as self-report and diagnosis codes, and 7 studies used a combination of both **(Table 2);** one study did not provide information about cognitive function assessment.[93] Mini-Mental State Examination (MMSE) was the most commonly used objective method to assess cognitive function (42 studies). Other methods used to objectively measure cognition included: Montreal-Cognitive Assessment test (MoCA),[28, 32, 34, 53, 94] Addenbrooke's Cognitive Examination-Revised (ACE-R),[45, 61, 80] Cognitive Performance Scale (CPS),[42, 49, 66] Blessed-Orientation-Memory-Concentration test (BOMC),[25, 63] Abbreviated Mental Test (AMT),[73, 75, 90, 101, 111, 120] Blessed Dementia Scale (BDS), [87] Digit Symbol Substitution Test (DSST),[22, 123] and Cambridge Cognitive test (CAMCOG). [89] For the studies that used other assessment methods, 4 used self-reported cognitive measures, [18, 22, 39, 106] and 9 used diagnostic codes to define cognitive decline and/or dementia. [19, 20, 21, 51, 70, 100, 104, 116, 118] Among the studies that used a combination of objective and subjective methods, 4 used self-reported cognitive function along with an objective measure. [22, 30, 48, 55]

Assessment of Visual Function

In order to assess visual function, 66 studies used objective assessments, 34 used other assessment methods such as self-report and diagnosis codes, and 8 studies used a combination of both **(Table 2)**; no information was available from two studies.[24, 117] Visual acuity (VA) was the most commonly measured visual function (42 studies), of which the Snellen acuity chart was the most commonly used method (18 studies). VA was also measured in combination with other visual functions, including: visual fields (VF) (6 studies), [20, 51, 71, 91, 103, 119] contrast sensitivity

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(CS) (8 studies), [29, 52, 87, 94, 95, 103, 119, 123] macular pigment optical density (MPOD) (2 studies), [14, 34] and fundus photography (2 studies).[64, 119] Other methods used to objectively measure visual functions included: color vision,[17] VF only,[28] CS only,[35, 77] fundus photo with grading,[38, 101] and autorefraction.[75] Other assessment methods included: self-reported vision (24 studies) and diagnostic codes or patient records to define VI (10 studies). [19, 21, 46, 70, 84, 100, 105, 116, 118, 122] The studies that used a combination of methods, 8 studies used self-report along with an objective measure of visual function.[22, 40, 53, 59, 69, 72, 108, 114]

Quality of Studies

The methodological quality of included studies was assessed using the EPHPP.[13] The tool assessed each study on 5 domains: 1) Selection bias, 2) Study design, 3) Confounders, 4) Data collection methods, and 5) Analysis. For each included study the five relevant domains were ranked on a three-point Likert scale with three representing a low risk of bias ("strong"), two a possible risk of bias ("moderate"), and one a high risk of bias ("weak"). An overall rating was derived following the EPHPP methodology. A study consisting of at least one "weak" rating in a domain received an overall rating of "moderate," while those with two or more domains with "weak" ratings were automatically classified as "weak" overall. We present our studies in three different tables which is categorized based on the overall ratings, with "strong", "moderate", and "weak" studies in **Tables 3, 4, and 5,** respectively. In our sample, 17 studies received a rating of strong, 70 moderate, and 23 weak.

Study Findings

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Of the 110 studies included (**Tables 3, 4 and 5**), 91 found a significant positive association between VI and cognitive decline, cognitive impairment or dementia, and 13 studies found no significant association. [26, 30, 43, 44, 60, 61, 69, 82, 83, 91, 95, 116, 118] There were 6 studies that were inconclusive. [23, 37, 68, 71, 80, 84] Of the 91 studies that found a significant association, 77 used objective methods to assess their vision or cognitive outcome. Of the 43 longitudinal studies, 35 found a significant association between VI and cognitive decline, cognitive impairment or dementia. The most commonly presented statistical measures were odds ratios (OR) and hazard ratios (HR). The random 10% of the study sample that was separately extracted by an independent author (LA) was found to be similar to elements from the primary extraction.

Discussion

In this systematic review, we evaluated and synthesized the literature examining the association between VI and cognitive function among older adults, and found strong agreement that VI is associated with cognitive impairment, cognitive decline, or dementia. Results from the longitudinal studies that found a positive association between vision and cognition supports our hypothesis that VI may be a risk factor for cognitive impairment, cognitive decline, or dementia.

Ninety-one studies reported associations between decline in visual and cognitive functions. Garin et al [40], who received a "moderate" rating in the quality assessment (QA), performed a cross-sectional analysis in a representative sample of Spanish population and measured cognition objectively. They also measured distance and near vision and found that objective and subjectively measured poor distance and near VA were associated with worse cognitive

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functioning. Lin et al [47], used data from a large longitudinal cohort study of older women and found that VI was associated with greater odds of cognitive and functional decline over 2 years This study used objective measures of assessment for both vision and cognition and received a "strong" rating in the bias assessment. Luo et al [48], who received a "moderate" rating in QA, performed a cross-sectional analysis on a large population sample from China. They reported that those with VI and Dual Sensory Impairment (DSI) were more likely to have severe to extremely severe dementia compared to those without any sensory impairment. Another longitudinal study that received a "moderate" rating in QA from Germany by Hajek et al [58] with a large sample size (n=2,394) showed that the onset of severe VI was associated with a decline in cognitive function scores. Uhlmann et al [65] in their paired case-control study between VI and dementia patients concluded that VI was associated with both an increased risk and an increased clinical severity of AD. Although Frost et al [38] found a strong association between early Age-related Macular Degeneration (AMD) and AD, their study was cross-sectional, and the sample size was too low to derive an inference. Both these studies received "moderate" rating in QA. Davies-Kershaw et al [96] in their longitudinal analysis using the ELSA wave 2 and wave 7 data found that individuals in the younger group (50–69 years) and with moderate and poor self-rated vision were at greater risk of developing dementia than those with normal self-rated vision. Hamedani et al [100] used Medicare claims data from 2014 consisting of 472,871 participants and concluded that blindness/low vision was associated with a greater odd of Alzheimer's disease and all-cause dementia. Both these studies also received "moderate" rating in QA. Soto-Perez-De-Celis et al in their cross-sectional study case-control study found that DSI was significantly associated with possible CI. However, the study received an overall "weak" rating in QA.

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Of the 91 studies that found an association between VI and cognitive function, 35 were longitudinal, 46 were cross-sectional and 10 were case-control studies. Of the 13 studies that found no association between VI and cognitive function, 6 were longitudinal, 5 were cross-sectional and 2 were case-control studies. Ihle et al [60] performed a cross-sectional analysis using a sample of 2,812 participants from Switzerland. They objectively measured cognition and vision and concluded that their data did not support an increased relation of cognitive and sensory abilities in old age. This study received a "weak" rating in QA. Hong T et al [83] used data from Blue Mountain Eye Study, a longitudinal study from Australia that studied associations between VI and a decline in MMSE scores over a duration of 10 years. The study concluded that VI was not associated with cognitive decline over 5 years or 10 years. Although the study included a large number of participants overall (n=2,334), only 152 individuals with VI were included in this analysis, suggesting that there may have been survival bias. Brenowitz et al [95] in their longitudinal study using the Health ABC data concluded that VA and CS independently were not significantly associated with incident dementia. However, Swenor et al [123] used data from the same study and found that impaired visual acuity, contrast sensitivity, and stereo acuity had a greater risk of incident cognitive impairment. This could be due to the different outcome measures assessed, i.e. dementia [95] vs cognitive impairment[123]. These three studies received a "moderate" rating in QA. Michalowsky et al [118], who received a "strong" rating in their case-control study concluded that VI was not significantly associated with dementia, a combination of both visual and hearing impairments was significantly associated with the risk of dementia.

There was considerable heterogeneity in the measurement and reporting of cognitive function. Studies measured cognitive function using a variety of instruments with the most common being

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MMSE. The MMSE is a paper-based test with a maximum score of 30, with lower scores indicating more severe cognitive impairment. A score of 24 is often used as a threshold to define "normal" cognitive function.[124] The MMSE has been found to be a valid and reliable tool as assesses by many studies.[124, 125] Several studies used self-report, diagnosis codes and data from existing records to define cognitive status. Similarly, visual function was also assessed by various methods including self-report. While VA was assessed most commonly, there was significant variation in the charts and tools used to assess it. The parameters used to define cognitive decline and VI may have impacted results across and within studies.

Our systematic review has found that there is a strong consensus in the literature that VI is associated with cognitive decline, cognitive impairment, or dementia. Two hypotheses may help explain this association. The first one is that a common pathological process (e.g. vascular disease) might be responsible for both the sensory and cognitive impairment in older adults. The second one is that by increasing cognitive load, sensory impairments such as VI might cause cognitive impairment. [126] Literature also suggests that vision rehabilitation in the form of cataract surgery slows the rate of cognitive decline, and therefore, early vision interventions could potentially reduce risk of dementia.[127]

Our review evaluated bias for all of the 110 included studies. The majority of studies included in our review were cross-sectional, and according to EPHPP guidelines, cross-sectional studies can only receive a low or moderate rating in the bias assessment. Cross-sectional studies are also prone to selection bias, thus yielding estimates that may not reflect true associations in the target population. Studies receiving a strong rating were all longitudinal. However, the tool penalizes

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longitudinal studies that lose >40% of participants due to dropouts/withdrawals. This may, perhaps unfairly, affect longer longitudinal studies to a greater extent since they collect data over many years and can have more dropouts due to deaths since they are conducted among older adults.

This review has several important implications. First, it highlights the need for standardized methods to assess and define both visual and cognitive function that will aid future research on these emerging public health issues. Second, it brings into focus the consistent association of VI with cognitive impairment in older adults and the need to better understand the mechanisms underlying this relationship. Third, as the longitudinal results support the sensory consequence theory, and suggest that VI may be a risk factor for cognitive decline, this points to a need for formulating preventive measures and vision rehabilitation models, such as prescription glasses, cataract surgery, low vision rehabilitation, etc., that could have the potential to improve overall health and well-being of older adults.

Limitations

Given the large number of studies included in this review and the heterogeneity of measures used to assess the outcome, it was not possible to compare and meta-analyze results across studies. Although 35 longitudinal studies found a positive association between VI and cognitive decline, we cannot establish temporality between this relationship due to the heterogenous nature of the studies. The studies included diverse populations, with different disease processes, and variation in definitions of both cognitive and visual impairment. There is also potential bias associated

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with studies that used different protocols for cognitive and sensory measurements. The MMSE, which was the most commonly used assessment method for testing cognition is sensory dependent and therefore one can argue that the results may be confounded with vision impairment.[128] Further studies should examine the impact of using vision independent cognitive tests on the vision-cognition relationship. Our review examined all cause vision impairment and dementia, and further study is needed to examine the vision-cognition relationship by dementia subtype and by different vision pathology. However, despite the heterogeneity in studies and assessment methods, we synthesized the evidence qualitatively and by taking into account study quality assessed using a validated tool. While our search strategy was robust, it may have been limited by the exclusion of studies that were not published in English.

Conclusion

The number of older adults with VI and dementia is increasing globally, and therefore the elucidation of the relationship between vision and cognition is of particular public health importance. This systematic review found that the positive association of VI with cognitive decline, cognitive impairment, or dementia is largely consistent across studies using different measures of vision and cognition, as well as between countries and cohorts. This overall agreement in the literature suggests that poor visual and cognitive function are associated, and that additional research is now needed to move beyond documenting these associations. The focus of this area of research should now turn to identifying the factors and strategies that mediate the vision-cognition relationship and identifying potential interventions, such as vision rehabilitation models and strategies tailored to people with VI, that may mitigate the cognitive implications of VI.

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Contributorship Statement:

Study concept and design: Bonnielin Swenor, Heather Whitson

Collection, management, analysis, and interpretation of the data: Bonnielin Swenor, Niranjani Nagarajan, Elizabeth Couser, Mina Motaghi, Yi Sun

Drafting of manuscript: Niranjani Nagarajan and Lama Assi

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Competing Interest:

a) Any associations with commercial entities that provided support for the work reported in the submitted manuscript (the time frame for disclosure in this section of the form is the lifespan of the work being reported): None
b) Any associations with commercial entities that could be viewed as having an interest in the general area of the submitted manuscript (the time frame for disclosure in this section is the 36 months before submission of the manuscript): None
c) Any similar financial associations involving author's spouse or children under 18 years of age: None
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Data sharing statement: All data relevant to the study are included in the article

Ethical Approval Statement: Not applicable. This systematic review did not require an ethics committee approval

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1

Demographics and Study Characteristics	**No.**
Total Number of studies	110
Total number of participants	9,799,329
Mean age, y, IQR	73.0, (50-93.06)
Study design	
Cross-sectional	53 (48%)
Longitudinal	43 (39%)
Case-Control	14 (13%)
Country	
US	30 (27%)
Japan	6 (5%)
United Kingdom	6 (5%)
China	6 (5%)
Australia	5 (4.5%)
France	5 (4.5%)
Germany	5 (4.5%)
Singapore	5 (4.5%)
Ireland	3 (3%)
Canada	3 (3%)
Spain	2 (2%)
Taiwan	2 (2%)

2 **Table 1: Patient Demographics and Study Characteristics**

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Sweden	1 (<1%)
Nordic Countries	1 (<1%)
New Zealand	1 (<1%)
Switzerland	1 (<1%)
Netherland	1 (<1%)
Others	27 (25%)

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Table 2: Measures of Vision and Cognition Assessed in Studies

Outcome Measures	Number of studies
Cognition (n, %)	
Objective assessment n=89 (81%)	
MMSE	42 (47%)
MoCA	5 (6%)
Global cognition scale	6 (7%)
Others	36 (40%)
Other assessment methods n=13 (12%)	
Self-report	4 (31%)
ICD diagnosis/from records	9 (69%)
Combination of Objective + Subjective (5%)	7
No info (<1%)	1
Vision (n, %)	
Objective n=66 (70%)	
Acuity	42 (65%)
Acuity + others	16 (23%)
Others	8 (12%)
Other assessment methods n=34 (22%)	
Self-report	24 (71%)

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ICD diagnosis/from records	10 (29%)
Combination of Objective + Subjective *(7.5%)*	8
No info (<1%)	2

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Table 3: Studies with a "strong" rating

Author, Title & Year	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Dearborn, PJ	US (Maine-Syracuse Longitudinal Study (MSLS)	Longitudinal cohort, wave 6&7	655	Normal VA 60.34	40%/60%	Cognition	Visual-Spatial Organization and Memory, Scanning and Tracking, Verbal Episodic & Working memory	Acuity: Snellen eye test (log transformed)	Cross-sectional (CSA) and prospective analysis (PA) using multiple linear and multiple logistic regression	Poorer VA associated with lower cognitive function and 5-yr decline	Strong
2018	compared sixth [2001-2006] wave and			VI 70.6	40%/60%		Global composite score-loaded equally across domains, z-transformed		Global OR: CSA 1.512 (0.71-3.23), PA 1.539 (0.74-3.21), VSMO OR: CSA 1.28 (0.63-2.59), PA 2.26 (1.16-4.4)	in cognitive function over a range of domains incl the global measurements	
	seventh [2006-2010] wave)								Working OR: CSA 1.73 (0.85-3.51), PA 1.55 (0.78-3.08), VEM OR: CSA 0.55 (0.25-1.24), PA 1.43 (0.73-2.81)	Visual-spatial organization, verbal episodic memory	
Diniz-Filho, A.; Delano-Wood, L.; Daga, F. B.;	US	Longitudinal cohort	115	67.4		Cognitive Decline	Montreal Cognitive Assessment test (MoCA); assesses atten and concentration, executive functions	Standard Automated Perimetry (SAP) using 24-2 SITA	Univariate: 5-point decline in MoCA=0.18 dB increase of residuals of SAP MD, R2 = 4.3% (0.06-0.30)p0.003	Statistically significant association between change in	Strong
2017							memory, language, visuoconstructional skills, conceptual thinking, calculations and orientation	<33% fixation loss and <15% FP only included	Multivariate: 5-point decline in MoCA=0.23 dB inc of residuals of SAP MD, (0.11-0.35) p<0.001	MoCA scores and visual field variability over time	
Fischer, M. E	US (The Epidemiology of Hearing Loss Study)	Longitudinal cohort. Baseline info was from EHLS-2 (1998-2000), CI data from EHLS-3 (2003-05)and EHLS-4 (2009-10)	1884	66.7	40.9/49.1	Cognition	MMSE	Contrast sensitivity using Pelli-Robson letter charts. VI was defined as	Cox discrete time proportional hazard analyses performed to model relationship btwn CI and Sensory Imp.	Hearing, Visual and Olfactory impairment were independently	Strong
2016								score <1.55 log units in the better eye	Vision: HR = 2.05 (1.24, 3.38); Olfaction: HR = 3.92 (2.45, 6.26), Hearing: HR= 1.90 (1.11, 3.26)	associated with cognitive impairment risk	
Hall, TA	US (Impact of Cataract on Mobility (ICOM) study)	Longitudinal cohort study. 3 groups identified: No cataract, with cataract and undergoing surgery, with cataract and declined surgery	No Cat: 92	66.8	48.9/51.1	Cognition	Mattis Organic Mental Syndrome Screening Examination (MOMSSE)- evaluates 14 domains (abstraction,	Acuity: Dist VA ETDRS chart, Log Contrast senstivity: Pelli-Robson chart	Cat+no Sx & cat+ surgery grps signifcantly less CI (P <.001 and P =.009, respectively) than at baseline.	Cataract surgery does not affect cognitive function	Strong
2005			W Cat+Sx: 122	70.9	41.8/58.2		orientation,memory,speech,comprehension). Scored 0 to 28 (lower score higher functioning)	Ophthalmologist/optometrist graded cataract at clinic visit	No cataract group=no change in Cog status, association between change in VA in the better and worse eye and		
			W cat+ no SX: 87	71.1	59.8/40.2				change in MOMSSE observed (p=0.003 &p=0.03, respectively)		
Lin MY, Gutierrez PR	US (Study of Osteoporotic Fractures (SOF))	Longitudinal cohort	Vision testing sample:1668	75.9	0/100	Cognition	3MS-Modified version of MMSE	Acuity: Bailey Lovie Target	VI (worse than 20/40): Cog decline - 1.78 (1.21-2.61), Functional decline- 1.79 (1.15-2.79)	Vision impairment is associated with greater odds of cognitive and	Strong
2004			Overall 6112	76.1					DSI: Cog decline - 2.19 (1.26-3.81), Functional decline- 1.87 (1.01-3.47)	functional decline over time in older women	
Jefferis JM, Taylor JP	UK	Longitudinal cohort.	112	Normal cog: 80 (3.8)	20/80	Vision	Addenbrooke's Cognitive Examination (ACE-R; which includes the MMSE)	Acuity: LogMAR, Other: VFQ-25	Normal cog group: Mean at baseline 0.13 (0.09), 1-year FU 0 (0.09), 95% CI for difference: 0.09 to 0.16 <0.001	Patients with impaired cognition benefit from cataract surgery, but not to the same extent as patients with normal cognition. Cognitive impairment may, however, limit visual improvements	Strong
2015				Impaired cog: 81.2 (3.9)	30/70				Impaired cog group: Mean at baseline 0.18 (0.14), 1-year FU 0.06 (0.11), 95% CI for difference: 0.08 to 0.16	following cataract surgery	
Reyes-Ortiz C, Kuo YF	US (Hispanic Established Populations for Epidemiologic Studies of the Elderly)	Longitudinal cohort.	2140	Near VI: 73.5	39.3/60.7	Cognition	MMSE-blind	Acuity: subjects hold cards at least 7 inches from their eyes and asking them to read the numbers	Near vision impairment, -0.62 +/- 0.29, .03, Interaction btwn near VI & time= -0.13 +/- 0.07, .045	Results showed that near vision impairment was predictive of	Strong
2005				Distance VI: 75.3	35/65			Each card had 7-digit "telephone numbers" of three different type sizes: 7, 10, and 23 points	Distance vision impairment = -0.06 +/- 0.36, .87, Interaction btwn distance VI & time, -0.12 +/- 0.08, .14	cog decline in older Mexican Americans independent of other health factors	

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Haan M, Espeland MA	US (Women's Health Initiative MS-MRI)	Longitudinal cohort	511	69	0/100	Cognition	Modified Mini Mental State (3MS)	Acuity: Snellen with pinhole	Acuity slope in model of 3MSE score (β slope = 0.33, p = 0.06). Covariate-adjusted mean 3MSE scores for women	Retinopathy as a marker of small vessel disease is a risk factor for cerebrovascular disease that may influence cognitive performance and related brain changes	Strong
2012									with retinopathy compared to others throughout follow-up: mean (SE) difference 1.01 (0.43), p = 0.019	Very mild association between acuity and 3MSE scores	
Trick GL, Trick LR	US	Case-control. Visually and cognitively normal controls (n=61), and senile dementia of Alzheimer's type (SDAT) n=61	122	SDAT: 73.4	NA	Cognition	each patient was assigned a clinical dementia rating (CDR) based upon a scale developed and tested at Washington Uni	Acuity: ETDRS. Visual field analysis using 30-2 HFA	Patient groups, Foveal threshold, Mean deviation, Pattern standard deviation, Corrected pattern standard deviation.	We found that the frequency of potentially unreliable visual fields was significantly higher (chisquare,	Strong
1995				Control: 72.8			i.e CDR scale ranges from 0.0 (individual without dementia) to 3.0 (severely demented).		Control (n=44): 33.2+/-3.05, -2.81+/-3.45, 3.48+/-2.05, 2.58+/-2.30	p = 0.0232) in SDAT patients (44.3%) than in controls (27.9%)	
									SDAT (n=34): 30.79+4.00 (p=0.0027), -5.11+/-3.17(p=0.0057), 4.461+/-2.49(NS), 3.91+/-2.73(p=0.0408)		
									CDR=0.5(n=25): 31.12+/-3.54, -4.25+/-2.23, 4.47+/-2.60, 3.83+/-2.89		
Helmer C, Malet F	France (The Three-City-Alienor Cohort)	Longitudinal, cohort	812	79.7	35/65	Cognition	MMSE; Free and Cued Selective Reminding Test Isaacs Set Test the Trail Making Test parts A and B Then suspected cases were assessed by a neurologist	Acuity: No chart name provided, IOP: pneumotonometer	Association of OAG and incident dementia: Model 1: OR 4.3 95%CI 1.7-10.8 p 0.0020, Model 2: OR 4.2 95%CI 1.6-10.9 p 0.0030	Yes. Results show that OAG participants were at an increased risk for developing dementia during the 3-year follow-up time period.	Strong
2013									Model 3: OR 3.9 95%CI 1.5-10.4 p 0.0054, Association of Vertical cup:disk ratio and incident dementia: ≥0.65: OR 3.7 95%CI 1.4-9.7 p 0.0083 ≥0.70: OR4.4 95%CI 1.5-12.5 p 0.0055		
									Association of Minimal rim:disk ratio =< 0.1 and incident dementia: OR 2.7 95%CI 1.005-7.1 p 0.0489		
Javier de la Fuente	UK (English Longitudinal Study of Ageing)	Longitudinal, cohort	3508	69	43.2/56.8	Cognition	Four measured tests of verbal fluency, processing speed,	Three self-reported items covering eyesight in far, near, and general vision	Visual (β = 0.140, p < .001) difficulties predicted cognitive difficulties 8 years later	Yes. Visual difficulties were identified as predictors of	Strong
2018							and short-term and long-term memory	Dichotomized, collapsing "Excellent," "Very good," and "Good" as "Absence of difficulties".	The latent increase in cognitive difficulties was steeper in people with visual impairment (*d* = 0.52, p < .001)	subsequent cognitive decline in the old age	
Anni Hamalainen	Canada (Canadian Longitudinal Study on Aging)	Cross-sectional cohort	30029	No info	No info	Cognition	5 cognitive tests: Mental Alternation Test, Animal Fluency test, Controlled Oral Word Association Test, Stroop test	Vision was measured as the better-seeing eye pinhole-corrected visual acuity (VA; reported in logMAR)	VA was a predictor of only executive function (b=-0.785, p<0.001) but not memory	Yes, greater executive function scores were associated with better vision	Strong
2019							and Rey Auditory Verbal Learning Test with immediate and 5-minute recall				
Chuanying Huang	UK (English Longitudinal Study of Ageing)	Cross-sectional cohort	4197	Optimal:70.21	58.6/41.4	Cognition	Memory (immediate and delayed), verbal fluency, and time orientation	Self-report: 'Is your eyesight (using glasses or corrective lens as usual) excellent, very good, good, fair, poor, or registered blind?'	poor vision (= –1.309, p = 0.004), and poor dual sensory function (= –2.442, p = 0.013) was associated with worse cognition	Yes.	Strong
2019				Good: 71.84	51.2/48.8		Global cognition was the sum of scores for the 3 domains, and higher score indicates better cognitive performance	Answer of 'excellent' or 'very good'was defined as optimal vision. Answer of 'fair' or 'poor' was			
				Poor: 74.75	47.2/52.8			defined as poor vision. Thus vision was classified as optimal, good, and poor vision			
Virginie Naël	France (The Three-City-Alienor Cohort)	Longitudinal, cohort	7736	Incident dem:76.9	34.8/65.2	Cognition	3-step procedure: 1. MMSE, the Isaacs set test and the Benton Visual Retention Test	Near VA assessed using the Parinaud scale. Mild near VI was classifed by Parinaud 3 or 4 (Snellen equivalent 20/30–20/60)	Moderate to severe near VI was associated with an increased risk of dementia in the frst 2 years (HR2.0, 95% CI 1.2–3.3)	Yes. Near VI may represent an indicator of dementia risk at	Strong

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2019				No inci dem: 73.6	39.2/60.8		2.senior neurologist to establish a clinical diagnosis, 3.an independent committee of	and moderate to severe near VI by Parinaud>4 (Snellen equivalent<20/60)	and from 2 to 4 years (HR 1.8, 95% CI 1.1–3.1). distance VF loss was associated with an increased risk beyond 4 years (HR 1.5, 95% CI 1.1–2.0)	short and middle-term, mostly in depressedelderly people	
							neurologists and geriatricians reviewed all potential cases of dementia obtain a consensus on the diagnosis and etiology, according to the DSM-IV and the NINCDS-ADRDA criteria	Dist VA was selfreported, defued as an inability or diffculty in recognizing a familiar face at 4 m	but the association was no longer signifcant after taking into account baseline cognitive performances		
Zhi Wei Lim	Singapore (Singapore Epidemiology of Eye Diseases (SEED))	Longitudinal cohort	2478	67.6	50.7/49.3	Cognition	A locally validated Abbreviated Mental Test (AMT)	VA was measured at 4 m using the logMAR number chart	Baseline VI was associated with a decrease in AMT score over 6 years (β = −0.27; 95% CI, −0.37 to −0.17; P < .001)	Yes, , poor vision was independently associated with a decline in cognitive function	Strong
2020								(Lighthouse International).VI was defined as presenting VA worse than 20/40.	When change in vision over 6 years was evaluated, unchanged or deteriorated VI was associated		
									with a decrease in AMT score over 6 years (β = −0.29; 95% CI, −0.40 to −0.18; P < .001)		
Bernard Michalowsky	Germany (NA)	Case-control study	122708	81	39/61	Cognition	ICD-10 diagnosis code for dementia	ICD-10 codes for visual impairment	Visual impairment was not significantly associated with dementia (OR = 0.97, CI = 95% 0.97–1.02, p = 0.219)	No, but DSI was significantly associated with risk of dementia	Strong
2019									Combination of both visual and hearing impairments and the risk of dementia (OR = 1.14, CI = 95% 1.04–1.24, p = 0.005)		
Jung-Yu Liao	Taiwan (National Health Insurance Research database)	Case-control study	9200	NA	LOAD 36.7/63.3	Cognition	LOAD cases diagnosed as dementia and prescribed any acetylcholinesterase inhibtiors (AChEIs)	ICD-9 codes	Disorders of refraction and accomodation with LOAD OR for Year 1,2,3,4: 1.2, Year 6:1.3	Yes, in the path analysis model, disorders of ref and acc	Strong
2020	1997-2013				Without LOAD 36.7/63.3				Total effect reflects an association between prior diseases and LOAD incidence via all paths in the model: 0.042	had a significant positive effect on LOAD incidence	

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Table 4: Studies with a "moderate" rating

Author&Title	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Ajana,S	France (ALIENOR study, 3 City-Bordeaux cohort) France	Cross-sectional study	184	82.3	31.5%/68.5%	Cognition	MMSE, Isaacs Set test (IST15), Benton Visual Retention test, Free and Cues Selective Reminding Test	Acuity: Measured but no mention of method	MPOD 0.5 degree, Z score (b)=0.12 (0.01, 0.23). MPOD 1 degree, Z score (b)= 0.11 (0.01, 0.22)	Yes- Between MPOD and Global cognitive z score	Moderate
2018							Summarized by a composite global cognitive Z-score	MPOD: Measured using confocal scanning laser Ophthalmoscope	Regression models		
Anstey, K. J	Australia (Australian Longitudinal Study of Aging)	Longitudinal	1823	77.77	51.2%/49.8%	Cognition and vision	MMSE, Verbal ability, Processing speed and Memory	Acuity: Distance using Snellen chart at 3m, Near at 20cm	N/A	Yes- significant moderate-sized association between rates of	Moderate
2003						Analysis of change using growth curves		using chart containg short passages in font size 5 to 18. Left and Right eye were tested separately. Score was the smallest font size	Latent growth curve analysis	change in Memory and Vision	
Anstey, K. J.; Luszcz, M. A.	Australia (Australian Longitudinal Study of Aging)	Longitudinal	3766	76.34	Wave 1: 56%/44%	Cognition	Verbal ability, Processing speed and Memory	Acuity: Distance using Snellen chart at 3m	Sensory status x group interaction for memory comparing participants who declined on Vision vs not declined	Yes- Decline in visual performance had a significant	Moderate
2001					Wave 3: 33%/67%				Wilks Λ=0.98, F=2.94, p<0.05, η2=0.054	effect on decline in memory performance	
									Two-factor repeated measures MANCOVA (sensory status*wave)		
Arnaoutoglou, N. A	Greece	Longitudinal, Case-control	Total 103 AD group n=32	AD 73.3	38%/62%	Vision	MMSE, Dementia blood screening, Reisberg AD scale, MRI and NINDS-AIREN to diagnose VaD	Colour vision using Ischihara color deficiency test	Difference in ischihara scores between two groups F(1,161)=9.558, p=0.003. AUC 0.819 (0.70, 0.93)	Yes- Ischihara a promising test to differentiate AD and	Moderate
2017			VaD group n=36	VaD 75.6				Consists of 38 pseudoisochromaric plates	ANOVA to estimate any sig variations between three groups within their age, MMSE, Resiberg and	VaD	
			Healthy controls n=35	Controls 74.2				6 errors suggests definite color blindness	total Ischihara score		
Arrighi, McLaughlin	US (National Health interview survey (NHIS)2001-05)	Cross-sectional study using nationally available survey data	23474	With dementia 80.72	With dementia 36%/64%	Vision	Self reported surveys. Was categorized as With dementia if patient reported :senile"	Self reported vision problems	Prevalence odds ratio 1.46 vision problems with funtional limitations among americans aged >60	Yes- Higher odds of vision problems being prevalent	Moderate
2010			Person with dementia 443	Without dementia 72.3	Without 39%/61%				with and without limitations related to dementia	among people with funtional limitation	
			Without dementia 23031								
Bauer K, Schwarzkopf L	German statutory health insurance fund data (2006)	Cross-sectional study using claims data from insurance company	37753	80.1	28.4%/71.6%	Comorbidity complex	ICD codes with diagnoses of dementia	ICD codes with diagnoses of severe vision reduction	OR 0.59 persons with severe vision problems were less likely to have dementia	Yes- significantly lesser odds of vison problems among	Moderate
2014			Dementia group 9139	81.6	25.4%/74.6%	Patient having a diagnosis of vision prob				cases with dementia	

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			Without dementia 28614	79.6	29.4%/70.6%	Using ICD codes					
Chen, S-C; Chang, Y-P	Taiwan National Health Insurance	Case-control. Randomized sample data of one million patients who made claims from insurance	Eye disease 4097	50 y	AMD 60.9%/39.1%	Alzheimer's disease	ICD codes with diagnoses of Alzheimer's Disease	ICD codes with diagnoses of AMD, Diabetic retinopathy and Glaucoma	Cumulative IR of 1.22% ADE among eye disease group vs 0.04% among controls	AMD, DR and Glaucoma were asso with an increased	Moderate
2016	Research database (HIRD)		Controls 20745		DR 52.9%/47.1%				HR for DR 39.31 (4.79, 332.67), HR for AMD 36.94 (4.62, 295.46), HR for Glau 34.08 (13.37, 86.84)	risk of Alzheimer's disease	
					Glaucoma 45.9%/54.1%						
Chen, S. P.; Bhattacharya, J	US (NHANES and NHATS)	Cross-sectional analysis of two datasets: 1999-2002 NHANES data and 2011-2015 NHATS data	NHANES 2975	72	48%/52%	Cognition	NHANES: Cog testing done using Digit symbol substitution test (DSST). </= 28 indicated poor cognition	NHANES: Dist-Snellen VA chart, Near- 5-line near vision card	NHANES: Dist VI b=-5.1 (-8.6, -1.6), OR 2.8 (1.1, 6.7), Near VI b= -3.8*, OR 1.7*, Subj VI b=-5.3*, OR 2.7*	Yes, VI significantly associated with worse cognitive fn	Moderate
2017			NHATS 30202	NA	42%/58%		NHATS: Probable/possible dementia vs No dementia based on NHATS classification scheme (self-report)	NHATS: Vision tested via questionnaires	NHATS: Subj Near VI: OR 2.6 (2.2-3.1), Subj Dist VI: OR 3.9 (3.4-4.4)		
Chriqui, E	Canada	Longitudinal cohort	150	86.7	27%/73%	Vision	MMSE	Dist vision- Snellen chart, 6m. VI was Vn <6/12	Presenting VI: 37.3 (29.1-46.1)	37.3% of dementia participants had VI.	Moderate
2017								Near vision- Topcon paragraph/lighthouse	VI after refraction: 20.1 (13.7-27.9)	Just prevalence study, no correlation tested	
De Celis, E. S. P	US (Hurria et al. JCO 2011 & 2016)	Cross-sectional analysis of 2 prospective studies	750	Median age 72 (65-94)	No info	Cognition	Blessed OMC test (>11 on this test)	self-reported visual impairment (fair, poor or blind)	OR for association between VI and Cognition deficit.	Older pts with cancer and hearing/visual impairment	Moderate
2017									Univariate model: 2.0 (0.8-4.7), Multivariate model: 1.9 (0.75-5)	at higher risk of funtional and cognitive deficits	
de Kok, D. S	New Zealand (Life and Living in Advanced Age: A	Cross-sectional baseline data	661	Maori: 82.3	40%/60%	Cognitive decline	MMSE-blind version, cognitive test for the visually impaired	Acuity: ETDRS	Generalized linear models and structural equation modelling	No direct association found in this population	Moderate
2017	Cohort Study in New Zealand (LiLACS NZ)			Non-Maori: 84.6	47%/53%				Maori: Dis VA (logmar) b=0.103(-0.21,0.42), Self reported VI b= 0.110 (-0.030, 0.251)		
									Non-Maori: Dis VA (logmar) b=0.197 (-0.12, 0.51), Self-report VI b= 0.065 (-0.056, 0.187)		
Elliott, A. F	US	Cross-sectional cohort	238	No info	26%/74%	Vision	MMSE	Dist VA: ETDRS, Near VA: Lighthouse near VA card. VI < 20/40 one eye	β MMSE in Multiple linear Regression for Predicting Vn; Dist VA: -0.015, Near VA: -0.013 both(p 0.00)	Cognitive status contributed to the prediction of VI	Moderate
2015								Mars contrast sensitivity to assess CS. Score worse than 1.50 in one eye	Contrast sen: 0.17 (p0.00)		
Elliott, A. F.; McGwin, G., Jr	US	Longitudinal case-control	78	REC: 79.2	23.1%/76.9%	Cognition	Self-report and MMSE	Distance: ETDRS	Intervention group: MMSE baseline 20.2 (4.4), follow-up 19.4 (4.8), p=0.05	Vision-enhancing interventions did not lead to short	Moderate
2009			64	Delayed corrc: 78	25%/75%			Near: Lighthouse Near Visual Acuity Test	Control group: MMSE baseline 21.7 (4.5), follow-up 20.5 (5.5), p=0.015	term improvements in physical function/cognitive status in sample.	

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			30	Cataract Sx: 81	26.7%/73.3%				Pre&post-intervention measures of Phy fn &cognitive status compared within treatmnt groups paired t-tests		
			15	No cat sx: 87	13.3%/86.7%						
Elyashiv, S. M	US (The Religious Orders Study and The Memory snd Aging Project)	Longitudinal cohort study of aging and AD	2716	78.13	28.6%/71.4%	Cognition	Global cognitive scale (GCS), summary of raw scored of 19 tests incl episodic, semantic and working mem	Tested with both eyes open with a card held at 14 inches and measured	Mixed model for assoc with global cog score: VA b=-0.03312 (p<0.0001)	Significant positive correlation between GCS and VA	Moderate
2014							Scores converted, avergd and normalised-categorised into no,mild,mild CI+other diag, AD and AD+diag	at 7 increments from 20/40 or better to 20/400 or worse	Pearson and spearmann correlation coeff to assess relationship btwn GCS and VA		
Elliott, A. F.; McGwin, G., Jr	US	Cross-sectional	382	Range: 60 to <=100	No Impairment 26.8/73.2	Vision	MMSE	Acuity- Lighthouse near visual acuity test	Multiple linear regression used to estimate assoc btwn vision and cognition	VI affected HRQoL both with and without CI	Moderate
2009					VI only 5.9%/94.1%				NHVQoL, mean: No VI+CI=72.8, VI only=75.2, CI only=76.5, VI+CI=65.2. p=0.02		
					CI only 16.7%/83.3%						
					VI+CI: 20.4%/79.6%						
Feeney, J.; Finucane, C	Ireland (The Irish Longitudinal Study on Aging)	Cross-sectional cohort	4453	4th quartile 62.8	42.5/57.5	Cognition	MMSE, MoCA, Neuropsyh battery, CAPI for word-recall, CAMDEX-R for picture memory	Acuity:ETDRS, MPOD measured using Macular metrics densitometer	Multivariate regression to analyse assoc btwn Cog test performance and MPOD(continuous variable)	Lower MPOD associated with poorer performance in MMSE & MoCA	Moderate
2013				3rd quartile 62.4	46.3/53.7			Pt was asked if diagnosed with ARMD and Diabetes	MMSE: b= 0.48 (0.06, 0.90) p=0.026, MoCA: b=0.83 (0.15, 1.50) p=0.016, Memory: OR=2.24 (1.2, 4.15)p0.01		
				2nd quartile 62.2	53.9/46.1						
				1st quartile 62.3	53.8/46.2						
Formiga, F	Spain (NonaSantfeliu population-based study)	Cross-sectional design	186	93.06	23.5/76.5	Cognition	MEC (Spanish version of the MMSE), Functional status: Lawton-Brody Index (LI) and Barthel Index (BI)	Acuity: Snellen Eye test	T-test and chi-square test. MEC (± SD): group with visual deficit:16.5 ± 11, w/o visual deficit:23.6 ± 11, p<0.001	Functional status and cognition significantly asso with sensory	Moderate
2006									Lower LI was asso with: VI OR= 1.86 (1.44, 2.39) and combined impairment OR=1.995 (1.32, 3.01)	Impairment	
Friedman, D	US (The Salisbury Eye Evaluation in Nursing Home	Cross-sectional	656	83.4	25/75	Vision	MMSE	ETDRS charts or Lea symbols and grating acuity tests using Teller cards	Bland-Altman plot to measure how strongly grating acuity and recognition acuity relate to one another.	Grating acuity testing appears to be useful in	Moderate
2002	Groups (SEEING))								MMSE (5-point decline)with Teller acuity cards: OR= 2.67 (2.14, 3.31). Teller and recognition ICC=0.79	cognitively impaired individuals	
Frost S, Guymer R	Australia (Australian Imaging Biomarkers and Lifestyle	Cross-sectional case control	123	AD 70.2	59.1/40.9	Vision (ARMD)	MMSE and PET scan	Digital, non-stereoscopic retinal colour photographs, both eyes with a	Logistic regression: Medium Drusen: [N (%)]: CN: 13 (13), AD: 9 (41), OR: 4.69 (1.67-13.13)	A logistic model for early AMD found a	Moderate
2016	study of aging (AIBL))			CN 71.3	39.6/60.4			Canon CR-1 Non-Mydriatic retinal camera with digital Canon camera at the back	Early AMD: [N (%)]: CN: 3 (3.0), AD: 8 (36), OR: 18.67 (4.42-78.80)	significant association with AD diagnosis (p < 0.0001)	
								Grader (masked) evaluated images for signs of ARMD			

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Fuller, S. D	US (The American Community Survey (ACS))	Cross-sectional survey data	721053 5	61.31	47.1/ 52.9	Cognitio n	Self-report: Because of a physical, mental or emotional condition, does this person have difficulty	Self report: serious visual impairment- is this person blind or does	Cog Imp: NSI reference group, SHI: 45-64y: 3.55-3.67, 65-79y: 2.86-2.95, 80y+: 1.81-1.86, SVI alone: 45-64y:	Cognitive difficulty greatly associated with sensory impairment when	M od er ate
2018							concentrating, remembering or making decisions? Yes/No	he/she have serious difficulty seeing even when wearing glasses?yes/no	4.42-4.57, 65-79y: 4.71-4.91, 80y+: 2.33-2.44, DSI: 45-64y: 12.03-13.77, 65-79y: 11.19-12.84, 80y+:4.69-5.35	compared to NSI, esp greatest with SVI alone.	
Garin, N	Spain (COURAGE)	Cross-sectional cohort	50-64: 1760	57	47.7/ 52.3	Vision	Global standardised score from 5 performance tests: learning and short-term mem, attention & working	Obj: Snellen, Subjective: two questions to the participant, 1. how much	Asso btwn poor VA and Cognition (OR): Dist VA: 1.47 (1.17-1.86), Near VA: 1.72 (1.46-2.02)	Objective and subjective poor distance and near visual acuity	M od er ate
2014			>65: 1865	74.9	44.9 5/55. 05		mem and language. Lower score indicates worse cognitive functioning	diffculty seeing object or recognising at 20 m 2. at arm's length	Subjective distance VA: 2.05 (1.44-2.90), Subjective near VA: 2.96 (1.92-4.55)	were associated with worse cognitive functioning.	
Gaynes, B. I	US (Rush memory and aging project (1997-2007))	Cross-sectional cohort	714	VI presen t: 82	26.1/ 73.9	Cognitio n	Global cog fn summary contructed from 19 neuropsychological tests that assessed episodic, semantic,	Near VA: Rosenbaum Pocket Vision Screener positioned 14 inches from	X2 test, pooled t-test for equal variance. Divided analytic cohort into VI present and VI not present. Linear reg	Each unit higher in neuroticism level worsened association	M od er ate
2014				VI not presen t: 78.8	23.6/ 76.4		working memory, perceptual speed and visuospatial ability. Raw scores converted to Z-scores & averag	the eye. VI was defined as BCVA < 20/40	Global cog fn: VI b=-0.147, SE=0.086, p=0.09, Neuroticism b=-0.001, p=0.04, VIxNeurot b=-0.017, SE=0.005, p0.001	between vision impairment and lower global cognitive func	
Guthrie DM, Davidson JGS	Canada (Resident Assessment Instrument for Home Care)	Cross-sectional study	Home care: 291,824	82.8	38.9/ 61.1	Physical , social and emotion al	Cognitive performance scale (CPS) which has 4 terms pertaining to short-term memory, independence	Data from RAI-HC, particpants interviewed on the use of glasses	HC: CI+VI highest rate of procedural mem problems (56.1%) and highest prevalence of depression (28.2%)	CI+DSI had highest rate of functional impairments, communication	M od er ate
2018	(RAI-HC)		Longter m care: 110,578	86.9	30.5/ 69.5	function ing	in eating, expressive communication and decision making. Scale of 0-6, intact to very severe impairmnt	0 adequate Vn and 4 severely impaired(can see only lights and colours)	LTC: All three impairments showed highest rates on indicators of delirium.	problems and difficulty understanding others	
Hofer, SM	Nordic Research on Aging Study (NORA)	Cross-sectional study	1041	75 y	42.7/ 57.3	Cognitio n, Vision	WAIS-R- Digit Symbol test, Raven's Progressive Matrices- test of inductive reasoning,	Acuity: Computerized refractometer (Topcon RMA2300).	Of the correlations with the set of cognitive variables, the highest correlations were with corrected vision	no consistent associations were found across sensory, balance,	M od er ate
2003							Word Fluency Task,The Digit Span Forward and Backward test, Reaction time to vis/audi stimuli	Composite scores derived using avg of right and left vision scores	between .20 and .40 (visual choice reaction time), with the highest asso on visuomotor performance tests.	strength, and cognitive domains.	
Jefferis, J. M; Taylor, JP	Uk (Newcastle 85+ Cohort study)	Cross-sectional study	839	Diag catara ct: 85.5	30/7 0	Cognitio n	Standardized MMSE (sMMSE), Mmblind (blind version)	Review of family practice records, any pre-existing diagnosis of glauco,	Diagnosed Glau: 1.76 (1.05, 2.95), Diagnosed Glau or drops: 1.73 (1.05, 2.85)	Association between recorded glaucoma diagnosis and cognitive	M od er ate
2013				Diag Glauc oma: 85.4	36/6 4			cataract, cataract surgery (uni/ or bilateral) sight impaired registration		scores and record of previous cataract surgery & better cognition	
Luo Y, He P	China (Second National Sample Survey on Disability)	Cross-sectional study	250752	Total: 72.9	47.5/ 52.5	Cognitio n	Dementia ascertained using combo of self-report/family member report and on-site psychiatrist eval	Acuity: Trained Ophthalmologist assessed VI according to WHO BCVA	Dementia: Only VI- 1.54 (1.27-1.86), p< 0.001, Only HI- 1.04 (0.88-1.22), DSI- 1.63 (1.25-2.11), p<0.001	VI & DSI were morc likely to have severe to extremely severe	M od er ate
2018				With demen tia: 77.2	38.6/ 61.4		Questionnaire tested: poor memory, diff concentrating, diff controlling emotions, strange language/beha	criteria (low vision: 0.05 ≤ BCVA ≤ 0.29; blindness: no light perception		dementia than those without sensory impairment.	
							Positive on any one- then tyested by psychiatrist. Disablity diagnosis based on WHO DASII (score >52)	≤ BCVA < 0.05, visual field less than 10 degrees; the betterseeing eye)			
Rovner B, Casten R	US (AMD Trial)	Cross-sectional study	241	82.8	36.5/ 63.5	Vision function	Animal Fluency Test-brief assessment of cognitive function relevant to the completion of daily activities	Acuity: BCVA using ETDRS chart, NEI-VFQ near vision subscale	Predictors of Visual function: VA b= -1.93(−2.59 to −1.27), <.001, Animal fluency b=0.054 (0.019 to 0.090), .003	Visual ability was highly correlated with visual acuity. Multivariate	M od er ate
2011				Range 65 and older			Name as many animals as possible in 60secs. Requires semantic knowledge, speed mental processing	Contrast sensitivity using Pelli-Robson chart	CS b= 0.20 (−0.210 to 0.610), .33, PHQ-9 severity: −0.032 (−0.100 to 0.035), .35, Age b= −0.008 (−0.033 to 0.016)	model revealed coping startegies and cog fn contributed to Near Vn	

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Spier O, Fischer N	US	Cross-sectional study	190	81.6	30.5/69.5	Cognition	MMSE for the visually impaired (removed 8 items from the original MMSE) max score of 22	Corrected near VA was tested using the Jaeger chart. Refraction using	Asso VA & High MMSE blind score: Bad near VA (>J3): 1.0, good near VA (<J3); OR=3.18 (1.57-6.43) p0.001	Good VA and Wearing reading glasses were significantly associated	Moderate
2016				Range 75-101				Autorefractometer		with high MMSE-blind score	
Varin M, Kergoat MJ	Canada	Cross-sectional case-cohort	303	AMD 83.6	30.2/69.8	Total cognitive activities	Victoria Activity Questionnaire, which is a 70-item survey assessing older adults' participation in various	Acuity: ETDRS chart, Visual field: Humphrey frequency doubling	Total cognitive activities: Normal vision b=0 (ref), AMD: −4.19 (−5.96, −2.43) *P<0.05.	Patients with AMD and glaucoma participated in fewer cognitive	Moderate
2017				Glaucoma 77.5	40.9/59.1		cognitive activities over the past two years. Ex. Gardening, exercise,prep meals, doing housework,etc	technology full-threshold N-30 testing each eye. Medical chart reviewed	Glaucoma: −1.80 (−3.34, −0.26) *P<0.05.	activities compared to older adults with normal vision potentially	
				Normal 72.8	48.3/51.7		Cognition was tested with the MMSE-blind version	for diagnosis of AMD and any co-existing eye disease		putting them at risk for cognitive impairment	
Gussekloo J, de Craen AJ	Netherland (Leiden 85+ study)	Cross-sectional cohort	459	85	34/66	Cognition	MMSE, The 12-Word Learning Test was used as a visual test of long-term memory. Cognitive speed	Acuity: ETDRS chart	MMSE and VI: beta = -1.2 (-1.6 to -0.84), Immediate word-learning test beta= -1.3 (-1.8 to -0.69)	VI was associated with lower MMSE scores. Increasing VI was asso with	Moderate
2005							was measured with the Letter Digit Coding Test (processing speed)		Delayed word-learning test beta= -0.62 (-0.9 to -0.35), Letter digit coding test beta = -2.1 (-2.8 to -1.4)	woth poorer scored on memory and cog speed.	
									Stroop test beta = 4.7 (1.1 to 8.4)		
Hajek A, Brettschneider C	Germany (German Study on Ageing, Cognition and Dementia in Primary Care Patients)	Longitudinal cohort	2,394	82.5	34.2/65.8	Cognition	Cognitive activities included reading, writing, solving crossword puzzles, memory training, games	Patient graded their vision on three level Likert scale	Onset of mild visual impairment: beta (SE)= -0.065 (0.026)	Linear fixed-effects regression showed that the onset of severe	Moderate
2016							(e.g. card games, board games), playing music, and social engagement (e.g., in church, volunteering, in a club)	(with optical aid if necessary) as no impairment, mild, severe or profound VI	Onset of severe or profound visual impairment: beta (SE)= -0.376 (0.062)	VI was associated with a decline in physical & cognitive function score in the total sample	
Hidalgo J, Martinez I	Spain	Cross-sectional cohort	1160	73.3	44.1/55.9	Vision	Short portable Mental Status questionnaire	Acuity: Essilor visiotest instrument with snellen optotypes	Visual acuity < 6/18 in the better eye −0.344 (−0.392 to −0.295), Self-reported vision: average/poor/very poor	Using multiple regression analysis, the variables associated with VF status	Moderate
2009								Self reported vision as very good, good, average, poor and very poor.	−0.077 (−0.101 to −0.054), Dependence in activities of daily living: −0.121 (−0.160 to −0.081), Cognitive impairment −0.073 (−0.117 to −0.029)	were: visual impairment, self-reported poor vision, dependence in daily activities, cognitive	
										impairment, depressive symptoms, female gender and older age	
Thuan QP, Kiffley A	Australia (The Blue Mountains Eye Study)	Cross-sectional case control	3509	Late AMD: 79	34/66	Cognition	Modified MMSE: omitted vision-dependent items, with a max score of 22.	Acuity: LogMAR chart, Macula photo using Zeiss FF3 fundus camera	OR for MMSE 24-27 and: NO AMD 1, Late AMD 2.3 (1.1-5.0), Early AMD 1.0 (0.7-1.4)	Significant association found between Late AMD and cognitive impairment	Moderate
2006				Early AMD: 75.1	38.1/61.9			Amd lesions were assesed from the photos based on Wisconsin AMD grading system	For MMSE 0-23 and: NO AMD 1, Late AMD3.7 (1.3-10.6), Early AMD 1.4 (0.8-2.6)		
				No AMD: 65.7	43.7/56.3						
Uhlmann RF, Larson EB	US	Cross-sectional case control	174	77	42/58	Cognition	MMSE	Acuity: was measured with the Snellen and Rosenbaum methods for far and near vision.	Relative Odds for Dementia of Far-vision Impairment: 1.9 (95% CI = 0.8-4.6), RR of dementia and near Vn:	Visual impairment is associated with both an increased	Moderate
1991									20/100-20/200 1.5 (0.5-4.7), >=20/200 1.8 (0.5-6.1), RR of dementia and far Vn: 20/70-20/100 1.5 (0.3-6.1),	risk and an increased clinical severity of Alzheimer's disease	
									>=20/200 1.5 (0.3-6.0)	Consistent with the hypotheses that VI exacerbates Cog dysfucntion in dementia	

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Guthrie DM, Declercq A	4 countries	Cross-sectional	Canada: Comparison: 138 650, DSI 48 210	NA	F 67.2, 67.9	Cognition	Cognitive performance scale (CPS), captures issues with memory, independence in eating and decision-making	inter RAI assessment inleudes items measuring the presence of impairments	individuals with DSI were more likely than all others to have moderate to severe cognitive impairment (between 3.8% and 9.0% higher rate),	Yes. DSI significantly associated with CI	Moderate
2016			US: comparison 60 971, DSI 6577		F 66.8, 68.6		(Established validity against MMSE)	in vision. The vision item captures the person's ability to see close objects in adequate light while using their typical assistive device. Scored on a five-point scale where 0 = adequate, 1 = impaired, 2 = moderately impaired, 3 = highly impaired and 4 = severely impaired	highly compromised independence with ADLs (5.2% to 11.4% higher), instrumental ADLs (8.8% to 21.4% higher). Those with DSI were not more likely to experience social isolation		
			Belgium: comparison 500, DSI 256		F 74.9, 78.9						
			Finland: comparison 4765 DSI 1358		F 70.9, 76.2						
Harrabi H, Kergoat MJ	Canada	Cross-sectional cohort	420	AMD 82.5	25/75	Cognition	MMSE	Acuity: ETDRS chart with illuminated light box at 2m, at 1m if patient could	Model 1*: Binocular Logmar acuity, per 0.1 unit, -0.11 (-0.18, -0.04),	People with AMD, Fuch's corneal dystrophy, and glaucoma had lower cognitive	Moderate
2015				Fuch's 78.5	17/83			not see at 2m. Letter by letter scoring done, converted to logMAR	Model 2*: Log contrast sensitivity, per 0.1 unit, 1.40, (0.72, 2.13)	scores, on average, than controls (P < 0.05).	
				Glaucoma 76.6	42/58			CS measured using Pelli-Robinson chart 1m	Model 3*: Visual field in better eye, per 1 dB, 0.04, (-0.003, 0.08)	Scores were between 0.7 and 0.8 units lower than the control group	
				Control 74.0	41/59			Visual field: Humphrey frequency doubling technology (FDT)			
Holmen K, Anderssonn L.	Sweden (Kungsholmen project)	Case-control	Cog imapired 55	82		Loneliness, but measured interaction between	MMSE	Acuity: Snellen, self-report questions: Do you have any problems with your sight? Yes or No" was used to assess the problems	There were no significant differences in the cognitive groups between subjects with slight to moderate	NO. In a multiple regression analysis, higher MMSE score and visual improvement were significantly related to lower levels of self-reported loneliness among the elderly with their cognition intact, but not	Moderate
1994			Cog intact 92	83		Cognition & vision			habitual visual acuity (i.e. those who could read newspapers with glasses and visually orient themselves	among the subjects with impaired cognition.	
									without difficulty), and subjects with pronounced to severe visual impairment with residual vision (x2 = 2.28, p = 0.13). The same differences were found		
									when corrected visual acuity was compared analytically between the cognitive groups (x2 = 1.15, p=0.28)		
Su C, Lin C	Taiwan (Taiwan National Health Insurance program)	Cross-sectional cohort	Glaucoma 6509	59.4	45.5/56.5	Cognition	Dementia patients enrolled using ICD codes	Glaucoma cohort formed enrolling patients who were newly diagnosed with it	HR=1.13 (1.01-1.27)	The patients with glaucoma exhibited a significantly higher risk of	Moderate
2016			Control 26036	59.2	42.4/54.6					dementia than the individuals without glaucoma did (hazard ratio	
										[HR]¼1.13, 95% confidence interval [CI]¼1.01-1.27)	
Miyata K, Yoshikawa T	Japan (Fujiwara-kyo Study)	Cross-sectional cohort	2764	76.3	52.6/47.4	Cognition	MMSE	Acuity, Landolt ring chart at 5m, converted to logMAR. refractive errors were measured by an autoref/keratometer	OR for MCI: no cataract surgery 1.00 (ref) / cataract surgery 0.79 (0.64-0.97) p 0.025	The subjects who had prior cataract surgery had significantly lower odds ratio (OR) of having MCI	Moderate
2018								A prior cataract surgery was determined by the selfadministered questionnaire	OR for dementia: no cataract surgery 1.00 (ref) / cataract surgery 1.20 (0.64-0.97) p 0.36	Cataract surgery may be important in reducing the risk of developing MCI but not for dementia independently	
Ong SY, Ikram MK	Singapore (Singapore Malay Eye Study (SiMES))	Cross-sectional cohort	1032	Cognitive dysfunction 70.5	24/76	Cognition	Abbreviated Mental Test (AMT)	Acuity: number chart at a distance of 4m converted to logMAR. VI logMAR > 0.3 in the better eye	Cog Dysfunction OR: Model 2 :Myopia 1.78 (1.02-3.10) Hyperopia 1.11 (0.68-1.80) Emmetropia 1.0	Yes. Compared with individuals with emmetropia, persons with myopia were	Moderate

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2013				No Cog dysfunction 67.7	66.5/33.5		[cognitive dysfunction was defined as a score less than or equal to 6 of 10 for those with 0 to 6 years of formal education		Model 3 : Myopia 1.86 (1.01-3.42) Hyperopia 0.92 (0.54-1.55) Emmetropia 1.0	almost twice as likely to have cognitive dysfunction	
							and less than or equal to 8 of 10 for those with more than 6 years of formal education.]				
Soler V, Sourdet S	France	Cross-sectional cohort	1648	82.6	35.6/64.4	Vision	MMSE	Acuity. Snellen decimal chart for distant vision, Parinaud chart for near vision	model 1 for distant vision: MMSE OR (CI) 0.94 (0.92-0.97) p<0.001	Yes. Results show that visual impairment is independently associated	Moderate
2016							ADL, IADL, Fried score, SPPB class (Short Physical Performance Battery),	Amsler grid testing	model 2 for near vision: MMSE OR (CI) 0.92 (0.89-0.94) p<0.001	with lower educational level, cognitive impairment, and lower ADL-assessed autonomy	
Sun HP, Liu H	China	Cross-sectional cohort	4123	67.5	48.8/51.2	Cognition	Abbreviated Mental Test (AMT)	Objective refraction was measured using an autorefractor (Canon RK-5)	OR of cog dysfunction for myopia: Before propensity score matching, OR 1.98 (1.61- 2.44)	Results of this PSM analysis support a small, but positive association of	Moderate
2016								Spherical equivalent (SE) [Myopia: a SE of less than -0.50 dioptre (D) in the right eye. High myopia: a SE of less than -6.00 D.]	After matching: OR 1.31 (1.00- 1.71). Multivariate logistic regression	myopia with cognitive dysfunction among elderly Chinese in China	
									model on the association between myopia and cognitive dysfunction, OR (95%CI):		
									Myopia (Yes vs No) 1.52 (1.23-2.06), Age-related cataract (Yes vs No) 2.06 (1.57-2.98)		
Zheng DD, Swenor BK	US (The Salisbury Eye Evaluation Study)	Longitudinal cohort	2520	73.5	42/58	Cognition	MMSE	Acuity. ETDRS chart refraction was performed on participantswith	Worse baseline VA was associated with worse baseline MMSE score	Worse baseline VA was associated with worse baseline MMSE score	Moderate
2018								20/32 visual acuity orworse using a forced-choice procedure.	(r = −0.226; 95%CI, −0.291 to −0.16; P < .001). The rate of worsening VA was associated with	The rate of worsening VA was associated with a significant	
								Presenting binocular distanceVAwas converted to logMAR with a higher score indicating worse VA.	the rate of declining MMSE score (r = −0.139; 95%CI, −0.261 to −0.017; P = .03).	the rate of declining MMSE score	
Ward ME, Gelfand JM	US (Study of Osteoporotic Fractures (SOF))	Longitudinal cohort	1352	77.7	0/100	Vision	Global cognition (3MS), verbal memory (California Verbal Learning Test II, short form [CVLT]),	Contrast sensitivity, using a VISTECH (Hartford, CT) VCTS 6500 wall chart and light mete	OR of MCI/dementia for lowest quartile of baseline contrast sensitivity	These data support an association between impaired contrast sensitivity and future MCI/ dementia.	Moderate
2018							executive function (Trails B, Digit Span Forward, Digit Span Backward), and semantic memory (category and verbal fluency)	Self-reported age-related macular degeneration (AMD), glaucoma, and cataract	compared with women in the highest quartile: OR 2.16 (95% CI =1.58-2.96)		
							Clinical adjudication if person was positive in the above tests.		OR of MCI/dementia for every 10 point decrease in contrast sensitivity:		
									OR=1.21, 95% CI=1.13-1.31		
Mitoku K, Masaki N	Japan (Gujo City Long-Term Care Insurance database)	Cross-sectional cohort	1754	80.89/82.42	34.5/65.5	Cognition	The assessment measures include communication, short-term memory,	Acuity: No chart name provided	Logistic regression: OR of CI was 1.389 (95% CI 1.04-1.85) in those with VI	Yes. Supports the relationship between DSI and CI, increased risk of mortality in thise with DSI & CI.	Moderate
2016							location awareness and understanding of daily tasks, remembering own name and date of birth, and recognizing the	Other: Vision at baseline 5 levels, "normal sight", "able to see chart at 1m"	1.581 (95 % CI 1.24-2.01) in those with HI, and 2.349 (95 % CI 1.79-3.09) in those with DSI		
							season of year in addition to 18 problem behaviors	able to see chart in front, "very little sight", "indeterminable"			
							CI classified as none, mild, moderate, severe				
Jefferis JM, Clarke MP	UK	Longitudinal, cohort	112	80.7	45/55	Cognition	The ACE-R (which includes the MMSE), is scored from 0 (worst cognition) to 100 (best cognition);	logMAR visual acuity in the better eye corrected with up-to-date refraction and/or pinhole	Group, Mean+/-SD:Baseline, Postop, 1 Year. P Value*: 3-Way†, Baseline Vs Postop, Baseline Vs 1 Year, Postop Vs 1 Year	Significant improvements in ACE-R scores were seen between baseline and 1 year	Moderate
2015							38 of these points pertain to vision-dependent items and 62 to vision-independent items		Normal (n =46): 92.5+/-2.8, 92.6+/-3.8, 93.7+/-3.4, 0.018, 0.9, 0.009, 0.004::: Mildly impaired cognition (n =22): 84.5+/-1.4, 84.8+/-4.3, 88.0+/-3.5, <0.001, 0.75, <0.001 0.002	postoperatively (95% confidence interval for improvement, 0.5-2.8; P Z .005). Improved	

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									Moderately impaired cognition (n =23): 72.6+/-8.7, 73.6+/-10.4, 73.3+/-14.4, 0.75, --, --,--	cognition did not correlate with improved visual acuity (r Z 0.13, P Z .22)	
									All participants (N=91): 85.6+/-9.5, 85.9+/-10.0, 87.2+/-11.4, 0.004, 0.45, 0.005, 0.006, *P < .05 statistically significant		
Lindenberger U	Germany (BASE (Berlin Aging Study))	Longitudinal, cohort	516	85	NA	Cognition	MMSE	Acuity: Snellen decimal units. Close visual acuity (close vision) was measured separately for the left and the right eye with a standard reading table presented at reading distance. Distance visual acuity (distant vision) was assessed binocularly with a reading table presented to the participants at a minimum distance of 2.5 meters	Multivariate Multilevel Modeling Results With Control for Age, Time to Death, and Risk of Dementia Construct, Mean (Level, Linear change, Quadratic change), Variance (Level, Linear change), Age, Dementia status, Age X Linear Change, Residual variance.	Contrary to expectations, the correlations between cognitive and sensory declines were only moderate in size, underscoring the need to delineate both domain-general and function-specific mechanisms ofbehavioral when controlling for age at first measurement, distance to death, and risk of dementia senescence	Moderate
2009									Close vision: 47.33 (0.30), -1.37 (0.11), ---, 34.17 (2.95), 0.83 (0.16), -0.44 (0.04), ---, -0.03 (0.01), 25.87 (1.42)		
									Distant vision: 48.47 (0.32), -0.56 (0.11), ---, 28.14 (3.17), 1.20 (0.25), -0.59 (0.04), ---, ---, 44.79 (2.45)		
Hong T, Mitchell P	Australia (Blue Mountain Eye Study)	Longitudinal, cohort	2334	No Sensory impairment 66.9 (7.4),	No SI 58.4/ 41.6	Cognition	MMSE	Acuity: Distance VA at 244 cm followed by pinhole acuity (VA was recorded as the number of letters read correctly) If no letters could be read count fingers, hand movements, light perception, or no light perception	association between Decline >=3 MMSE blind score and visual impairment (adjusted for age and sex). OR after 5 years: Visual impairment 0.84 95%CI 0.40-1.79 Dual sensory impairment 1.41 95%CI 0.54-3.72 after 10 years: Visual impairment 1.09 95%CI 0.52-2.30 Dual sensory impairment 1.15 95%CI 0.28-4.73	No.The presence of VI, HL or DSI was not associated with possible cognitive decline over 5 years or 10years.	Moderate
2016				Visual impairment 74.3 (8.4),	VI 65.8/ 34.2					There were no changes to these findings after adjustment for other potential confounder	
				Dual impairment 80.4 (7.0)	Dual impairment 57/43						
Mandas A, Mereu RM	Italy	Case-Control	1168	78.4	29/71	Cognition	MMSE	Acuity: Snellen Chart	Total(n =1168), Control1 (n =436), All types of dementia (n =732), MCI (n =181), AD (n =230), MD (n =126), VD (n =195)		Moderate
2014								IOP, fundus examination under slit-lamp and fundu photograph	All vision disorders: No, n=297, n=433, n=117, n=135, n=68, n=113 yes, n=139, n=299, n=64, n=95, n=58, n=82 χ2, 9.37, 0.70, 5.87, 8.60,,6.12 P-value, 0.002, 0.40, 0.015,,0.003, 0.013 OD (95%CI), 1.5(1.1-1.9), 1.2(0.8-1.7), 1.5(1.1-2.1), 1.8(1.2-2.7), 1.5(1.1-2.2)	Subjects with any type of age-related vision disorders(n D438), cataract,	
										macular degeneration,glaucoma,and diabetic retinopathy,were more likely be depressed	
										and had significantly lower mini mental state examination scores than Control 2	
Mendola JD, Cronin-Golomb A	US	Case-Control	188	NA	NA	Vision	Blessed Dementia Scale (BDS)	Acuity: Snellen, Contrast sensitivity: two-alternative forced-choice format,	Test, Prevalence in AD, Correlation with BDS, % variance from BDS.	The results indicate that visual dysfunction, especially on Backward Masking, is a common sign of AD.	Moderate
1995								Stereopsis: binocular disparity stereopsis, as a clue to depth perception(Stereo Optical)	Backward Pattern Masking: 58% r=0.22; p=0.106; df=55, 13%, Gollin Incomplete Pictures: 30%, r= 0.33; p= 0.03; df= 17, 6%		
								funduscopy, evaluation of fixation, pupillary function, extraocular movement,	Luria Mental Rotation: 39%, rb= 0.74; p= 0.004; df= 36, 13%		
								Backward Masking, Gollin Incomplete Pictures. City University Colour Vision.	Log Contrast Sensitivity at 1 cpd: 39%, rb= 0.69; p= 0.002; df= 17, 52%, Log Contrast Sensitivity at 2cpd: 6%, rb= 0.55; p= 0.022; df= 17, 7%		
								Luria Mental Rotation. Moeny Road Map. Local speed discrimination and Global motion detection. Critical Flicker fusion.	Money Road Map: 29%, rb= 0.37; p=0.028 ; df= 35, 17%		
Magalhães, M. O. D. C	Brazil	Cross-sectional cohort	466	71.4	44.4/ 55.6	Cognition	CAMCOG (Cambridge Cognitive Examination)	Self-report	logistic regression model for dementia:	Yes. Study showed strong association between dementia and sight-impairment	Moderate

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2008							Dementia was diagnosed in individuals who fulfilled CAMDEX criteria for definitive, probable or possible dementia through a combination of clinical criteria, organicity indices ≥5 and CAMCogG<80.		sight impairment OR (95%CI): 1.83 (1.15-2.91)		
									Pvalue 0.011		
Ong SY, Cheung CY	Singapore (Singapore Malay Eye Study (SiMES))	Cross-sectional cohort	1179	with cognitive dysfunction : 71.2	21/79	Cognition	The Abbreviated Mental Test (AMT) is a 10-question test of general cognitive function	Acuity: LogMAR number chart (lighthouse international)	Age-related eye disease, model 1, model 2, model 3.		Moderate
2012				Without cognitive dysfunction : 68.0	67.7/33.3		Items assess orientation (3 points), semantic knowledge (1point), episodic memory (3 points), delayed recall	Cataracts were assessed from lens photographs, Glaucoma diagnosed and classified using International Society Geographical and Epidemiological Ophthalmology scheme based on gonioscopy	Cataract: 1.52 (1.02-2.27), 1.42 (0.93-2.16), 1.44 (0.94-2.22), AMD: 1.71 (0.89-3.29), 1.44 (0.71-2.91), 1.36 (0.34-5.45)	Yes. VI due to cataract (OR=2.75; 95% CI, 1.35-5.63)	
							(1 point), picture naming (1 point), and attention (1 point)		Moderate or severe DR: 2.43 (1.13-5.20), 2.26 (1.02-5.00), 5.57 (1.56-19.91)	were more likely to have cognitive dysfunction. Only moderate to severe diabetic retinopathy was independently associated with cognitive dysfunction	
									Glaucoma: 1.68 (0.81-3.46), 1.61 (0.75-3.45), 2.09 (0.52-8.31)		
Mangione CM	US	Cross-sectional cohort	472	73	40/60	Cognition	TICS questionnaire	Acuity: Snellen (results were transformed into percentages of functional central	multivariate linear regression , percent increase in ODDS of abnormal TICS score (95% CI):		Moderate
1993								vision loss following the formula outlined in the Physician's Desk Reference for Ophthalmology)	binocular vision loss(per 10% decline in vision) -9 (-23,9), percent field loss (per 1 category decline) 13 (-13,48) cataract (per eye) 18 (-12,58) ocular miotics 37 (-17,126)	No. An association between low TICS score and visual disorders were not seen	
								Visual field: Humphrey field analyser			
Walter Wittich	Canada (COMPASS-ND study)	Longitudinal, cohort	109	72.94	No info	Cognition	Montreal Cognitive Assessment (MoCA)	Reading acuity (MNRead) and Contrast Sensitivity (Mars Test)	Individuals with AD had significantly lower contrast sensitivity than those with MCI and SCI (p = .04, $\omega 2$ = .04;), after adjusting for age, sex and education. No differences in visual acuity were observed p = .46, $\omega 2$ = .004.	Yes. Declines seen in contrast sensitivity in individuals with AD relative to those with MCI and SCI.	Moderate
2019										Study also observed higher rates of reduced reading acuity in the AD and MCI groups relative to those reported the general population.	
Brenowitz	US (Health ABC Study)	Longitudinal, cohort	1810	76.7	39.4/60/6	Cognition	Dementia was defined as meeting one or more of the following criteria through study	Bailey-Lovie distance VA test, Pelli-Robson contrast sensitivity	Multivariable model for incident dementia with VA: IRR= 1.26 (0.90-1.77), CS: HR=1.11 (0.88-1.38)	No. Vision impairment and contrast sensitivity independently was not significantly associated with incident dementia	Moderate
2019							Year 15 (2011–2012):(a) hospitalization with dementia as a primary or secondary diagnosis,				
							(b) documented use of dementia medication, or (c) clinically meaningful Modified Mini-Mental State Examination (3MS)				
							decline from Health ABC baseline (1.5 SD, race-stratified)				
Davies-Kershaw	UK (English Longitudinal Study of Ageing)	Longitudinal cohort	7685	NA	44/56	Cognition	3 methods to identify individuals with dementia: a physician diagnosis of dementia	Rate their eyesight (using glasses or corrective lenses as usual)	Cross-sectional analysis using Wave 7 measures adjusted for covariates: compared to normal VA, moderate OR = 2.04 (1.36–3.07)	Yes. Older adults with vision impairment have higher rates of dementia crosssectionally (all ages) and are at greater risk of incident dementia longitudinally (<70 only)	Moderate
2918	Wave 7						that the participant or a caregiver reported between Wave 3 (2006–7) and 7 (2014– 15):	as excellent, very good, good, fair, poor, or registered blind	poor OR = 4.02 (2.64–6.13) p<0.001		
							a score less than 3.5 on the adaptive Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE)	Combined groups into 3 categories for analysis (excellent or very good=normal,	Longitudinal analysis using Wave 2 measures individuals in the younger group (50–69) and with moderate (HR=1.78, 95% CI=1.04–3.04) and poor (HR=3.60, 95%CI=1.10–11.78) selfrated vision were at greater risk of developing dementia than those with normal self-rated vision		

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							or prescriptions for anticholinesterase inhibitors, N-methyl-D-aspartic acid receptor antagonists,	good or fair=moderate, poor or registered or legally blind=poor or blind)			
							and other relevant medication (galantamine, rivastigmine, memantine, donepezil, tacrine) to indicate dementia	Self-rated vision measure from Wave 7 (2014–15) for cross-sectional analysis and from Wave 2 (2004–05) for the longitudinal analysis			
Gui-Ying Cao	China (China Health and Retirement Longitudinal Study)	Longitudinal, cohort	7269	60.2	56.2/43.8	Cognition	Episodic memory (measured as the average of immediate and delayed recall scores of ten Chinese nouns),	VI was assessed by two self-reported questions, about distance visual acuity for seeing faces on the other side of the street	Compared with older people with no VI, (1) those with DVI only were associated with poor episodic memory (β=–0·076, p<0·0001)	Yes. VI is associated with an increased risk of poor cognitive function	Moderate
2018							mental intactness (measured using some components of the Telephone Interview of Cognitive Status (TICS) battery),	and about near visual acuity for reading newspapers. Individuals were assigned to one of four categories: no VI,	mental intactness (β=–0·074, p<0·0001) and global cognition ((β=–0·089, p<0·0001)		
							and global cognition (the sum of episodic memory and mental intactness scores)—was evaluated and followed up every 2 years	distance vision impairment (DVI), near vision impairment (NVI), or both distance and near vision impairment (DNVI)	Those with NVI only were associated with poor mental intactness (β=–0·031, p=0·0001) and global cognition (β=–0·032, p=0·0224)		
									Those with DNVI were associated with poor episodic memory (β=–0·106, p<0·0001), mental intactness (β=–0·107, p<0·0001), global cognition (β=–0·105, p<0·0001)		
Ali. G. Hamedani	US (Medicare Claims data from 2014)	Cross-sectional cohort	472,871	>65	34/66	Cognition	ICD-9 diagnosis codes for dementia or cognitive impairment	ICD-9 diagnosis codes for blindness/ low vision	Among the low vision/blindness group, the prevalence of dementia or CI was 50%.	Yes. There was a significant association between low vision/blindness and dementia or AD	Moderate
2019									Blindness/low vision was also associated with a greater odds of Alzheimer's disease (AOR 1.44 95% CI: 1.415–1.464)		
									and all-cause dementia (AOR 1.239, 95% CI: 1.223–1.254)		
Preeti Gupta	Singapore (Singapore Epidemiology of Eye Disease Study)	Longitudinal, cohort	682	67.3	55.6/44.4	Cognition	CI was assessed using the validated Abbreviated Mental Test (AMT), defined as scores of	Fundus images using Canon DGI camera, graded by trained graders at University of Sydney	Those with any DR had higher odds of incident CI (OR=2.32, 95% CI 1.07 to 5.03), and those with moderate or worse DR also had a higher likelihood of	Yes, DR, particularly at the more severe stages, is associated with	Moderate
2019							≤6 and ≤8 for those with 0–6 and >6 years of formal education, respectively	using the modified Airlie House classification system and categorised as none (Early Treatment of Diabetic Retinopathy Study (ETDRS) level 10),	developing CI (OR=3.41, 95% confidence interval 1.06 to 11.00, p-trend=0.021), compared with those without DR	increased risk of developing CI, independent of vision and other risk factors	
								minimal/mild (level 20–35) and moderate or worse DR, (level 43–90) using data from the better eye			
Jost. B. Jonas	China (The Beijing Eye Study)	Cross-sectional cohort	3127	64.2	43.4/56.6	Cognition	MMSE, assessed as cognitive function score (cfs). Categorised as mild (cfs 23-19), moderate (cfs 18-10)	Autorefractometry, Presenting, uncorrected and best corrected VA,	Better cognition (i.e., higher CFS) was significantly associated with better best corrected visual acuity (r2=0.38),	Yes	Moderate
2018							severe (cfs <9)	fundus photo, SD-OCT	smaller amount of undercorrected visual acuity, lower prevalence of primary angle-closure glaucoma, and		
									thicker subfoveal choroidal thickness		
Allen T.C. Lee	China	Longitudinal, cohort	15576	74.5	36.2/63.8	Cognition	ICD-10 codes or Clinical dementia rating 1 to 3	Acuity: Snellen E chart, converted to LogMAR	Participants with incident dementia had poorer visual acuity at baseline than those without (adjusted HR= 5.88, 95% CI=4.04–8.57).	Yes, Moderate-to-severe visual impairment could be a	Moderate
2020									Incident Dementia at years 4- 6 of Follow-up in Participants With Mild (HR=1.19, 95% CI 0.86–1.65, p= 0.31)	potential predictor and possibly a risk factor for dementia	
									Moderate (HR= 2.09, 95% CI=1.47–2.97) or Severe VI (HR= 8.66, 95% CI=4.60–16.30) at baseline		
Cecilia. S. Lee	US (Adult Changes in Thought (ACT))	Longitudinal, cohort	3877	NA	42/58	Cognition	Cognitive Abilities Screening Instrument scores 85 underwent a standardized diagnostic evaluation, including physical and neurologic examinations and a neuropsychological test battery	ICD-9 codes for diagnosis of glaucoma, AMD, DR and cataract	The recent and established hazard ratio were 1.46 (P 5.01) and 0.87 (P 5.19) for glaucoma,	Yes, Increased AD risk was found for recent glaucoma diagnoses,	Moderate

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2019							Dementia diagnoses using NINCDS-ARDRA criteria, Our primary outcome was either probable or possible late-onset clinical AD		1.20 (P 5.12) and 1.50 (P .001) for AMD, and 1.50 (P 5 .045) and 1.50 (P 5 .03) for DR	established AMD diagnoses, and both recent and established DR	
Moon Jeong Lee	US (National Health and Nutrition Examination Survey (1999-2006 cycles))	Cross-sectional cohort	5795	Non-VI 70.3	45/55	Cognition	Memory or confusion complaints were defined as present if participants responded "yes" to the question—	VA was assessed using the built-in chart in an auto-refractor (NIDEK ARK-760; Nidek Co Ltd, Tokyo, Japan)	Individuals with VI were more likely (OR = 1.3, p = .049) to report cognitive complaints as compared with those without VI	Yes	Moderate
2019				VI 77.8	39/61		"(Are you/is survey participant) limited in any way because of difficulty	VI was defined as autorefractor corrected VA worse than 20/40 in the better-seeing eye			
							remembering or because (you/s/he) experience(s) periods of confusion?"				
Ruby Yu	China	Longitudinal, cohort	1949	76.1	23.1/76.9	Cognition	5-item Abbreviated Memory Inventory for Chinese (AMIC). Scored range from 0–5 (1 point for each item; 0—best to 5—worse). An AMIC score ≥ 3 is predictive of MCI	Questionnaire: "Do you have any difficulty seeing things?""very good" and "good" = "robust"; "fair", "not too well", "poor", and "very poor" = "poor"	Poor vision (OR 2.2 95% CI 1.8–2.7) at baseline was significantly associated with an increased risk of at least 3 SMCs at follow-up	Yes	Moderate
2019											
Michio Maruta	Japan	Longitudinal cohort	2190	78.9	20.6/79.4	Cognition	Dementia scale labelled Level 0 to Level IV and level M based on symptoms and the	VA assessed at 5 levels: normal vision (there is no hindrance in daily life),' 'able to see vision testing chart at a distance of about 1 m,'	DSI associated with a higher cumulative dementia incidence compared to no sensory impairment (log-rank χ2 = 39.92; P < 0.001)	Yes.	Moderate
2020							necessity for care. Level II and greater during the 8-year follow-up period was considered	'able to see vision testing chart at a distance of in front,' 'very poor eyesight,'	DSI is the greatest risk factor for developing dementia among sensory impairments (HR, 1.45; 95% CI, 1.22–1.71; P < 0.001)	Older adults with sensory impairments have a high incidence of dementia, with DSI presenting the greatest risk	
							"incident dementia"	and 'undecidable due to difficulty in communication.	Older adults with VI were found to be more likely to have day-night reversal symptoms when dementia occurs		
Ann E. M. Liljas	UK (English Longitudinal Study of Ageing)	Longitudinal cohort	4621	64.9	45/55	Cognition	Working memory, Executive function. Results from the three cognitive tests available were summed,	VA: Self-report asking participants whether their eyesight was excellent, very good, good, fair, or poor using glasses or corrective lens if they normally do so	Compared with people with good vision, poor vision was asso with worse cognitive function	Yes, aging adults with individual and combined impairments in hearing and vision had greater risks of worse cognitive performance at 6-year follow-up compared with those with good sensory function	Moderate
2020							providing a cognitive score (range 0-80), with a lower score indicating worse cognitive function		B = 1.61, 95% CI = [0.92, 2.29] adjusted for age, sex, baseline cognition		
									Compared with no sensory impairment, dual sensory impairment was associated with worse cognition (B = 2.30, 95% CI = [1.21, 3.39]		
Stephanie Chen	US (National Health and Aging Trends Study (NHATS))	Longitudinal cohort	7075	NA	40/60	Cognition	1. Doctor told the sample person that he/she had dementia or AD, 2. A score that indicates probable dementia on the AD8 Dementia Screening Interview	Self-reported distance and near VI	Participants with self-reported visual impairment were at significantly higher risk of developing	Yes, self-reported visual impairment in the US Medicare	Moderate
2019							3. Cognitive tests that evaluate the sample person's memory and orientation		probable or possible dementia over subsequent followup (HR=4.4, CI: 3.9-5.0, p<0.001), compared to those without visual impairment	population may be associated with greater risk of cognitive decline	
									This association persisted after full adjustments for covariates (HR=2.1, CI: 1.8-2.5, p<0.001)		
Ali G. Hamedani	US (Medicare beneficiaries)	Cross-sectional cohort	773905	NA	NA	Cognition	ICD-9 codes and procedure code-based algorithms	ICD-9 codes and procedure code-based algorithms	Low vision was associated with greater odds of incident hip fracture (AOR 1.13, 95% CI: 1.04–1.22	No.	Moderate
2019									and incident anxiety (AOR 1.11, 95% CI: 1.05–1.18 but not incident depression or dementia.		

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TIEN Y. WONG	Singapore (SEED)	Cross-sectional cohort	10020	58.9	49.3/ 50.7	Vision	Abbreviated Mental Test (AMT), which consists of 10 questions of general cognitive function	VA ETDRS chart at 4m converted to LogMAR	For best-corrected bilateral visual loss, cognitive impairment (OR=2.07; 95% CI=1.60–2.68)	Yes	Moderate
2019									was a significantly associated risk factor. CI was also significantly associated with higher risk of presenting bilateral VI or blindness (OR=2.15; 95% CI= 1.75-2.63)		
Bonnielin Swenor	US (Health ABC study)	Longitudinal cohort	2444	74	47.8/ 52.2	Cognition	Digit symbol substitution test and 3MS. Incident cognitive impairment was defined as	VA: Bailey-love chart, CS: Pelli-Robson chart	VA impairment HR=1.55, 95% CI 1.12, 2.14 vs No VA impairment, CS impairment HR=1.33	Yes, visual acuity, contrast sensitivity, and stereo acuity are	Moderate
2019							a 3MS score <80 or a decline in 3MS >5 points following Year 3		95% CI 1.13, 1.55 vs No CS impairment	risk factors for cognitive decline	

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Table 5: Studies with a "weak" rating

Author&Title	Country (Study name if applicable)	Study design	No. of participants	Mean age (M/F)	Gender (M/F)	Outcome variable	CI: type of measurement/evaluation	VI: Type of measurement/evaluation	Point estimates and Analysis performed	Summary of association	Quality of study
Bayer, A. U.; Ferrari, F	Germany	Case control	228	With AD+ Glau 72.9	10%/ 19%	Glaucoma	Diagnosis of Alzheimer's Disease according to the NINADS-ADRDA classification	Acuity - Snellen	25.9% occurrence rate of probable glaucoma among AD	Yes- Significantly higher occurrence rate of glaucoma	Weak
2002			AD 112	With AD+ no Glau 71.4	28%/ 55%			Visual fields - Humphrey Field Analyser	5.2% occurrence rate of probable glaucoma among healthy controls	among AD patients	
			Control 116	W/o AD + No glau 68.1	38%/ 72%			Optic disc evaluation			
				W/o AD + Glau 70.1	2%/4 %						
Cigolle, C	US (1995-2010 of Health and Retirement Study)	Longitudinal cohort, Adults >65y	8847	No info	No info	Cognition	Performance based measure (Telephone interview) to determine cognitive function (0-27)	No info	Vison and hearing predicted cognitive decline (p<0.001)	Cognitive function declines in an accelerating fashion, with older age	Weak
2013	Abstract Only									and visual impairment predicting decline over time	
Feeney, J.; Finucane, C	Ireland (The Irish Longitudinal study on Ageing)	Cross-sectional study	4281	No info	No info	Cognition	MMSE, MoCA, CTT, CRT, SART, Prospective memory, picture memory, word recall and visual reasoning	heterochromatic flicker photometry (HFP) - a non-invasive method of	Linear, Negative binomial and logistic regression. MPOD men=0.20, SD=0.15.	MPOD is significantly associated with cognitive function	Weak
2013	Abstract only							gauging the density of Macular Pigment	One SD inc in MPOD assoc with few errors on MoCA: b=-0.03 p<0.01 & MMSE: b= -0.05 p=<0.05	among older adults.	
									Faster time to complete CTT2: b= -0.02 p<0.01, faster CRT: b= -0.06 p<0.01, better word recall:b=.07 p<0.05		
									Fewer SRT omission errors: b= -0.04 p<0.05, success on prospective memory task: b= 0.14 p<0.01		
Jefferis, J. M; Clarke MP	UK	Cross-sectional study	112	80.7	NA	Cognition	Revised Addenbrook's Cognitive examination (ACE-R)	Acuity: LogMar chart, Cataract density graded using LOC III system	ACE-R scores were split into those items requiring vision and those items not requiring vision, both subscores	Better general cognition (ACE-R total score) was associated with	Weak
2013	Abstract Only								were significantly and similarly associated with visual acuity (p=0.008 and p=0.001 respectively).	better vision (p=0.001)	
Maharani A, Pendleton N	Survey of Health, Ageing and Retirement in Europe 2002-2014	Cross-sectional study	24116	NA	NA	Cognition	Using a summary cognitive score from the questions on episodic memory, verbal fluency, and numeracy.	Self-reported hearing and visual quality. Sensory function recoded into 2:	Older adults with single SI: b=-0.064 (-0.072 to -0.056), with DSI: b=-0.229 (-0.247 to -0.211)	Older adults with single and dual SI showed significantly lower	Weak
2017	Conference abstract							Excellent,v good & good into Good sens fn and fair & poor into Poor sens		cognitive function compared to those without SI	
Miyata K, Mine M	Japan (Fujiwara-Kyo study)	Cross-sectional study	668	76.3	NA	Cognition	MMSE	Cataract surgery history- obtained using a self-reported questionnaire	CI for Cataract surgery group OR: 0.82 (0.68, 0.99) p=0.042 when compared to no cataract sx group	Cat Sx group significantly lower odds for CI than the no Cat Sx group	Weak
2016	Conference abstract										
Nael V, Delcourt C	France (Three-City Study)	Longitudinal cohort study	7722	Range: 65 & older	NA	Cognition	Incident dementia over the 12-year follow-up was actively screened for (diagnosis code)	Dist VF:loss self-reported, as inability/difficulty recognizing a face@ 4 m.	Cox regression models: Near VI: HR=1.30 (1.05, 1.61), Dist VF loss: HR= 1.47 (1.02, 2.11) compared to those	Both near VI and distant VF loss were associated with an increased	Weak
2017	Short oral presentation							Near VI:presenting binoc visual acuity <20/30 @ reading distance of 33cm.	without visual loss	risk of dementia	

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Setti A, Lough man J	Ireland (The Irish Longitudinal Study on Ageing (TILDA))	Cross-sectional study	5021	NA	NA	Cog nitio n	MMSE and MoCA	Vision measured subjectively and objectively (no other info)	Poor hearing and immediate recall of orally presented words: b = -0.46, poor hearing and delayed recall b= -0.60	Vision was a significant predictor of visually presented tasks	We ak
2013	Abstract only								Poor hearing and category fluency b =-1.96, poor hearing & memory (OR 0.23)& absent-mindedness (OR2.18).	but also of category fluency.	
Zheng DD, Christ SL	US (The Salisbury Eye Evaluation Study)	Longitu dinal cohort	2520	Range 65-84		Visi on	MMSE	Acuity: ETDRS chart	VA and MMSE score worsened over time (VA (Log MAR) intercept=.004, slope=.022 for VA; MMSE intercept=27.3, slope=-.59; all P<.001)	Both presenting VA and MMSE score worsened over time	We ak
2017	Conference abstract								The intercept of VA trajectory is statistically significantly associated with the intercept of MMSE trajectory (r=-.267, se=.029, p <.001)		
									suggesting that worse baseline VA is associated with worse baseline MMSE score.		
Ihle A, Oris M	Switzerland (LIVES study)	Cross-sectional cohort	2812	77.9	52.7/ 47.3	Cog nitio n	Processing speed- Was assessed by the Trail Making Test part A, Cognitive flexibility- Was assessed by the Trail Making Test part B	Self-report on a 3-point Likert-type scale whether their	Stability in (inter)relations of the other cognitive and the sensory abilities across the age tranches in old age were not moderated by general cognitive ability, educational level, nor general health status (all ps > .225)	No.Present data do not support the view of a generally increased relation of cognitive and sensory abilities in old age.	We ak
2015				range 65-101			Verbal abilities- Was assessed by administering the Mill Hill vocabulary scale	current eyesight allowed them to read a newspaper by choosing one of the following			
								0 D 'no'; 1 D 'yes, but with difficulties'; or 2 D 'yes, without difficulties.'			
Soto-Perez-de-Celis E	US	Cross-sectional case control	750	72	56/44	Cog nitio n	Score of 11 in the Blessed Orientation-Memory-Concentration test (BOMC).	Self-reported visual impairment based on their rating of their eyesight	OR for VI and: IADL Dependence: 1.9 (1.2-3.2) <.01, Poor Physical Function: 1.9 (1.1-3.3) .03 , Possible CI: 1.9	VI was associated with IADL dependence, poor physical function and	We ak
2018				Range 65-94				(with glasses) as "excellent," "good," "fair," "poor", "totally blind"	(0.7-4.8) 0.20, Depression: 2.5 (1.4-4.3) <.01, Anxiety: 1.4 (0.8-2.4) .19. OR for DSI and: IADL dep 2.8 (1.5-5.3) <0.01	depression. Dual SI was assicated with IADL dependence, anxiety, depression and CI	
									Poor Physical Function: 1.7 (0.9-3.4) .10, Possible CI: 3.2 (1.3-8.1) .01, Depression: 2.5 (1.3-4.8) <.01, Anxiety: 2.3 (1.2-4.2) .01		
Mahar ani A, Dawes P	HRS, ELSA, SHARE	Cross-sectional cohort	13,1 23 HRS	67.8/3 2.2	41.8/ 58.2	Cog nitio n	Episodic memory: In all surveys, the interviewer reads a list of 10 common nouns to the respondent	In HRS and ELSA, self-reported vision quality was collected in all seven waves using the question: 'Is your eyesight [using glasses or corrective lens as usual] excellent (1), very good (2), good (3), fair (4) or poor (5)?'.	HRS: Single impairment β −0.15 (0.02)*	Yes. Those with sensory impairment are at a greater risk of developing cognitive impairment	We ak
2018			11,4 17 ELS A	64.8/3 5.2	45.6/ 54.4		then asks the respondent to recall as many words as possible from the list in any order twice: immediately after the respondent heard the complete list (immediate recall) and at the end of the cognitive function module	In SHARE, we used the two self-reported measures of visual function	Dual impairment β −0.25 (0.04)*	and may show a faster trajectory of cognitive decline that those without sensory impairment	
			21,2 65 SHA RE	64.8/3 5.2	45.5/ 54.5		(delayed recall).	that are present in all waves: distance eyesight and reading eyesight.	ELSA: Single impairment β−0.14(0.02)*		
									Dual impairment β −0.35 (0.05)*		
									SHARE: Single impairment β −0.26 (0.01)*		
									Dual impairment β −0.68 (0.03)*		
Jefferi s JM, Coller ton J	UK (Newcastle 85+ study)	Cross-sectional cohort	839	no info	no info	Cog nitio n	sMMSE	Data collection of general practice records on whether the participant was	Median (inter-quartile range) sMMSE scores were 25 (22-29) for SI and 28 (25-29) for non-SI participants (P= 0.006).	It is important to consider the possibility of vision impairment in older	We ak
2012								registered as blind (severly sight impaired) or partially sighted (sight impaired), or neither (by a consultant ophthalmologist)	SI participants had lower subscale scores on tasks requiring vision (P< 0.001 for each) but also for some subscale	people when carrying out the MMSE and to consider using the MMblind.	
									scores not obviously requiring vision: orientation (P= 0.018) and repetition (P= 0.030).	Beyond this however, visual impairment may be a marker for cognitive impairment.	
MacD onald SWS	Canada	Longitu dinal, cohort	408	NIC 75.44	NIC 35.5/ 64.5	Cog nitio n	Episodic memory-The word recall task (Hultsch, Hertzog, & Dixon, 1990) was used to assess episodic memory	Acuity: Vision-Binocular-corrected distance visual acuity was measured at 3 meters using the Snellen chart.	Age-related change in sensory function.	Decline in visual acuity was also a notable predictor	We ak
2018				SA-MCI :76.71	SA-MCI : 40.5/ 59.5		Inductive reasoning-The Letter Series test (Thurstone, 1962) was used to assess inductive reasoning.	Snellen acuity fractions ranged from 0.20 to 1.	Variables, Intercept γ00, Slope γ10, SE Slope, p	of being classified as SA-MCI or MA-MCI	

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				MA-MCI: 75.68	MA-MCI : 48.2/ 51.8		Perceptual speed-The WAIS-R Digit Symbol Substitution task (Wechsler, 1958) was used to assess perceptual processing speed		Olfaction: 7.05, -0.043, 0.008, < .001		
							Verbal fluency-The Controlled Associations test from the ETS kit of factor-referenced cognitive tests (Ekstrom, French, Harman, & Dermen, 1976) was used to assess verbal fluency.		Distance Vision: 0.946, -0.004, 0.001, < .001		
							Vocabulary-A recognition vocabulary test, combining three 18-item multiple-choice tests from the ETS kit of factor-referenced cognitive tests (Ekstrom et al., 1976), was used to assess vocabulary		Audition: 32.84, 0.796, 0.059, < .001, γ00 = Average sensory function centered at the grand mean of age (74.17 years; SD=9.20); γ10 = slope reflecting the average rate of linear change per additional year of age		
Miyata K, Obayashi K	Japan (HEIJO-KYO Cohort)	Cross-sectional cohort	945	71.7	46.8/ 53.2	Cognition	MMSE	Acuity: using the Landolt ring chart. A better value of LogMAR was used for analysis.	Logistic Regression Analysis for the Association Between Ocular Status and Cognitive Impairment	Yes. ORs for cognitive impairment were significantly lower in the pseudophakic group than in the phakic group	Weak
2015								Questionnaire asking participants about pseudophakia/phakia	Age-adjusted, Multivariate OR for cognitive impairment: Model 1, Model 2, Model 3		
								Confirmed by ophthalmologist using slit-lamp	Phakia (no previous cataract surgery): 1.00 (ref), 1.00 (ref), 1.00 (ref), 1.00 (ref),		
									Pseudophakia (previous cataract surgery): 0.66 (0.45, 0.98), 0.66 (0.44, 0.98), 0.64 (0.43, 0.95), 0.64 (0.43, 0.96)		
									p: 0.038, 0.039, 0.026, 0.031		
Mine M, Miyata K	Japan (Fujiwara-kyo Study)	Cross-sectional cohort	2818	76.3	52.7/ 47.3	Cognition	MMSE; analyzed the MMSE excluding the following vision-related five items: "naming two objects," "following a 3-step command," "reading and following instruction," "writing a sentence," and "visual reconstruction" and the maximum score for this was 22 points.	Acuity: Landolt ring chart at 5m (converted to logMAR) [mild visual impairments (>0.2 logMAR)]	Associations Between Visual Acuity and Cognitive Impairment (multivariate regression model):	Yes. Subjects with mild visual impairments had 2.4 times higher odds of	Weak
2016									OR (CI) for different BCVA in the better eye (logMAR) groups:	having cognitive impairment than those without visual impairment	
									≤0: 1.0 (reference), 0-0.1: 1.8 (1.2-2.7) p 0.007 0.1-0.2: 1.7 (0.8-3.5) p 0.187 >0.2: 3.3 (2.1-5.4) p 0.005		
Steffi G. Riedel-Heller	Germany (LEILA 75+ and AgeCoDe)	Longitudinal, cohort	3,199	79.3	34.7/ 65.3	Cognition	No data	Self-report at baseline	Vision impairment (HR 1.19, 95%CI 1.01-1.42, p=0.043) and the combination of both (HR 1.47, 95%CI 1.18-1.83, p=0.001)	Yes. Vision impairment is independently associated with incident dementia. Looking at the combination we could show that individuals suffering from both are at highest risk for developing dementia	Weak
2019									on dementia incidence adjusted for baseline gender, age, education, marital status, depression, diabetes and cardiovascular risks (smoking, hypertension)		
Asri Maharani	US (Health and Retirement Study)	Longitudinal cohort	19618	57.8	44.6/ 55.4	Cognition	Telephone Interview for Cognitive Status (TICS): include episodic memory, serial of 7 subtraction, and counting backward tests	Self report: "Is your eyesight [using glasses or corrective lens as usual] excellent (1), very good (2), good (3), fair (4) or poor (5)?"	VI and risk of possible CIND HR 1.351 (1.267, 1.441) p< 0.001 when compared to no VI. VI and risk of probable	Yes. Self-assessed sensory impairment is independently associated with cognitive decline and incident possible CIND and probable dementia	Weak
2019	Waves 3 (1996) to 12 (2014)						categorized those scoring 0 to 6 points on the 27-point TICS scale as having probable dementia	Further categorized sensory impairment in the simultaneous model into no impairment,	dementia HR 1.255 (1.074, 1.466) p=0.004 when compared to no VI		

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							7 to 11 as having possible CIND, and 12 to 27 as having normal cognitive function	single (vision or hearing) sensory impairment, and dual sensory impairment (impairment in both senses).			
Phillip Hwang	US (GEM study)	Longitudinal cohort	2827	NA	NA	Cognition	Incident dementia over seven years of follow-up was based on a clinical diagnosis of dementia using DSM-IV criteria	Self-report	For all-cause dementia, the adjusted hazard ratio was 1.27 (95% CI: 1.02, 1.59) for single sensory visual or hearing impairment,	Yes, Dual visual and hearing impairment was strongly associated with increased risk of all-cause dementia and AD	Weak
2019							Alzheimer's disease was determined using NINCDS-ADRDA criteria		and 1.70 (95% CI: 1.18, 2.45) for dual visual and hearing impairment, compared to no sensory impairment		
Virginie Naël	France (The Three-City-Alienor Cohort)	Longitudinal cohort	7460	NA	NA	Cognition	4 cognitive domains (global cognition (MMSE and MMSE-blind), verbal fluency (IST),	At baseline, near VI was measured using the Parinaud scale	Participants with near VI and distance VF loss had lower baseline performances in verbal fluency,	Yes but only baseline. Participants with	Weak
2019							executive function (TMT) and visuospatial abilities (BVRT)) were assessed up to 6 times over 12 years of follow-up	at a standardized reading distance of 33 cm and distance Visual Function (VF) loss was	global cognition, executive function and visuospatial abilities. Regarding changes over time,	visual loss had lower baseline performances in several cognitive tests	
								self-reported, defined as inability or difficulty in recognizing a familiar face at 4m	MMSE where participants with mild near VI exhibited a faster cognitive decline (b= -0.02, p=0.04)		
M.Q.Li	China	Cross-sectional cohort	10116	NA	NA	Vision	AD8 questionnaire	No info	Risk factor for cognitive impairment: visual acuity loss 1.383 (1.188, 1.610)	Yes, The old people who suffered from a variety of chronic diseases including, poor eyesight were prone to cognitive impairment	Weak
2019											
Melanie Varin	US	Cross-sectional cohort	365	AMD: 83.1	32.4/67.6	Cognition	6 cognitive tests orally: the 1-minute verbal fluency test (letter and category versions)	Binocular visual acuity was measured using the ETDRS chart at 2 m	People with glaucoma showed lower scores on 3 cognitive tests than the group with normal vision:	Yes	Weak
2019				Glaucoma:78.1	43.7/56.3		the digit span test (forward and backward versions), and the logical memory test (immediate and 30-minute delayed recall)	VF measured using HFA, CS measured using the Pelli Robson chart at 1 m	the digit span forward and backward tests (b = -0.8, 95% CI -1.5, -0.2 and b= -0.7, 95% CI -1.3, -0.1, respectively)	People with glaucoma showed lower scores on cognitive tests	
Peiyuan Qiu	China (Chinese Longitudinal Healthy Longevity Survey)	Longitudinal cohort	3859	74.5	48.7/51.3	Cognition	Chinese Mini-Mental State Examination (C-MMSE)	Self-report	NA	Yes, visual impairment was a risk factor for cognitive decline	Weak
2019											

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2 & 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	e-supplement 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7, 8 & 9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10, 11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11, 12, 13, 14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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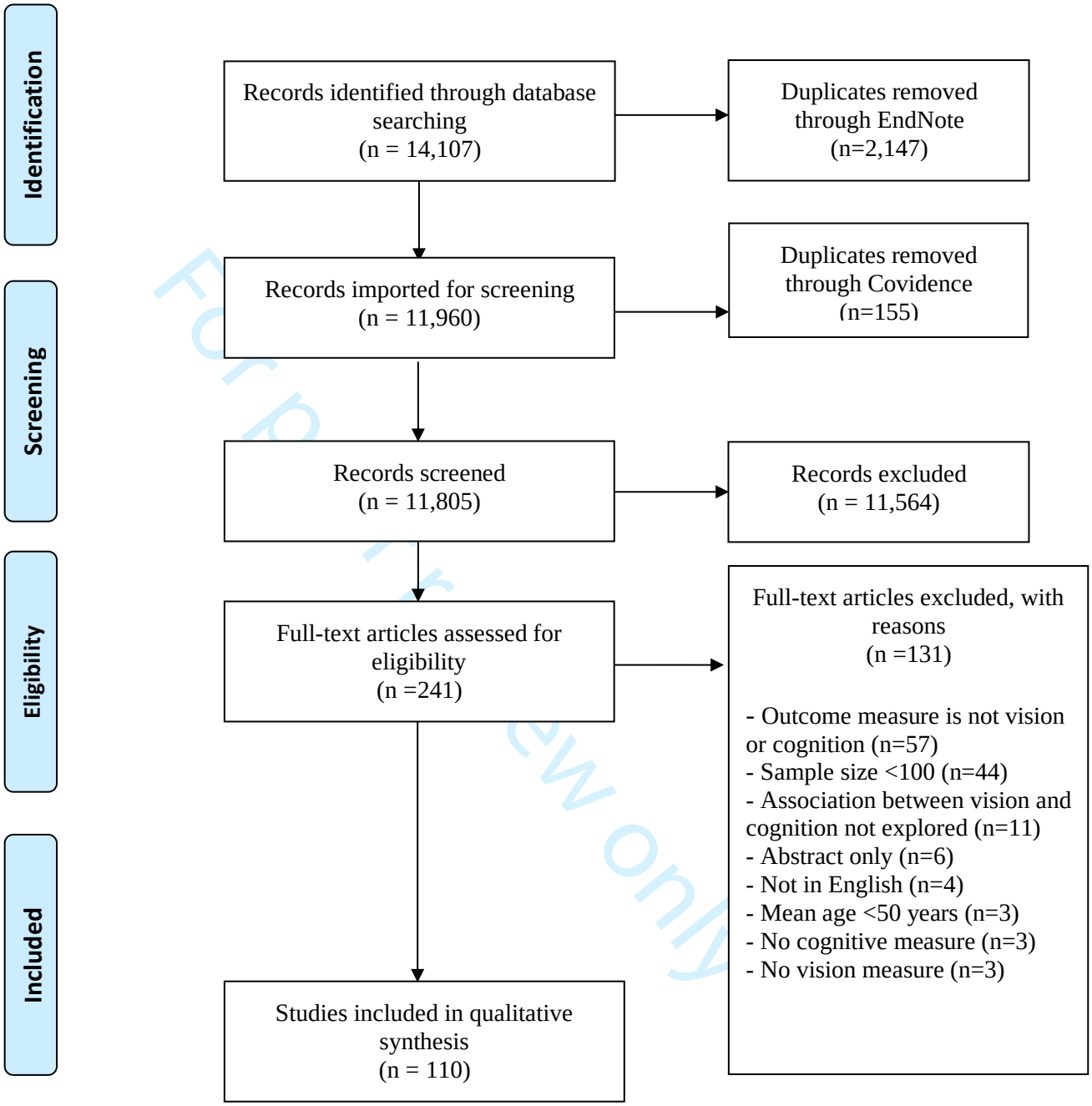
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e-Supplement 2 – Ovid Medline Search Strategy

1. exp Aged/ or exp Geriatrics/
2. (ageing or latelife or laterlife or "late life" or "later life" or elder* or Geron* or Geriatr* or senior*).tw.
3. (old* adj3 (age* or adult* or population* or communit* or people* or person*)).tw.
4. ((aging or aged) adj3 (adult* or population* or communit* or people* or person*)).tw.
5. 1 or 2 or 3 or 4
6. exp Cognitive Aging/ or Cognition/ or exp Cognition Disorders/ or exp Executive Function/ or Memory Disorders/ or exp Neuropsychological Tests/ or Dementia/ or exp Alzheimer Disease/
7. (cognitive or cognition or executive function* or executive control* or executive dysfunction* or executive impairment* or memory or dementia* or alzheimer*).tw.
8. 6 or 7
9. exp Vision, Ocular/
10. exp Vision Disorders/
11. exp Diagnostic Techniques, Ophthalmological/
12. exp Visual Perception/
13. exp Visually Impaired Persons/
14. (vision or visual or visually or occular or Ophthalmol*).tw.
15. 9 or 10 or 11 or 12 or 13 or 14
16. 5 and 8 and 15
17. Animals/ not humans/
18. 16 not 17
19. Epidemiologic Studies/ or exp Observational Study/ or exp Observational Studies as Topic/ or exp Cohort Studies/
20. ("epidemiologic study" or "epidemiologic studies" or "cohort study" or "cohort studies" or longitudinal or prospective* or "observational study" or "observational studies").tw.
21. 19 or 20
22. 18 and 21

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e-supplement 3: FIGURE DESCRIBING A PRISMA FLOWCHART



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The PRISMA for Abstracts Checklist

TITLE	CHECKLIST ITEM	REPORTED ON PAGE #
1. Title:	Identify the report as a systematic review, meta-analysis, or both.	1
BACKGROUND		
2. Objectives:	The research question including components such as participants, interventions, comparators, and outcomes.	2
METHODS		
3. Eligibility criteria:	Study and report characteristics used as criteria for inclusion.	2
4. Information sources:	Key databases searched and search dates.	
5. Risk of bias:	Methods of assessing risk of bias.	
RESULTS		
6. Included studies:	Number and type of included studies and participants and relevant characteristics of studies.	3
7. Synthesis of results:	Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals.	
8. Description of the effect:	Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients.	
DISCUSSION		
9. Strengths and Limitations of evidence:	Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence)	4
10. Interpretation:	General interpretation of the results and important implications	
OTHER		
11. Funding:	Primary source of funding for the review.	4
12. Registration:	Registration number and registry name.	